

WAVE REPORT

The Forrester WaveTM: Integrated Software Delivery Platforms, Q2 2023

The 13 Providers That Matter Most And How They
Stack Up

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Summary

In our 26-criterion evaluation of integrated software delivery platform (ISDP) providers, we identified the most significant ones and researched, analyzed, and scored them. This report shows how each provider measures up and helps modern application development leaders and DevOps platform team leaders select the right one for their needs.

Additional resources are available in the [online version](#) of this report.

ISDPs Provide A Strong Platform Foundation But Still Need Best-Of-Breed Tools

In our reports, [The Future Of DevOps](#) and [The Forrester Guide To DevOps](#), Forrester interviewed dozens of enterprise software development leaders from diverse companies ranging from Netflix to TransUnion to understand the current state of DevOps, its relevance, and next steps enterprise leaders were taking to further advance their practices. In addition, we asked which tools they were using, how effective the tools were, which were using [ISDPs](#), and how well those ISDPs were performing. We learned that many enterprises have shifted to platform teams to support their DevOps efforts and many platform leaders are evaluating and using ISDPs, but few are using all of their ISDP vendor's capabilities. They explained that while these ISDP tools hold great potential for a complete end-to-end toolchain, many individual capabilities fell short of enterprise needs, forcing these leaders to source those capabilities from separate best-of-breed vendors.

Our reference customer survey backs this up by indicating that enterprises are using (on average) only 53% of the capabilities they are paying for, with the remaining capabilities either being delivered by a separate vendor or simply not used. This means that ISDP vendors are not only competing with each other but also competing with the best-of-breeds, which is why the bar has risen so high for this evaluation. ISDP vendors recognize these challenges and are responding with a number of innovative solutions that will make them more attractive than a traditional DIY toolchain but open and extensible enough that end users can add the tools they want without sacrificing capability. The challenge is striking the right balance to create overall value.

As a result of these trends, ISDP customers should look for providers that:

- **Offer a strong core of CI/CD capabilities.** Continuous integration (CI) and continuous delivery (CD) are the core capabilities that the rest of the software delivery toolchain is built around. Several vendors offered build optimization algorithms that slash build and unit test times by intelligently building and testing only the code that changed. Deployment automation is also a differentiating capability, with only a few vendors offering advanced capabilities out of the box (OOTB) such as blue/green, canary, and feature management deployment strategies.
- **Provide connected platforms that best-of-breed tools can augment.** ISDPs provide engineering leverage in the form of preintegrated toolchains that enable work to easily flow from one process to the next, where administration occurs at

the platform level versus the tool level, and reduced context switching improves the developer experience. Often, platform leaders want to override the default capability with third-party tools that they believe offer superior capability, which can lead to a bit of a balancing act. That's why some ISDP vendors are heavily investing in improvements to their end-to-end platform, while others are banking on investments in third-party connectivity and partnerships.

- **Help enterprise leaders shift from DevOps to DevSecOps.** Forrester fields calls every day from enterprise development leaders that are transforming their DevOps practice into a development, security, and operations (DevSecOps) practice to make security a first-class concern for the entire development organization. ISDP vendors are responding with capabilities that better secure software delivery pipelines, including integrated secrets management, signed pipelines, and signed artifacts. Reference customers noted that not all of these tools are direct replacements for best-of-breed (best-of-breed security tools often having a significant advantage over bundled ones), but end users like them as forms of additional assurance and coverage.

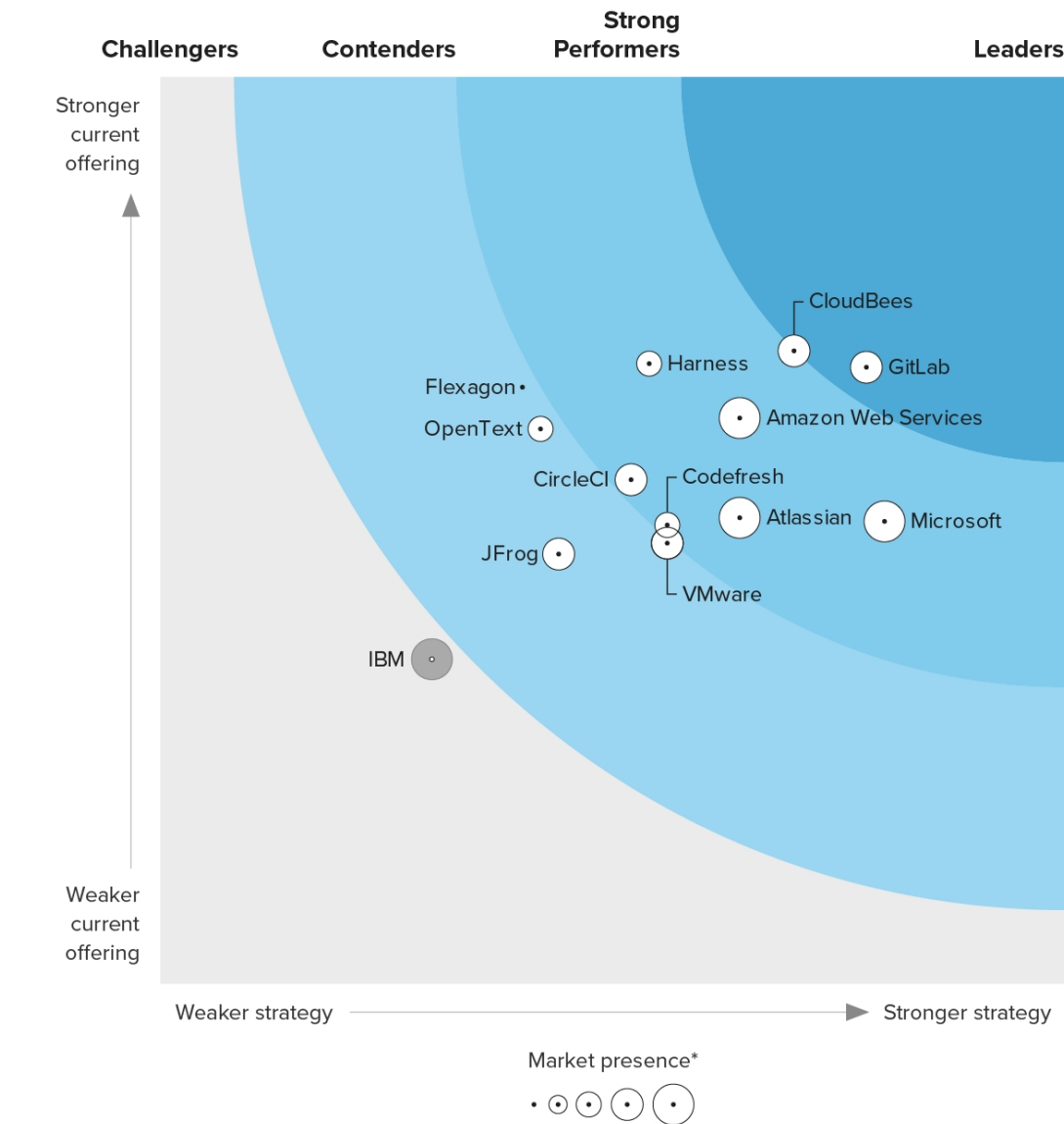
Evaluation Summary

The Forrester Wave™ evaluation highlights Leaders, Strong Performers, Contenders, and Challengers. It's an assessment of the top vendors in the market; it doesn't represent the entire vendor landscape. You'll find more information about this market in Forrester's [Now Tech: Integrated Software Delivery Platforms, Q2 2022](#).

We intend this evaluation to be a starting point only and encourage clients to view product evaluations and adapt criteria weightings using the Excel-based vendor comparison tool (see Figures 1 and 2). Click the link at the beginning of this report on [Forrester.com](#) to download the tool.

Figure 1
Forrester Wave™: Integrated Software Delivery Platforms, Q2 2023

THE FORRESTER WAVE™
Integrated Software Delivery Platforms
Q2 2023



*A gray bubble or open dot indicates a nonparticipating vendor.

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Figure 2**Forrester Wave™: Integrated Software Delivery Platforms Scorecard, Q2 2023**

	Forrester's weighting	Amazon Web Services	Atlassian	CircleCI	CloudBees	Codefresh	Flexagon	GitLab
Current offering	50%	2.57	3.12	2.78	3.49	2.53	3.29	3.40
Platform capabilities	20%	3.50	3.00	2.00	2.50	3.00	2.50	3.50
CI/CD capabilities	25%	2.00	2.60	3.00	4.00	4.00	3.20	3.40
Development, security, and operations	10%	1.00	3.00	2.00	2.00	2.00	3.00	4.00
Extensibility	10%	3.00	3.40	3.00	3.40	1.00	3.00	3.00
Governance and compliance	10%	3.00	5.00	3.00	5.00	1.00	3.00	3.00
Infrastructure management	10%	2.20	3.80	3.80	5.00	1.80	3.40	3.00
Product support	10%	3.00	3.00	3.00	3.00	3.00	5.00	3.00
Support for test automation	5%	3.00	1.00	3.00	3.00	1.00	5.00	5.00
Strategy	50%	3.20	3.20	2.60	3.50	2.80	2.00	3.90
Vision	20%	3.00	3.00	1.00	3.00	3.00	3.00	3.00
Roadmap	20%	3.00	3.00	3.00	3.00	3.00	3.00	5.00
Partner ecosystem	10%	5.00	5.00	1.00	5.00	1.00	1.00	3.00
Innovation	25%	3.00	3.00	3.00	3.00	3.00	1.00	3.00
Community	15%	3.00	3.00	3.00	5.00	3.00	1.00	5.00
Pricing flexibility and transparency	10%	3.00	3.00	5.00	3.00	3.00	3.00	5.00
Market presence	0%	5.00	4.50	3.50	3.50	2.50	1.00	3.50
ISDP revenue	50%	5.00	5.00	4.00	4.00	2.00	1.00	4.00
Number of enterprise customers	50%	5.00	4.00	3.00	3.00	3.00	1.00	3.00

All scores are based on a scale of 0 (weak) to 5 (strong).

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	Forrester's weighting	Harness	IBM*	JFrog	Microsoft	OpenText	VMware
Current offering	50%	3.42	1.79	2.37	2.55	3.06	2.43
Platform capabilities	20%	4.00	1.50	2.00	3.00	2.50	2.00
CI/CD capabilities	25%	4.20	2.80	2.40	2.40	3.40	2.40
Development, security, and operations	10%	3.00	1.00	4.00	4.00	1.00	3.00
Extensibility	10%	1.00	1.40	4.20	4.20	3.40	3.00
Governance and compliance	10%	3.00	3.00	1.00	1.00	5.00	3.00
Infrastructure management	10%	4.20	1.00	1.00	1.80	2.20	1.80
Product support	10%	3.00	1.00	3.00	1.00	3.00	3.00
Support for test automation	5%	3.00	1.00	1.00	3.00	5.00	1.00
Strategy	50%	2.70	1.50	2.20	4.00	2.10	2.80
Vision	20%	3.00	1.00	1.00	3.00	1.00	3.00
Roadmap	20%	3.00	1.00	1.00	3.00	3.00	3.00
Partner ecosystem	10%	3.00	3.00	3.00	5.00	3.00	3.00
Innovation	25%	3.00	1.00	3.00	5.00	3.00	3.00
Community	15%	1.00	3.00	3.00	5.00	1.00	3.00
Pricing flexibility and transparency	10%	3.00	1.00	3.00	3.00	1.00	1.00
Market presence	0%	3.00	4.50	4.00	5.00	2.50	4.00
ISDP revenue	50%	3.00	5.00	4.00	5.00	4.00	4.00
Number of enterprise customers	50%	3.00	4.00	4.00	5.00	1.00	4.00

All scores are based on a scale of 0 (weak) to 5 (strong).

*Indicates a nonparticipating vendor

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Vendor Offerings

Forrester evaluated the offerings listed below (see Figure 3).

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Figure 3
Evaluated Vendors And Product Information

Vendor	Product evaluated
Amazon Web Services	AWS Code Services
Atlassian	Atlassian Open DevOps
CircleCI	CircleCI
CloudBees	CloudBees Platform
Codefresh	Codefresh
Flexagon	Flexagon FlexDeploy
GitLab	GitLab
Harness	Harness Software Delivery Platform
IBM	IBM UrbanCode Deploy
JFrog	JFrog Software Supply Chain Platform
Microsoft	Microsoft Azure DevOps and GitHub Enterprise
OpenText (formerly Micro Focus)	Micro Focus ValueEdge
VMware	VMware Tanzu Application Platform

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Vendor Profiles

Our analysis uncovered the following strengths and weaknesses of individual vendors.

Leaders

- **GitLab leads with a balance between solid platform capabilities and overall strategy.** GitLab is known for being the vendor that made end-to-end OOTB toolchains hip again and has led the industry toward consolidated ISDPs. GitLab’s strategy includes an on-par vision to deliver an excellent developer experience without sacrificing security or compliance. Its innovation is also good, going beyond traditional developers to include AI/ML engineering. GitLab is an open core product that not only invests heavily in the open source software (OSS) community but also enables its customers to contribute to the product, earning it

high scores for community. Pricing is completely transparent and includes a free tier. Its roadmap gets leading scores and includes enhanced supply chain security, enhanced UI, granular security and compliance controls, and pipeline security — all things enterprises need.

GitLab's platform cohesion received high scores for simplicity of setup and operation as both an on-premises product and a software-as-a-service (SaaS) product. Though GitLab comes with its own GitLab version control system (VCS), it supports complete integration with all major Git providers, earning it high scores for version control. GitLab gets high scores for DevSecOps by providing OOTB security tools for static application security testing (SAST), dynamic application security testing (DAST), and Secret Detection. GitLab is one of the few ISDP vendors to incorporate a test automation dashboard, earning a top score. Dev experience is on par, as are administration and management, deployment automation, and release readiness. Build automation, governance and compliance, and infrastructure as code (IaC) were on par with others. One reference customer commented, "The CI/CD experience using secrets, environments, runners, and SAST/DAST/license scans/etc. is unparalleled in my opinion." GitLab is great for enterprises wishing to consolidate their best-of-breed toolchain into one high-performing ISDP.

Strong Performers

- **CloudBees delivers on its core promise of CI/CD, but platform cohesion needs work.** CloudBees is primarily associated with Jenkins; this well-known open source CI tool that it donated to the Continuous Delivery Foundation is one of the original best-of-breed tools that accompanied the beginning of the DevOps era more than 10 years ago. CloudBees leverages open source Jenkins and augments that with enterprise-scale deployment automation, release readiness, and value stream management (VSM) analytics to provide an enterprise-class ISDP. It got high scores for community, with significant contributions to open source projects and cultivation of its own user community. Its strategy features a solid vision and one of the larger partner ecosystems in this market. Innovation was on par and includes efforts to improve codification of policies. A planned SaaS offering should address its platform cohesion weakness and will help CloudBees serve clients that want a more integrated and maintenance-free ISDP.

CloudBees deployment automation is above par, covering blue/green, canary, and powerful feature management capabilities that are third-party add-ons for most

other platforms. CloudBees release readiness, governance, and compliance are also top-notch with complete integration with popular IT service management (ITSM) tools such as ServiceNow that bring those UIs directly into the ISDP experience. Build automation requires a bit of effort to manage fleets of build runners, and pipeline security needs to go beyond branch security to earn higher scores. When asked what they liked about working with CloudBees, reference customers had good things to say: “The vendor relationship has been a true partnership. It’s listening to our internal customers.” CloudBees is a great fit for enterprise platform teams that value open source and open choice.

- **Microsoft differentiates with innovation, but governance and compliance need attention.** Microsoft acquired GitHub, the industry’s largest open source registry, in 2018. Since that time, Microsoft’s Azure DevOps and GitHub have slowly been moving toward a unified offering. However, both the product vision and the roadmap could have been clearer on how this will shape future offerings, resulting in an on-par score. Innovation strategy is tops because GitHub Copilot has set new standards for the development tools industry and planned investments will enable it to continue this leadership position. Its partner ecosystem also scored high, with Microsoft having one of the most extensive. GitHub also supports numerous open source projects, and its online community is one of the biggest, earning a top score. The roadmap heavily weighted developer productivity but was missing key enterprise needs for governance.

Microsoft’s developer experience gets top scores for GitHub Actions, which turned GitHub into a CI leader almost overnight; Codespaces, a virtual dev environment that requires no setup by developers to get started; and Copilot, the AI-powered coding assistant. It also received top scores for OOTB security tools such as Dependabot. Support for APIs and third-party tools was also top scoring, with numerous ways to incorporate not only software but also actions. Pipeline security and platform cohesion were on par, while administration and management were missing the granular permissions of those scoring higher. Microsoft data and analytics were mostly operational, resulting in a below-par score. Deployment automation was also below par — it lacked blue/green and canary deployment scenarios. Reference customers were mostly positive about Microsoft, noting, “Generally, adoption of standard features is quick and doesn’t require a great deal of time to become productive.” Microsoft is great for organizations looking to boost developer effectiveness with cutting-edge tools.

- **Atlassian gets top scores for cohesiveness, but pipeline security capabilities are lacking.** Atlassian is best known for Jira, one of the most popular agile planning

tools on the market serving 80% of the Fortune 500. Atlassian's product strategy is on par and strives to serve two audiences, teams of developers and platform teams, by building out a data layer to connect the tools together. Its partner ecosystem is top-notch. Innovation is on par and includes platform capabilities and a data lake component to capture and aggregate data for greater analytical insight. The roadmap reflects enterprise concerns with plans to aggregate third-party security data, automatically infuse issue tickets with context, provide greater VSM insights, and expand support plans for DevOps and platform teams.

Atlassian's Confluence, which enables teams to create custom developer onboarding content and a hub for team collaboration, enhances the developer experience. This plus integrated development environment plug-ins give Atlassian a solid on-par score for developer experience. The OOTB build and CI were on par, with reusable build templates and parallel build execution to reduce build times. Deployment automation was below par, as were capabilities to secure pipelines. OOTB security tools are few, earning a below-par score, because Atlassian prefers to leverage its partner ecosystem for those. A reference customer said, "We chose Atlassian for the product features and the maturity of the well-integrated suites of products and plug-ins, which help simplify using the well-designed solution." Atlassian is great fit for software teams and enterprise platform teams that are looking for a standard ISDP approach but want to leverage best-of-breed CD and security components.

- **Harness ISDP delivers on platform cohesion, but community contributions are nascent.** Harness has a SaaS ISDP with core CI/CD modules and cloud cost management capabilities. Its vision is to simplify software development with an end-to-end software delivery platform that offers a modern developer experience and automated governance and guardrails. The strategy has mostly on-par elements such as a growing partner ecosystem and investments in innovation. The only weak area was the smaller-than-average community program — those scoring higher had greater support for internal and external communities. The roadmap includes support for database development, test optimization, end-to-end visibility, and infusion of AI/ML throughout the platform.

Harness gets top scores for platform cohesion, delivering a great OOTB experience for first-time users. It also earned top scores in build automation that includes pipelines as code, reusable pipelines, resource management, support for test automation, and intelligent test automation that can drastically reduce build times. It supports multiple deployment strategies OOTB, including blue/green and

canary. In addition, Harness can pinpoint code that's caused pipeline errors, helping speed remediation. Analytics was also top scoring and offered dev, process, and operational metrics. Customization of the UI and some ISDP components is possible and easy, but the limited scope of those changes resulted in a below-par score. Other areas for improvement include expanding the number of third-party integrations and its footprint in the user and open source community. Customers appreciate Harness' "willingness to engage directly with our development teams." Harness is great for development teams that want to spend less time managing tools and more time developing code.

- **AWS is strong in administration and management, but dev metrics are a miss.**

Amazon Web Services (AWS) pioneered infrastructure as a service (IaaS) and offers nearly every possible form of it. Its ISDP product includes CodeBuild, CodeStar, CloudFormation, and several other SaaS offerings that are integrated on demand when users create their account. The overall strategy score was solid. Innovation scored on par for current delivery and upcoming features for helping developers code faster. The vision was on par, focusing mainly on CodeCatalyst that will enable AWS to make its developer platform more accessible to a broader set of end users with different levels of technical ability. AWS actively supports its user community and open source projects, getting an on-par score for community. The roadmap is also on par and includes the aforementioned CodeCatalyst, a product currently in beta that offers a preassembled and more opinionated ISDP to help customers accelerate project setup and collaborate more fluidly.

AWS shines with excellent administration and management capabilities that enable user permissions to be managed down to the API level. AWS supports all major authentication protocols including emerging standards such as SCIM. AWS gets top scores for governance and compliance that include OOTB compliance templates. Platform cohesion and developer experience are on par. Build automation would have scored higher had it offered test execution optimization. AWS got low scores for analytics and reports; other vendors offered better dashboards with more dev-related metrics. Release readiness also scored low. Reference customer sentiment was mostly positive, saying, "Security is a first-class citizen." AWS is great for companies that want speed, scale, and security hosted in the same environment as their public cloud infrastructure.

- **CircleCI provides outstanding build automation, but its vision does not go beyond CI/CD.** CircleCI was one of the first vendors to offer CI as SaaS. Over time, it added CD and extensibility features to make a more well-rounded ISDP offering. CircleCI's on-par vision to speed time to value while simplifying the ISDP

experience is well articulated and strikes a pragmatic balance between individual developer concerns, the needs of platform teams, and support for enterprises in regulated industries. CircleCI's strategic vision is to remain a best-of-breed CI/CD tool vendor and not extend beyond that into other ISDP areas, resulting in a below-par score. The on-par roadmap aims at enterprise teams and includes features to help platform teams better manage an ISDP at scale with pervasive validation at each step of software delivery.

CircleCI has invested in optimized build processes, giving it a top score for build automation and CI. Its support for a wide range of platforms, including all public clouds and even mainframe, earns above-par scores. IaC capabilities were also top scoring. Product support leverages dedicated customer success engineers for each account, but average reference customer scores kept it on par. CircleCI user identities are derived from third-party VCS identities (e.g., those managed by GitHub or Bitbucket) and thus received a low score for administration and management. However, a new identity management system is on the roadmap to address this shortcoming. CircleCI has a good set of OOTB metrics and analytics, including DORA metrics, but the dashboards and capabilities were not as extensive as those scoring higher. One reference customer listed the following capabilities as most appreciated: "parallelization of workflows and rerunning only failed parts of workflows." CircleCI is a great fit for dev teams that value team autonomy and superfast build automation.

- **Codefresh has differentiating CI/CD but needs to improve governance and compliance.** Codefresh provides high-scoring CI/CD capabilities that emphasize the GitOps methodology, which requires all changes to infrastructure to be stored and managed as code in a Git repository prior to deployment. This methodology gained popularity from the Argo CD open source project, which powers Codefresh. Codefresh is also a registered project maintainer and contributor, which factored into its on-par community score. The Codefresh vision and roadmap focus on improvements to the developer experience and deployment capabilities such as Kubernetes context injection, improved disk utilization, and better support for teams. Innovation is confined to Argo and GitOps improvements, earning on-par scores. The one below-par score was for its smaller-than-average partner ecosystem.

Codefresh earned above-par scores for build automation, CI, and deployment automation featuring unique pipeline debugging capabilities that enable users to step in and out of containerized build steps, which no other vendor demonstrated.

Product support was strong, earning on-par scores. IaC provides strong support for Kubernetes configurations, but support for other legacy platforms lacks, keeping this score at par. Developer experience is on par; vendors scoring higher had integrated knowledge management. Other on-par scores include pipeline security, administration and management, and analytics that incorporate OOTB DORA metrics. Codefresh earned below-par scores for governance and compliance (providing limited support for ITSM integration) and support for functional test automation. When asked what was best about Codefresh, one customer said, “easy, standardized deployments for a hybrid universe.” Codefresh is a great fit for organizations that have adopted GitOps and place high value on developer autonomy.

Contenders

- **Flexagon has excellent customer support, but developer experience was below par.** Flexagon builds an end-to-end ISDP that supports traditional forms of software and infrastructure (e.g., Java and Linux) as well as business platforms such as Salesforce, Oracle, and SAP. The product vision is clear and focuses on delivering enterprise value through security and lowering risk. Planned updates include managed Oracle Cloud services, SAP blueprints, and a compliance and policy engine. Innovation is below par, investment in leading-edge areas is lacking, and the partner ecosystem was small, compared with others in this evaluation. Flexagon pricing features multiple tiers and full transparency of pricing options. Flexagon got above-par scores for test automation because it was one of the few vendors that featured native functional test automation capabilities and support.

Flexagon offers good platform cohesion with its offering that includes everything from agile planning to release automation in one package. It would have scored higher if not for some complexity in setting up and operating fleets of build runners. Flexagon got high scores for version control, with its support for the largest number of VCSes including Git and non-Git systems, and environment management that enabled teams to easily operate and manage any number of environments to support engineering teams. Deployment automation, release readiness, and platform support were also solid. Dev experience was not a focus area and resulted in a low score for that capability. Reference customers gave Flexagon perfect high scores for customer support, with one stating, “Flexagon continues to knock our socks off.” Flexagon is a great fit for enterprises looking for DevOps capabilities that span both open systems and business systems.

- **VMware delivers strong pipeline security but is light on data and analytics.**

VMware is best known for its infrastructure virtualization software. Tanzu Application Platform is VMware's ISDP and has a vision that is broad and incorporates elements of sociological factors to help developers be happier and more productive. VMware scored on par for community with strong support for open source tools and its user community. Innovation includes investments in autoremediation and AI/ML, earning an on-par score. The on-par roadmap will help users simplify the product lifecycle, improve developer collaboration, provide cost governance, and have more secure supply chains.

VMware pipeline security is one of the best in the evaluation with signed pipelines, signed commits, and additional security capabilities throughout the pipeline. Governance and compliance were on par, with OOTB compliance policies. Platform cohesion was also good with minimal setup required. Developer experience included a built-in knowledge management system to help onboard new developers and scored on par. Deployment automation included blue/green and canary deployment strategies, earning an on-par score. Administration and management were below par because they were missing features of those scoring higher. Data and analytics were few, and platform support seemed limited to Kubernetes workloads. One reference customer remarked, "The vendor is always willing to accept and act on feedback." VMware is a great fit for enterprises operating on Kubernetes and places high value on pipeline security.

- **OpenText has top-rated platform support, but DevSecOps capabilities score**

low. OpenText is best known for its portfolio of software development tools ranging from portfolio to test automation. OpenText's ValueEdge is an open platform that bundles these capabilities together in one ISDP. Its roadmap is on par, featuring elements that will better connect business and development. Innovation features AI/ML infused in risk management and testing, and OpenText boasts a healthy partner ecosystem. Both areas scored on par, with good evidence of progress from its AI/ML investments. But its vision focused more on business concerns than stronger CI/CD capabilities. In addition, pricing is opaque, compared with those scoring higher.

OpenText has one of the strongest governance and compliance offerings of this evaluation because nearly everything connects to change management systems. It has superior support for test automation capabilities that include autoclassification of test failures. In addition to providing its own VCS, users can connect with virtually any other VCS, earning above-par scores for that capability. Support for a

multitude of platforms including mainframe was a plus. Administration, management, and developer experience were on par, as were data and analytics that featured DORA and flow metrics. Deployment automation was also on par with blue/green and canary deployment support. Pipeline security lagged others evaluated. IaC, an area of growing importance to developers, also scored low. OpenText reference customers were positive about their experience, with one stating, “We don’t have any challenges at this stage.” OpenText is a great fit for enterprises with varied platforms in need of strong ties to test automation and compliance.

- **JFrog provides differentiated pipeline security capabilities, but the core ISDP falls short.** JFrog is best known for its flagship product, Artifactory, a binary artifact repository with a name that is so well known that it is synonymous with artifact repositories in general. JFrog’s strategy includes strong on-par scores for innovation based on investments in helping secure enterprise software supply chains. JFrog has a strong user community and actively participates in open source projects. Pricing flexibility was also on par, offering several payment tiers. JFrog scored below par for vision because it seemed overly focused on security at the expense of other ISDP capabilities. The roadmap was also below par due to its limited scope.

JFrog has superior version control due to the strength of Artifactory. Pipeline security is also top-notch with signed artifacts and signed pipelines that provide some of the strongest supply chain security measures available. The JFrog platform has good cohesion and requires few steps to set up and get started. Administration and management are on par and offer LDAP, OAuth, and support for SCIM to manage users at scale. Xray, a security product that scans artifacts, comes bundled and offers container and image scanning along with OSS supply chain management. To make this a more competitive ISDP offering, JFrog needs to invest in release readiness, data, and analytics and expand policy as code to all aspects of the ISDP while doing more to support test automation. Customers like how they are treated by JFrog, saying: It’s a “partner, not a vendor — they bring this mentality to everything.” JFrog is great for enterprises that place high value on supply chain security.

Challengers

- **IBM UrbanCode Deploy offers solid build and deployment capabilities but looks dated.** IBM UrbanCode offers UrbanCode Build and UrbanCode Deploy as two components that work together to deliver an enterprise-scale ISDP. IBM has a

solid development community because IBM Garage has thousands of contributors. IBM's partner ecosystem is also solidly on par with a number of top systems integrators listed as partners. The strategy has several weak elements including a roadmap that emphasizes maintenance versus new features, and innovation proof points are lacking.

IBM has solid build automation, deployment automation, release readiness, and support for a large variety of target platforms, which are core ISDP capabilities valued by enterprise dev teams. IBM's UrbanCode Velocity module offers strong analytics and graphics, earning it on-par scores for analytics, and there's a healthy number of third-party integrations for an on-par score in that category. But platform cohesion suffers from a complicated installation process, administration and management offer limited capabilities such as LDAP, and the developer experience is bare bones. Pipelines are present but don't offer the same level of capability as more modern tools. IBM is a good fit for enterprises that need to build an ISDP bridge between legacy and modern infrastructure. IBM declined to participate in the full Forrester Wave evaluation process.

Evaluation Overview

We grouped our evaluation criteria into three high-level categories:

- **Current offering.** Each vendor's position on the vertical axis of the Forrester Wave graphic indicates the strength of its current offering. Key criteria for these solutions include platform capabilities to facilitate management at scale, CI/CD to perform core ISDP functions, extensibility to enable platform operators to incorporate third-party tools as needed, and integrated systems to secure the software delivery pipeline.
- **Strategy.** Placement on the horizontal axis indicates the strength of the vendors' strategies. We evaluated vision, roadmap, and innovation to determine which vendor was placing the best bets to fulfill long-term product promises.
- **Market presence.** Represented by the size of the markers on the graphic, our market presence scores reflect each vendor's ISDP revenue and number of enterprise customers.

Vendor Inclusion Criteria

Each of the vendors we included in this assessment has:

- **Integrated CI/CD (e.g., deployment and release automation) capabilities.** The vendor provides integrated CI/CD capabilities OOTB that enable teams to rapidly establish build and deploy processes with minimal configuration.
- **Extensibility via APIs or plug-ins.** The vendor provides APIs to support third-party tool integration to extend the platform to incorporate performance, test automation, and security.
- **A system for centralized administration.** A single dashboard helps platform administrators operate and maintain an ISDP at enterprise scale.
- **Strong client interest.** Analyst inquiry logs indicate the vendor is top of mind among Forrester's end-user clientele with regard to ISDPs.
- **General-purpose programming languages and platforms.** The vendor supports build and deployment automation for open source programming languages like C# and Java as well as deployment to general-purpose runtime platforms such as Linux or Windows. It is also not exclusively tied to any particular business platform such as Salesforce or SAP.
- **High market performance.** The vendor's market performance demonstrates that it is a significant player in the ISDP market.

Supplemental Material

Online Resource

We publish all our Forrester Wave scores and weightings in an Excel file that provides detailed product evaluations and customizable rankings; download this tool by clicking the link at the beginning of this report on Forrester.com. We intend these scores and default weightings to serve only as a starting point and encourage readers to adapt the weightings to fit their individual needs.

The Forrester Wave Methodology

A Forrester Wave is a guide for buyers considering their purchasing options in a technology marketplace. To offer an equitable process for all participants, Forrester follows [The Forrester Wave™ Methodology](#) to evaluate participating vendors.

In our review, we conduct primary research to develop a list of vendors to consider for the evaluation. From that initial pool of vendors, we narrow our final list based on the inclusion criteria. We then gather details of product and strategy through a detailed questionnaire, demos/briefings, and customer reference surveys/interviews. We use those inputs, along with the analyst's experience and expertise in the marketplace, to

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score vendors, using a relative rating system that compares each vendor against the others in the evaluation.

We include the Forrester Wave publishing date (quarter and year) clearly in the title of each Forrester Wave report. We evaluated the vendors participating in this Forrester Wave using materials they provided to us by March 2023 and did not allow additional information after that point. We encourage readers to evaluate how the market and vendor offerings change over time.

In accordance with [our vendor review policy](#), Forrester asks vendors to review our findings prior to publishing to check for accuracy. Vendors marked as nonparticipating vendors in the Forrester Wave graphic met our defined inclusion criteria but declined to participate in or contributed only partially to the evaluation. We score these vendors in accordance with [our vendor participation policy](#) and publish their positioning along with those of the participating vendors.

Integrity Policy

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WAVE REPORT

The Forrester WaveTM: Integrated Software Delivery Platforms, Q2 2023

The 13 Providers That Matter Most And How They
Stack Up

May 31, 2023

By Christopher Condo with Chris Gardner, Sara Sjoblom, Kara Hartig

FORRESTER®

Summary

In our 26-criterion evaluation of integrated software delivery platform (ISDP) providers, we identified the most significant ones and researched, analyzed, and scored them. This report shows how each provider measures up and helps modern application development leaders and DevOps platform team leaders select the right one for their needs.

Additional resources are available in the [online version](#) of this report.

ISDPs Provide A Strong Platform Foundation But Still Need Best-Of-Breed Tools

In our reports, [The Future Of DevOps](#) and [The Forrester Guide To DevOps](#), Forrester interviewed dozens of enterprise software development leaders from diverse companies ranging from Netflix to TransUnion to understand the current state of DevOps, its relevance, and next steps enterprise leaders were taking to further advance their practices. In addition, we asked which tools they were using, how effective the tools were, which were using [ISDPs](#), and how well those ISDPs were performing. We learned that many enterprises have shifted to platform teams to support their DevOps efforts and many platform leaders are evaluating and using ISDPs, but few are using all of their ISDP vendor's capabilities. They explained that while these ISDP tools hold great potential for a complete end-to-end toolchain, many individual capabilities fell short of enterprise needs, forcing these leaders to source those capabilities from separate best-of-breed vendors.

Our reference customer survey backs this up by indicating that enterprises are using (on average) only 53% of the capabilities they are paying for, with the remaining capabilities either being delivered by a separate vendor or simply not used. This means that ISDP vendors are not only competing with each other but also competing with the best-of-breeds, which is why the bar has risen so high for this evaluation. ISDP vendors recognize these challenges and are responding with a number of innovative solutions that will make them more attractive than a traditional DIY toolchain but open and extensible enough that end users can add the tools they want without sacrificing capability. The challenge is striking the right balance to create overall value.

As a result of these trends, ISDP customers should look for providers that:

- **Offer a strong core of CI/CD capabilities.** Continuous integration (CI) and continuous delivery (CD) are the core capabilities that the rest of the software delivery toolchain is built around. Several vendors offered build optimization algorithms that slash build and unit test times by intelligently building and testing only the code that changed. Deployment automation is also a differentiating capability, with only a few vendors offering advanced capabilities out of the box (OOTB) such as blue/green, canary, and feature management deployment strategies.
- **Provide connected platforms that best-of-breed tools can augment.** ISDPs provide engineering leverage in the form of preintegrated toolchains that enable work to easily flow from one process to the next, where administration occurs at

the platform level versus the tool level, and reduced context switching improves the developer experience. Often, platform leaders want to override the default capability with third-party tools that they believe offer superior capability, which can lead to a bit of a balancing act. That's why some ISDP vendors are heavily investing in improvements to their end-to-end platform, while others are banking on investments in third-party connectivity and partnerships.

- **Help enterprise leaders shift from DevOps to DevSecOps.** Forrester fields calls every day from enterprise development leaders that are transforming their DevOps practice into a development, security, and operations (DevSecOps) practice to make security a first-class concern for the entire development organization. ISDP vendors are responding with capabilities that better secure software delivery pipelines, including integrated secrets management, signed pipelines, and signed artifacts. Reference customers noted that not all of these tools are direct replacements for best-of-breed (best-of-breed security tools often having a significant advantage over bundled ones), but end users like them as forms of additional assurance and coverage.

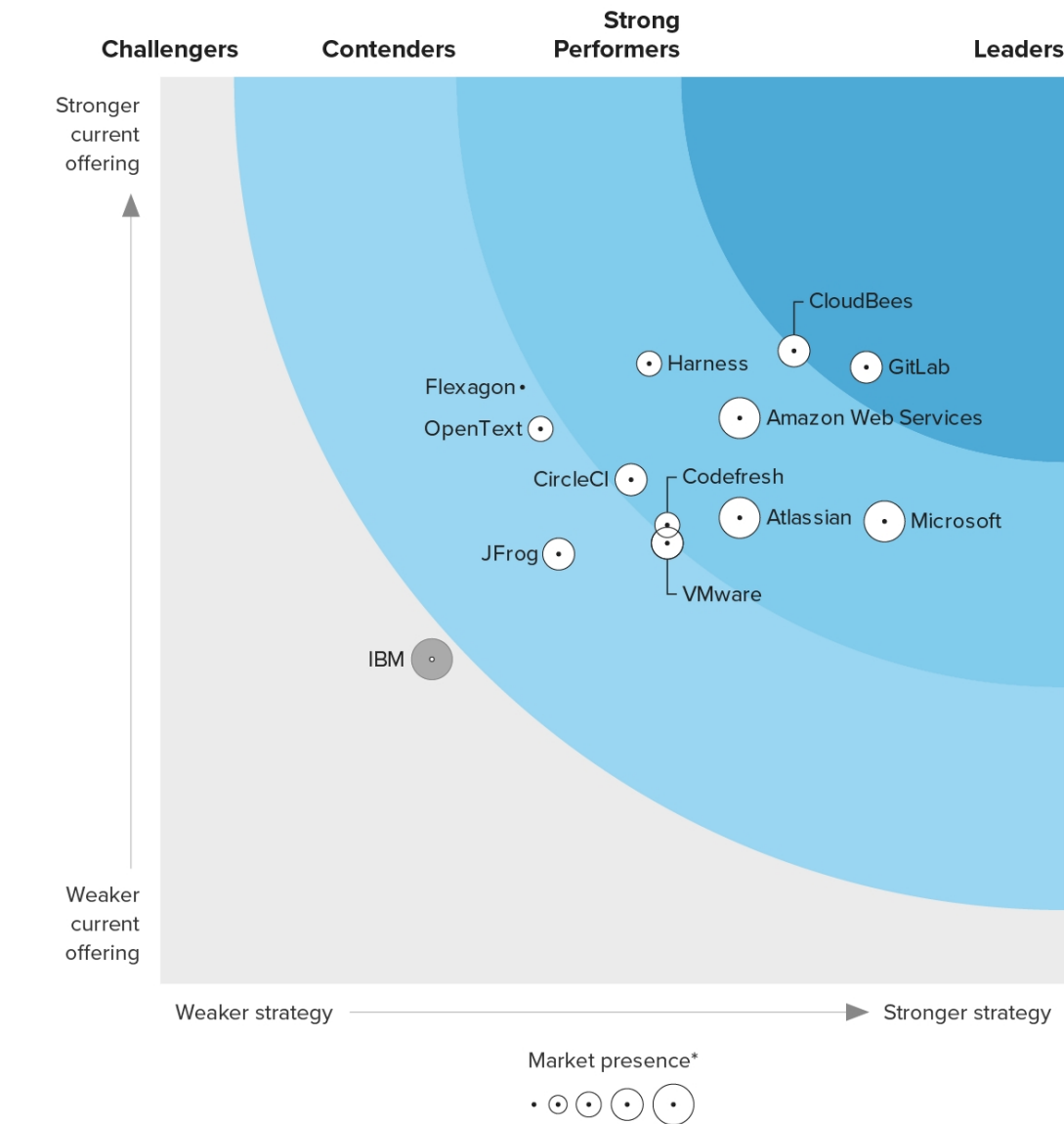
Evaluation Summary

The Forrester Wave™ evaluation highlights Leaders, Strong Performers, Contenders, and Challengers. It's an assessment of the top vendors in the market; it doesn't represent the entire vendor landscape. You'll find more information about this market in Forrester's [Now Tech: Integrated Software Delivery Platforms, Q2 2022](#).

We intend this evaluation to be a starting point only and encourage clients to view product evaluations and adapt criteria weightings using the Excel-based vendor comparison tool (see Figures 1 and 2). Click the link at the beginning of this report on [Forrester.com](#) to download the tool.

Figure 1
Forrester Wave™: Integrated Software Delivery Platforms, Q2 2023

THE FORRESTER WAVE™
Integrated Software Delivery Platforms
Q2 2023



*A gray bubble or open dot indicates a nonparticipating vendor.

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Figure 2**Forrester Wave™: Integrated Software Delivery Platforms Scorecard, Q2 2023**

		Forrester's weighting	Amazon Web Services	Atlassian	CircleCI	CloudBees	Codefresh	Flexagon	GitLab
Current offering		50%	2.57	3.12	2.78	3.49	2.53	3.29	3.40
Platform capabilities		20%	3.50	3.00	2.00	2.50	3.00	2.50	3.50
CI/CD capabilities		25%	2.00	2.60	3.00	4.00	4.00	3.20	3.40
Development, security, and operations		10%	1.00	3.00	2.00	2.00	2.00	3.00	4.00
Extensibility		10%	3.00	3.40	3.00	3.40	1.00	3.00	3.00
Governance and compliance		10%	3.00	5.00	3.00	5.00	1.00	3.00	3.00
Infrastructure management		10%	2.20	3.80	3.80	5.00	1.80	3.40	3.00
Product support		10%	3.00	3.00	3.00	3.00	3.00	5.00	3.00
Support for test automation		5%	3.00	1.00	3.00	3.00	1.00	5.00	5.00
Strategy		50%	3.20	3.20	2.60	3.50	2.80	2.00	3.90
Vision		20%	3.00	3.00	1.00	3.00	3.00	3.00	3.00
Roadmap		20%	3.00	3.00	3.00	3.00	3.00	3.00	5.00
Partner ecosystem		10%	5.00	5.00	1.00	5.00	1.00	1.00	3.00
Innovation		25%	3.00	3.00	3.00	3.00	3.00	1.00	3.00
Community		15%	3.00	3.00	3.00	5.00	3.00	1.00	5.00
Pricing flexibility and transparency		10%	3.00	3.00	5.00	3.00	3.00	3.00	5.00
Market presence		0%	5.00	4.50	3.50	3.50	2.50	1.00	3.50
ISDP revenue		50%	5.00	5.00	4.00	4.00	2.00	1.00	4.00
Number of enterprise customers		50%	5.00	4.00	3.00	3.00	3.00	1.00	3.00

All scores are based on a scale of 0 (weak) to 5 (strong).

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	Forrester's weighting	Harness	IBM*	JSFrog	Microsoft	OpenText	VMware
Current offering	50%	3.42	1.79	2.37	2.55	3.06	2.43
Platform capabilities	20%	4.00	1.50	2.00	3.00	2.50	2.00
CI/CD capabilities	25%	4.20	2.80	2.40	2.40	3.40	2.40
Development, security, and operations	10%	3.00	1.00	4.00	4.00	1.00	3.00
Extensibility	10%	1.00	1.40	4.20	4.20	3.40	3.00
Governance and compliance	10%	3.00	3.00	1.00	1.00	5.00	3.00
Infrastructure management	10%	4.20	1.00	1.00	1.80	2.20	1.80
Product support	10%	3.00	1.00	3.00	1.00	3.00	3.00
Support for test automation	5%	3.00	1.00	1.00	3.00	5.00	1.00
Strategy	50%	2.70	1.50	2.20	4.00	2.10	2.80
Vision	20%	3.00	1.00	1.00	3.00	1.00	3.00
Roadmap	20%	3.00	1.00	1.00	3.00	3.00	3.00
Partner ecosystem	10%	3.00	3.00	3.00	5.00	3.00	3.00
Innovation	25%	3.00	1.00	3.00	5.00	3.00	3.00
Community	15%	1.00	3.00	3.00	5.00	1.00	3.00
Pricing flexibility and transparency	10%	3.00	1.00	3.00	3.00	1.00	1.00
Market presence	0%	3.00	4.50	4.00	5.00	2.50	4.00
ISDP revenue	50%	3.00	5.00	4.00	5.00	4.00	4.00
Number of enterprise customers	50%	3.00	4.00	4.00	5.00	1.00	4.00

All scores are based on a scale of 0 (weak) to 5 (strong).

*Indicates a nonparticipating vendor

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Vendor Offerings

Forrester evaluated the offerings listed below (see Figure 3).

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Figure 3
Evaluated Vendors And Product Information

Vendor	Product evaluated
Amazon Web Services	AWS Code Services
Atlassian	Atlassian Open DevOps
CircleCI	CircleCI
CloudBees	CloudBees Platform
Codefresh	Codefresh
Flexagon	Flexagon FlexDeploy
GitLab	GitLab
Harness	Harness Software Delivery Platform
IBM	IBM UrbanCode Deploy
JFrog	JFrog Software Supply Chain Platform
Microsoft	Microsoft Azure DevOps and GitHub Enterprise
OpenText (formerly Micro Focus)	Micro Focus ValueEdge
VMware	VMware Tanzu Application Platform

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Vendor Profiles

Our analysis uncovered the following strengths and weaknesses of individual vendors.

Leaders

- **GitLab leads with a balance between solid platform capabilities and overall strategy.** GitLab is known for being the vendor that made end-to-end OOTB toolchains hip again and has led the industry toward consolidated ISDPs. GitLab’s strategy includes an on-par vision to deliver an excellent developer experience without sacrificing security or compliance. Its innovation is also good, going beyond traditional developers to include AI/ML engineering. GitLab is an open core product that not only invests heavily in the open source software (OSS) community but also enables its customers to contribute to the product, earning it

high scores for community. Pricing is completely transparent and includes a free tier. Its roadmap gets leading scores and includes enhanced supply chain security, enhanced UI, granular security and compliance controls, and pipeline security — all things enterprises need.

GitLab's platform cohesion received high scores for simplicity of setup and operation as both an on-premises product and a software-as-a-service (SaaS) product. Though GitLab comes with its own GitLab version control system (VCS), it supports complete integration with all major Git providers, earning it high scores for version control. GitLab gets high scores for DevSecOps by providing OOTB security tools for static application security testing (SAST), dynamic application security testing (DAST), and Secret Detection. GitLab is one of the few ISDP vendors to incorporate a test automation dashboard, earning a top score. Dev experience is on par, as are administration and management, deployment automation, and release readiness. Build automation, governance and compliance, and infrastructure as code (IaC) were on par with others. One reference customer commented, "The CI/CD experience using secrets, environments, runners, and SAST/DAST/license scans/etc. is unparalleled in my opinion." GitLab is great for enterprises wishing to consolidate their best-of-breed toolchain into one high-performing ISDP.

Strong Performers

- **CloudBees delivers on its core promise of CI/CD, but platform cohesion needs work.** CloudBees is primarily associated with Jenkins; this well-known open source CI tool that it donated to the Continuous Delivery Foundation is one of the original best-of-breed tools that accompanied the beginning of the DevOps era more than 10 years ago. CloudBees leverages open source Jenkins and augments that with enterprise-scale deployment automation, release readiness, and value stream management (VSM) analytics to provide an enterprise-class ISDP. It got high scores for community, with significant contributions to open source projects and cultivation of its own user community. Its strategy features a solid vision and one of the larger partner ecosystems in this market. Innovation was on par and includes efforts to improve codification of policies. A planned SaaS offering should address its platform cohesion weakness and will help CloudBees serve clients that want a more integrated and maintenance-free ISDP.

CloudBees deployment automation is above par, covering blue/green, canary, and powerful feature management capabilities that are third-party add-ons for most

other platforms. CloudBees release readiness, governance, and compliance are also top-notch with complete integration with popular IT service management (ITSM) tools such as ServiceNow that bring those UIs directly into the ISDP experience. Build automation requires a bit of effort to manage fleets of build runners, and pipeline security needs to go beyond branch security to earn higher scores. When asked what they liked about working with CloudBees, reference customers had good things to say: “The vendor relationship has been a true partnership. It’s listening to our internal customers.” CloudBees is a great fit for enterprise platform teams that value open source and open choice.

- **Microsoft differentiates with innovation, but governance and compliance need attention.** Microsoft acquired GitHub, the industry’s largest open source registry, in 2018. Since that time, Microsoft’s Azure DevOps and GitHub have slowly been moving toward a unified offering. However, both the product vision and the roadmap could have been clearer on how this will shape future offerings, resulting in an on-par score. Innovation strategy is tops because GitHub Copilot has set new standards for the development tools industry and planned investments will enable it to continue this leadership position. Its partner ecosystem also scored high, with Microsoft having one of the most extensive. GitHub also supports numerous open source projects, and its online community is one of the biggest, earning a top score. The roadmap heavily weighted developer productivity but was missing key enterprise needs for governance.

Microsoft’s developer experience gets top scores for GitHub Actions, which turned GitHub into a CI leader almost overnight; Codespaces, a virtual dev environment that requires no setup by developers to get started; and Copilot, the AI-powered coding assistant. It also received top scores for OOTB security tools such as Dependabot. Support for APIs and third-party tools was also top scoring, with numerous ways to incorporate not only software but also actions. Pipeline security and platform cohesion were on par, while administration and management were missing the granular permissions of those scoring higher. Microsoft data and analytics were mostly operational, resulting in a below-par score. Deployment automation was also below par — it lacked blue/green and canary deployment scenarios. Reference customers were mostly positive about Microsoft, noting, “Generally, adoption of standard features is quick and doesn’t require a great deal of time to become productive.” Microsoft is great for organizations looking to boost developer effectiveness with cutting-edge tools.

- **Atlassian gets top scores for cohesiveness, but pipeline security capabilities are lacking.** Atlassian is best known for Jira, one of the most popular agile planning

tools on the market serving 80% of the Fortune 500. Atlassian's product strategy is on par and strives to serve two audiences, teams of developers and platform teams, by building out a data layer to connect the tools together. Its partner ecosystem is top-notch. Innovation is on par and includes platform capabilities and a data lake component to capture and aggregate data for greater analytical insight. The roadmap reflects enterprise concerns with plans to aggregate third-party security data, automatically infuse issue tickets with context, provide greater VSM insights, and expand support plans for DevOps and platform teams.

Atlassian's Confluence, which enables teams to create custom developer onboarding content and a hub for team collaboration, enhances the developer experience. This plus integrated development environment plug-ins give Atlassian a solid on-par score for developer experience. The OOTB build and CI were on par, with reusable build templates and parallel build execution to reduce build times. Deployment automation was below par, as were capabilities to secure pipelines. OOTB security tools are few, earning a below-par score, because Atlassian prefers to leverage its partner ecosystem for those. A reference customer said, "We chose Atlassian for the product features and the maturity of the well-integrated suites of products and plug-ins, which help simplify using the well-designed solution." Atlassian is great fit for software teams and enterprise platform teams that are looking for a standard ISDP approach but want to leverage best-of-breed CD and security components.

- **Harness ISDP delivers on platform cohesion, but community contributions are nascent.** Harness has a SaaS ISDP with core CI/CD modules and cloud cost management capabilities. Its vision is to simplify software development with an end-to-end software delivery platform that offers a modern developer experience and automated governance and guardrails. The strategy has mostly on-par elements such as a growing partner ecosystem and investments in innovation. The only weak area was the smaller-than-average community program — those scoring higher had greater support for internal and external communities. The roadmap includes support for database development, test optimization, end-to-end visibility, and infusion of AI/ML throughout the platform.

Harness gets top scores for platform cohesion, delivering a great OOTB experience for first-time users. It also earned top scores in build automation that includes pipelines as code, reusable pipelines, resource management, support for test automation, and intelligent test automation that can drastically reduce build times. It supports multiple deployment strategies OOTB, including blue/green and

canary. In addition, Harness can pinpoint code that's caused pipeline errors, helping speed remediation. Analytics was also top scoring and offered dev, process, and operational metrics. Customization of the UI and some ISDP components is possible and easy, but the limited scope of those changes resulted in a below-par score. Other areas for improvement include expanding the number of third-party integrations and its footprint in the user and open source community. Customers appreciate Harness' "willingness to engage directly with our development teams." Harness is great for development teams that want to spend less time managing tools and more time developing code.

- **AWS is strong in administration and management, but dev metrics are a miss.**

Amazon Web Services (AWS) pioneered infrastructure as a service (IaaS) and offers nearly every possible form of it. Its ISDP product includes CodeBuild, CodeStar, CloudFormation, and several other SaaS offerings that are integrated on demand when users create their account. The overall strategy score was solid. Innovation scored on par for current delivery and upcoming features for helping developers code faster. The vision was on par, focusing mainly on CodeCatalyst that will enable AWS to make its developer platform more accessible to a broader set of end users with different levels of technical ability. AWS actively supports its user community and open source projects, getting an on-par score for community. The roadmap is also on par and includes the aforementioned CodeCatalyst, a product currently in beta that offers a preassembled and more opinionated ISDP to help customers accelerate project setup and collaborate more fluidly.

AWS shines with excellent administration and management capabilities that enable user permissions to be managed down to the API level. AWS supports all major authentication protocols including emerging standards such as SCIM. AWS gets top scores for governance and compliance that include OOTB compliance templates. Platform cohesion and developer experience are on par. Build automation would have scored higher had it offered test execution optimization. AWS got low scores for analytics and reports; other vendors offered better dashboards with more dev-related metrics. Release readiness also scored low. Reference customer sentiment was mostly positive, saying, "Security is a first-class citizen." AWS is great for companies that want speed, scale, and security hosted in the same environment as their public cloud infrastructure.

- **CircleCI provides outstanding build automation, but its vision does not go beyond CI/CD.** CircleCI was one of the first vendors to offer CI as SaaS. Over time, it added CD and extensibility features to make a more well-rounded ISDP offering. CircleCI's on-par vision to speed time to value while simplifying the ISDP

experience is well articulated and strikes a pragmatic balance between individual developer concerns, the needs of platform teams, and support for enterprises in regulated industries. CircleCI's strategic vision is to remain a best-of-breed CI/CD tool vendor and not extend beyond that into other ISDP areas, resulting in a below-par score. The on-par roadmap aims at enterprise teams and includes features to help platform teams better manage an ISDP at scale with pervasive validation at each step of software delivery.

CircleCI has invested in optimized build processes, giving it a top score for build automation and CI. Its support for a wide range of platforms, including all public clouds and even mainframe, earns above-par scores. IaC capabilities were also top scoring. Product support leverages dedicated customer success engineers for each account, but average reference customer scores kept it on par. CircleCI user identities are derived from third-party VCS identities (e.g., those managed by GitHub or Bitbucket) and thus received a low score for administration and management. However, a new identity management system is on the roadmap to address this shortcoming. CircleCI has a good set of OOTB metrics and analytics, including DORA metrics, but the dashboards and capabilities were not as extensive as those scoring higher. One reference customer listed the following capabilities as most appreciated: "parallelization of workflows and rerunning only failed parts of workflows." CircleCI is a great fit for dev teams that value team autonomy and superfast build automation.

- **Codefresh has differentiating CI/CD but needs to improve governance and compliance.** Codefresh provides high-scoring CI/CD capabilities that emphasize the GitOps methodology, which requires all changes to infrastructure to be stored and managed as code in a Git repository prior to deployment. This methodology gained popularity from the Argo CD open source project, which powers Codefresh. Codefresh is also a registered project maintainer and contributor, which factored into its on-par community score. The Codefresh vision and roadmap focus on improvements to the developer experience and deployment capabilities such as Kubernetes context injection, improved disk utilization, and better support for teams. Innovation is confined to Argo and GitOps improvements, earning on-par scores. The one below-par score was for its smaller-than-average partner ecosystem.

Codefresh earned above-par scores for build automation, CI, and deployment automation featuring unique pipeline debugging capabilities that enable users to step in and out of containerized build steps, which no other vendor demonstrated.

Product support was strong, earning on-par scores. IaC provides strong support for Kubernetes configurations, but support for other legacy platforms lacks, keeping this score at par. Developer experience is on par; vendors scoring higher had integrated knowledge management. Other on-par scores include pipeline security, administration and management, and analytics that incorporate OOTB DORA metrics. Codefresh earned below-par scores for governance and compliance (providing limited support for ITSM integration) and support for functional test automation. When asked what was best about Codefresh, one customer said, “easy, standardized deployments for a hybrid universe.” Codefresh is a great fit for organizations that have adopted GitOps and place high value on developer autonomy.

Contenders

- **Flexagon has excellent customer support, but developer experience was below par.** Flexagon builds an end-to-end ISDP that supports traditional forms of software and infrastructure (e.g., Java and Linux) as well as business platforms such as Salesforce, Oracle, and SAP. The product vision is clear and focuses on delivering enterprise value through security and lowering risk. Planned updates include managed Oracle Cloud services, SAP blueprints, and a compliance and policy engine. Innovation is below par, investment in leading-edge areas is lacking, and the partner ecosystem was small, compared with others in this evaluation. Flexagon pricing features multiple tiers and full transparency of pricing options. Flexagon got above-par scores for test automation because it was one of the few vendors that featured native functional test automation capabilities and support.

Flexagon offers good platform cohesion with its offering that includes everything from agile planning to release automation in one package. It would have scored higher if not for some complexity in setting up and operating fleets of build runners. Flexagon got high scores for version control, with its support for the largest number of VCSes including Git and non-Git systems, and environment management that enabled teams to easily operate and manage any number of environments to support engineering teams. Deployment automation, release readiness, and platform support were also solid. Dev experience was not a focus area and resulted in a low score for that capability. Reference customers gave Flexagon perfect high scores for customer support, with one stating, “Flexagon continues to knock our socks off.” Flexagon is a great fit for enterprises looking for DevOps capabilities that span both open systems and business systems.

- **VMware delivers strong pipeline security but is light on data and analytics.**

VMware is best known for its infrastructure virtualization software. Tanzu Application Platform is VMware's ISDP and has a vision that is broad and incorporates elements of sociological factors to help developers be happier and more productive. VMware scored on par for community with strong support for open source tools and its user community. Innovation includes investments in autoremediation and AI/ML, earning an on-par score. The on-par roadmap will help users simplify the product lifecycle, improve developer collaboration, provide cost governance, and have more secure supply chains.

VMware pipeline security is one of the best in the evaluation with signed pipelines, signed commits, and additional security capabilities throughout the pipeline. Governance and compliance were on par, with OOTB compliance policies. Platform cohesion was also good with minimal setup required. Developer experience included a built-in knowledge management system to help onboard new developers and scored on par. Deployment automation included blue/green and canary deployment strategies, earning an on-par score. Administration and management were below par because they were missing features of those scoring higher. Data and analytics were few, and platform support seemed limited to Kubernetes workloads. One reference customer remarked, "The vendor is always willing to accept and act on feedback." VMware is a great fit for enterprises operating on Kubernetes and places high value on pipeline security.

- **OpenText has top-rated platform support, but DevSecOps capabilities score**

low. OpenText is best known for its portfolio of software development tools ranging from portfolio to test automation. OpenText's ValueEdge is an open platform that bundles these capabilities together in one ISDP. Its roadmap is on par, featuring elements that will better connect business and development. Innovation features AI/ML infused in risk management and testing, and OpenText boasts a healthy partner ecosystem. Both areas scored on par, with good evidence of progress from its AI/ML investments. But its vision focused more on business concerns than stronger CI/CD capabilities. In addition, pricing is opaque, compared with those scoring higher.

OpenText has one of the strongest governance and compliance offerings of this evaluation because nearly everything connects to change management systems. It has superior support for test automation capabilities that include autoclassification of test failures. In addition to providing its own VCS, users can connect with virtually any other VCS, earning above-par scores for that capability. Support for a

multitude of platforms including mainframe was a plus. Administration, management, and developer experience were on par, as were data and analytics that featured DORA and flow metrics. Deployment automation was also on par with blue/green and canary deployment support. Pipeline security lagged others evaluated. IaC, an area of growing importance to developers, also scored low. OpenText reference customers were positive about their experience, with one stating, “We don’t have any challenges at this stage.” OpenText is a great fit for enterprises with varied platforms in need of strong ties to test automation and compliance.

- **JFrog provides differentiated pipeline security capabilities, but the core ISDP falls short.** JFrog is best known for its flagship product, Artifactory, a binary artifact repository with a name that is so well known that it is synonymous with artifact repositories in general. JFrog’s strategy includes strong on-par scores for innovation based on investments in helping secure enterprise software supply chains. JFrog has a strong user community and actively participates in open source projects. Pricing flexibility was also on par, offering several payment tiers. JFrog scored below par for vision because it seemed overly focused on security at the expense of other ISDP capabilities. The roadmap was also below par due to its limited scope.

JFrog has superior version control due to the strength of Artifactory. Pipeline security is also top-notch with signed artifacts and signed pipelines that provide some of the strongest supply chain security measures available. The JFrog platform has good cohesion and requires few steps to set up and get started. Administration and management are on par and offer LDAP, OAuth, and support for SCIM to manage users at scale. Xray, a security product that scans artifacts, comes bundled and offers container and image scanning along with OSS supply chain management. To make this a more competitive ISDP offering, JFrog needs to invest in release readiness, data, and analytics and expand policy as code to all aspects of the ISDP while doing more to support test automation. Customers like how they are treated by JFrog, saying: It’s a “partner, not a vendor — they bring this mentality to everything.” JFrog is great for enterprises that place high value on supply chain security.

Challengers

- **IBM UrbanCode Deploy offers solid build and deployment capabilities but looks dated.** IBM UrbanCode offers UrbanCode Build and UrbanCode Deploy as two components that work together to deliver an enterprise-scale ISDP. IBM has a

solid development community because IBM Garage has thousands of contributors. IBM's partner ecosystem is also solidly on par with a number of top systems integrators listed as partners. The strategy has several weak elements including a roadmap that emphasizes maintenance versus new features, and innovation proof points are lacking.

IBM has solid build automation, deployment automation, release readiness, and support for a large variety of target platforms, which are core ISDP capabilities valued by enterprise dev teams. IBM's UrbanCode Velocity module offers strong analytics and graphics, earning it on-par scores for analytics, and there's a healthy number of third-party integrations for an on-par score in that category. But platform cohesion suffers from a complicated installation process, administration and management offer limited capabilities such as LDAP, and the developer experience is bare bones. Pipelines are present but don't offer the same level of capability as more modern tools. IBM is a good fit for enterprises that need to build an ISDP bridge between legacy and modern infrastructure. IBM declined to participate in the full Forrester Wave evaluation process.

Evaluation Overview

We grouped our evaluation criteria into three high-level categories:

- **Current offering.** Each vendor's position on the vertical axis of the Forrester Wave graphic indicates the strength of its current offering. Key criteria for these solutions include platform capabilities to facilitate management at scale, CI/CD to perform core ISDP functions, extensibility to enable platform operators to incorporate third-party tools as needed, and integrated systems to secure the software delivery pipeline.
- **Strategy.** Placement on the horizontal axis indicates the strength of the vendors' strategies. We evaluated vision, roadmap, and innovation to determine which vendor was placing the best bets to fulfill long-term product promises.
- **Market presence.** Represented by the size of the markers on the graphic, our market presence scores reflect each vendor's ISDP revenue and number of enterprise customers.

Vendor Inclusion Criteria

Each of the vendors we included in this assessment has:

- **Integrated CI/CD (e.g., deployment and release automation) capabilities.** The vendor provides integrated CI/CD capabilities OOTB that enable teams to rapidly establish build and deploy processes with minimal configuration.
- **Extensibility via APIs or plug-ins.** The vendor provides APIs to support third-party tool integration to extend the platform to incorporate performance, test automation, and security.
- **A system for centralized administration.** A single dashboard helps platform administrators operate and maintain an ISDP at enterprise scale.
- **Strong client interest.** Analyst inquiry logs indicate the vendor is top of mind among Forrester's end-user clientele with regard to ISDPs.
- **General-purpose programming languages and platforms.** The vendor supports build and deployment automation for open source programming languages like C# and Java as well as deployment to general-purpose runtime platforms such as Linux or Windows. It is also not exclusively tied to any particular business platform such as Salesforce or SAP.
- **High market performance.** The vendor's market performance demonstrates that it is a significant player in the ISDP market.

Supplemental Material

Online Resource

We publish all our Forrester Wave scores and weightings in an Excel file that provides detailed product evaluations and customizable rankings; download this tool by clicking the link at the beginning of this report on Forrester.com. We intend these scores and default weightings to serve only as a starting point and encourage readers to adapt the weightings to fit their individual needs.

The Forrester Wave Methodology

A Forrester Wave is a guide for buyers considering their purchasing options in a technology marketplace. To offer an equitable process for all participants, Forrester follows [The Forrester Wave™ Methodology](#) to evaluate participating vendors.

In our review, we conduct primary research to develop a list of vendors to consider for the evaluation. From that initial pool of vendors, we narrow our final list based on the inclusion criteria. We then gather details of product and strategy through a detailed questionnaire, demos/briefings, and customer reference surveys/interviews. We use those inputs, along with the analyst's experience and expertise in the marketplace, to

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score vendors, using a relative rating system that compares each vendor against the others in the evaluation.

We include the Forrester Wave publishing date (quarter and year) clearly in the title of each Forrester Wave report. We evaluated the vendors participating in this Forrester Wave using materials they provided to us by March 2023 and did not allow additional information after that point. We encourage readers to evaluate how the market and vendor offerings change over time.

In accordance with [our vendor review policy](#), Forrester asks vendors to review our findings prior to publishing to check for accuracy. Vendors marked as nonparticipating vendors in the Forrester Wave graphic met our defined inclusion criteria but declined to participate in or contributed only partially to the evaluation. We score these vendors in accordance with [our vendor participation policy](#) and publish their positioning along with those of the participating vendors.

Integrity Policy

We conduct all our research, including Forrester Wave evaluations, in accordance with the [integrity policy](#) posted on our website.



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WAVE REPORT

The Forrester WaveTM: Integrated Software Delivery Platforms, Q2 2023

The 13 Providers That Matter Most And How They
Stack Up

May 31, 2023

By Christopher Condo with Chris Gardner, Sara Sjoblom, Kara Hartig

FORRESTER®

Summary

In our 26-criterion evaluation of integrated software delivery platform (ISDP) providers, we identified the most significant ones and researched, analyzed, and scored them. This report shows how each provider measures up and helps modern application development leaders and DevOps platform team leaders select the right one for their needs.

Additional resources are available in the [online version](#) of this report.

ISDPs Provide A Strong Platform Foundation But Still Need Best-Of-Breed Tools

In our reports, [The Future Of DevOps](#) and [The Forrester Guide To DevOps](#), Forrester interviewed dozens of enterprise software development leaders from diverse companies ranging from Netflix to TransUnion to understand the current state of DevOps, its relevance, and next steps enterprise leaders were taking to further advance their practices. In addition, we asked which tools they were using, how effective the tools were, which were using ISDPs, and how well those ISDPs were performing. We learned that many enterprises have shifted to platform teams to support their DevOps efforts and many platform leaders are evaluating and using ISDPs, but few are using all of their ISDP vendor's capabilities. They explained that while these ISDP tools hold great potential for a complete end-to-end toolchain, many individual capabilities fell short of enterprise needs, forcing these leaders to source those capabilities from separate best-of-breed vendors.

Our reference customer survey backs this up by indicating that enterprises are using (on average) only 53% of the capabilities they are paying for, with the remaining capabilities either being delivered by a separate vendor or simply not used. This means that ISDP vendors are not only competing with each other but also competing with the best-of-breeds, which is why the bar has risen so high for this evaluation. ISDP vendors recognize these challenges and are responding with a number of innovative solutions that will make them more attractive than a traditional DIY toolchain but open and extensible enough that end users can add the tools they want without sacrificing capability. The challenge is striking the right balance to create overall value.

As a result of these trends, ISDP customers should look for providers that:

- **Offer a strong core of CI/CD capabilities.** Continuous integration (CI) and continuous delivery (CD) are the core capabilities that the rest of the software delivery toolchain is built around. Several vendors offered build optimization algorithms that slash build and unit test times by intelligently building and testing only the code that changed. Deployment automation is also a differentiating capability, with only a few vendors offering advanced capabilities out of the box (OOTB) such as blue/green, canary, and feature management deployment strategies.
- **Provide connected platforms that best-of-breed tools can augment.** ISDPs provide engineering leverage in the form of preintegrated toolchains that enable work to easily flow from one process to the next, where administration occurs at

the platform level versus the tool level, and reduced context switching improves the developer experience. Often, platform leaders want to override the default capability with third-party tools that they believe offer superior capability, which can lead to a bit of a balancing act. That's why some ISDP vendors are heavily investing in improvements to their end-to-end platform, while others are banking on investments in third-party connectivity and partnerships.

- **Help enterprise leaders shift from DevOps to DevSecOps.** Forrester fields calls every day from enterprise development leaders that are transforming their DevOps practice into a development, security, and operations (DevSecOps) practice to make security a first-class concern for the entire development organization. ISDP vendors are responding with capabilities that better secure software delivery pipelines, including integrated secrets management, signed pipelines, and signed artifacts. Reference customers noted that not all of these tools are direct replacements for best-of-breed (best-of-breed security tools often having a significant advantage over bundled ones), but end users like them as forms of additional assurance and coverage.

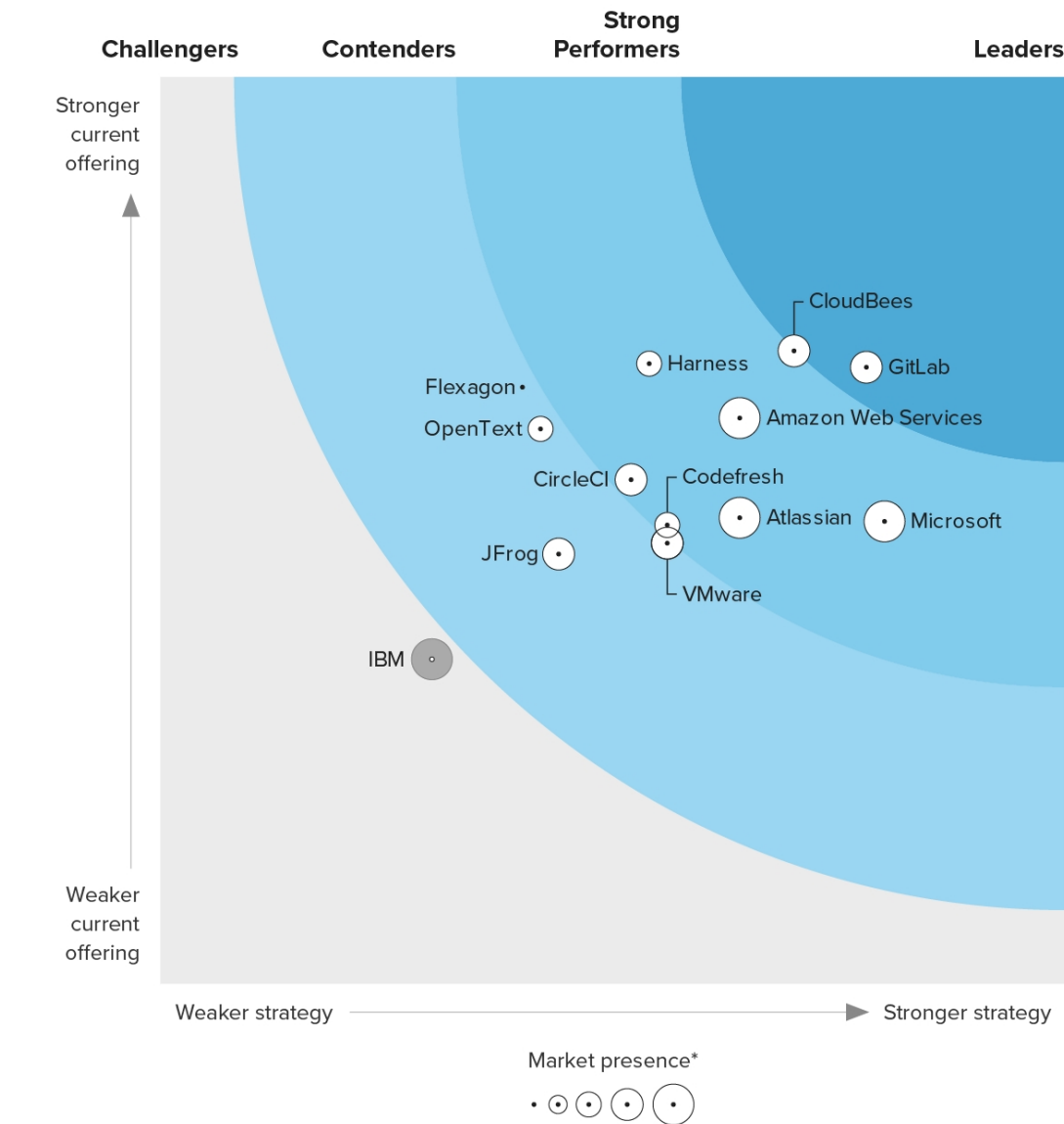
Evaluation Summary

The Forrester Wave™ evaluation highlights Leaders, Strong Performers, Contenders, and Challengers. It's an assessment of the top vendors in the market; it doesn't represent the entire vendor landscape. You'll find more information about this market in Forrester's [Now Tech: Integrated Software Delivery Platforms, Q2 2022](#).

We intend this evaluation to be a starting point only and encourage clients to view product evaluations and adapt criteria weightings using the Excel-based vendor comparison tool (see Figures 1 and 2). Click the link at the beginning of this report on [Forrester.com](#) to download the tool.

Figure 1
Forrester Wave™: Integrated Software Delivery Platforms, Q2 2023

THE FORRESTER WAVE™
Integrated Software Delivery Platforms
Q2 2023



*A gray bubble or open dot indicates a nonparticipating vendor.

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Figure 2**Forrester Wave™: Integrated Software Delivery Platforms Scorecard, Q2 2023**

	Forrester's weighting	Amazon Web Services	Atlassian	CircleCI	CloudBees	Codefresh	Flexagon	GitLab
Current offering	50%	2.57	3.12	2.78	3.49	2.53	3.29	3.40
Platform capabilities	20%	3.50	3.00	2.00	2.50	3.00	2.50	3.50
CI/CD capabilities	25%	2.00	2.60	3.00	4.00	4.00	3.20	3.40
Development, security, and operations	10%	1.00	3.00	2.00	2.00	2.00	3.00	4.00
Extensibility	10%	3.00	3.40	3.00	3.40	1.00	3.00	3.00
Governance and compliance	10%	3.00	5.00	3.00	5.00	1.00	3.00	3.00
Infrastructure management	10%	2.20	3.80	3.80	5.00	1.80	3.40	3.00
Product support	10%	3.00	3.00	3.00	3.00	3.00	5.00	3.00
Support for test automation	5%	3.00	1.00	3.00	3.00	1.00	5.00	5.00
Strategy	50%	3.20	3.20	2.60	3.50	2.80	2.00	3.90
Vision	20%	3.00	3.00	1.00	3.00	3.00	3.00	3.00
Roadmap	20%	3.00	3.00	3.00	3.00	3.00	3.00	5.00
Partner ecosystem	10%	5.00	5.00	1.00	5.00	1.00	1.00	3.00
Innovation	25%	3.00	3.00	3.00	3.00	3.00	1.00	3.00
Community	15%	3.00	3.00	3.00	5.00	3.00	1.00	5.00
Pricing flexibility and transparency	10%	3.00	3.00	5.00	3.00	3.00	3.00	5.00
Market presence	0%	5.00	4.50	3.50	3.50	2.50	1.00	3.50
ISDP revenue	50%	5.00	5.00	4.00	4.00	2.00	1.00	4.00
Number of enterprise customers	50%	5.00	4.00	3.00	3.00	3.00	1.00	3.00

All scores are based on a scale of 0 (weak) to 5 (strong).

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	Forrester's weighting	Harness	IBM*	JSFrog	Microsoft	OpenText	VMware
Current offering	50%	3.42	1.79	2.37	2.55	3.06	2.43
Platform capabilities	20%	4.00	1.50	2.00	3.00	2.50	2.00
CI/CD capabilities	25%	4.20	2.80	2.40	2.40	3.40	2.40
Development, security, and operations	10%	3.00	1.00	4.00	4.00	1.00	3.00
Extensibility	10%	1.00	1.40	4.20	4.20	3.40	3.00
Governance and compliance	10%	3.00	3.00	1.00	1.00	5.00	3.00
Infrastructure management	10%	4.20	1.00	1.00	1.80	2.20	1.80
Product support	10%	3.00	1.00	3.00	1.00	3.00	3.00
Support for test automation	5%	3.00	1.00	1.00	3.00	5.00	1.00
Strategy	50%	2.70	1.50	2.20	4.00	2.10	2.80
Vision	20%	3.00	1.00	1.00	3.00	1.00	3.00
Roadmap	20%	3.00	1.00	1.00	3.00	3.00	3.00
Partner ecosystem	10%	3.00	3.00	3.00	5.00	3.00	3.00
Innovation	25%	3.00	1.00	3.00	5.00	3.00	3.00
Community	15%	1.00	3.00	3.00	5.00	1.00	3.00
Pricing flexibility and transparency	10%	3.00	1.00	3.00	3.00	1.00	1.00
Market presence	0%	3.00	4.50	4.00	5.00	2.50	4.00
ISDP revenue	50%	3.00	5.00	4.00	5.00	4.00	4.00
Number of enterprise customers	50%	3.00	4.00	4.00	5.00	1.00	4.00

All scores are based on a scale of 0 (weak) to 5 (strong).

*Indicates a nonparticipating vendor

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Vendor Offerings

Forrester evaluated the offerings listed below (see Figure 3).

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Figure 3
Evaluated Vendors And Product Information

Vendor	Product evaluated
Amazon Web Services	AWS Code Services
Atlassian	Atlassian Open DevOps
CircleCI	CircleCI
CloudBees	CloudBees Platform
Codefresh	Codefresh
Flexagon	Flexagon FlexDeploy
GitLab	GitLab
Harness	Harness Software Delivery Platform
IBM	IBM UrbanCode Deploy
JFrog	JFrog Software Supply Chain Platform
Microsoft	Microsoft Azure DevOps and GitHub Enterprise
OpenText (formerly Micro Focus)	Micro Focus ValueEdge
VMware	VMware Tanzu Application Platform

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Vendor Profiles

Our analysis uncovered the following strengths and weaknesses of individual vendors.

Leaders

- **GitLab leads with a balance between solid platform capabilities and overall strategy.** GitLab is known for being the vendor that made end-to-end OOTB toolchains hip again and has led the industry toward consolidated ISDPs. GitLab’s strategy includes an on-par vision to deliver an excellent developer experience without sacrificing security or compliance. Its innovation is also good, going beyond traditional developers to include AI/ML engineering. GitLab is an open core product that not only invests heavily in the open source software (OSS) community but also enables its customers to contribute to the product, earning it

high scores for community. Pricing is completely transparent and includes a free tier. Its roadmap gets leading scores and includes enhanced supply chain security, enhanced UI, granular security and compliance controls, and pipeline security — all things enterprises need.

GitLab's platform cohesion received high scores for simplicity of setup and operation as both an on-premises product and a software-as-a-service (SaaS) product. Though GitLab comes with its own GitLab version control system (VCS), it supports complete integration with all major Git providers, earning it high scores for version control. GitLab gets high scores for DevSecOps by providing OOTB security tools for static application security testing (SAST), dynamic application security testing (DAST), and Secret Detection. GitLab is one of the few ISDP vendors to incorporate a test automation dashboard, earning a top score. Dev experience is on par, as are administration and management, deployment automation, and release readiness. Build automation, governance and compliance, and infrastructure as code (IaC) were on par with others. One reference customer commented, "The CI/CD experience using secrets, environments, runners, and SAST/DAST/license scans/etc. is unparalleled in my opinion." GitLab is great for enterprises wishing to consolidate their best-of-breed toolchain into one high-performing ISDP.

Strong Performers

- **CloudBees delivers on its core promise of CI/CD, but platform cohesion needs work.** CloudBees is primarily associated with Jenkins; this well-known open source CI tool that it donated to the Continuous Delivery Foundation is one of the original best-of-breed tools that accompanied the beginning of the DevOps era more than 10 years ago. CloudBees leverages open source Jenkins and augments that with enterprise-scale deployment automation, release readiness, and value stream management (VSM) analytics to provide an enterprise-class ISDP. It got high scores for community, with significant contributions to open source projects and cultivation of its own user community. Its strategy features a solid vision and one of the larger partner ecosystems in this market. Innovation was on par and includes efforts to improve codification of policies. A planned SaaS offering should address its platform cohesion weakness and will help CloudBees serve clients that want a more integrated and maintenance-free ISDP.

CloudBees deployment automation is above par, covering blue/green, canary, and powerful feature management capabilities that are third-party add-ons for most

other platforms. CloudBees release readiness, governance, and compliance are also top-notch with complete integration with popular IT service management (ITSM) tools such as ServiceNow that bring those UIs directly into the ISDP experience. Build automation requires a bit of effort to manage fleets of build runners, and pipeline security needs to go beyond branch security to earn higher scores. When asked what they liked about working with CloudBees, reference customers had good things to say: “The vendor relationship has been a true partnership. It’s listening to our internal customers.” CloudBees is a great fit for enterprise platform teams that value open source and open choice.

- **Microsoft differentiates with innovation, but governance and compliance need attention.** Microsoft acquired GitHub, the industry’s largest open source registry, in 2018. Since that time, Microsoft’s Azure DevOps and GitHub have slowly been moving toward a unified offering. However, both the product vision and the roadmap could have been clearer on how this will shape future offerings, resulting in an on-par score. Innovation strategy is tops because GitHub Copilot has set new standards for the development tools industry and planned investments will enable it to continue this leadership position. Its partner ecosystem also scored high, with Microsoft having one of the most extensive. GitHub also supports numerous open source projects, and its online community is one of the biggest, earning a top score. The roadmap heavily weighted developer productivity but was missing key enterprise needs for governance.

Microsoft’s developer experience gets top scores for GitHub Actions, which turned GitHub into a CI leader almost overnight; Codespaces, a virtual dev environment that requires no setup by developers to get started; and Copilot, the AI-powered coding assistant. It also received top scores for OOTB security tools such as Dependabot. Support for APIs and third-party tools was also top scoring, with numerous ways to incorporate not only software but also actions. Pipeline security and platform cohesion were on par, while administration and management were missing the granular permissions of those scoring higher. Microsoft data and analytics were mostly operational, resulting in a below-par score. Deployment automation was also below par — it lacked blue/green and canary deployment scenarios. Reference customers were mostly positive about Microsoft, noting, “Generally, adoption of standard features is quick and doesn’t require a great deal of time to become productive.” Microsoft is great for organizations looking to boost developer effectiveness with cutting-edge tools.

- **Atlassian gets top scores for cohesiveness, but pipeline security capabilities are lacking.** Atlassian is best known for Jira, one of the most popular agile planning

tools on the market serving 80% of the Fortune 500. Atlassian's product strategy is on par and strives to serve two audiences, teams of developers and platform teams, by building out a data layer to connect the tools together. Its partner ecosystem is top-notch. Innovation is on par and includes platform capabilities and a data lake component to capture and aggregate data for greater analytical insight. The roadmap reflects enterprise concerns with plans to aggregate third-party security data, automatically infuse issue tickets with context, provide greater VSM insights, and expand support plans for DevOps and platform teams.

Atlassian's Confluence, which enables teams to create custom developer onboarding content and a hub for team collaboration, enhances the developer experience. This plus integrated development environment plug-ins give Atlassian a solid on-par score for developer experience. The OOTB build and CI were on par, with reusable build templates and parallel build execution to reduce build times. Deployment automation was below par, as were capabilities to secure pipelines. OOTB security tools are few, earning a below-par score, because Atlassian prefers to leverage its partner ecosystem for those. A reference customer said, "We chose Atlassian for the product features and the maturity of the well-integrated suites of products and plug-ins, which help simplify using the well-designed solution." Atlassian is great fit for software teams and enterprise platform teams that are looking for a standard ISDP approach but want to leverage best-of-breed CD and security components.

- **Harness ISDP delivers on platform cohesion, but community contributions are nascent.** Harness has a SaaS ISDP with core CI/CD modules and cloud cost management capabilities. Its vision is to simplify software development with an end-to-end software delivery platform that offers a modern developer experience and automated governance and guardrails. The strategy has mostly on-par elements such as a growing partner ecosystem and investments in innovation. The only weak area was the smaller-than-average community program — those scoring higher had greater support for internal and external communities. The roadmap includes support for database development, test optimization, end-to-end visibility, and infusion of AI/ML throughout the platform.

Harness gets top scores for platform cohesion, delivering a great OOTB experience for first-time users. It also earned top scores in build automation that includes pipelines as code, reusable pipelines, resource management, support for test automation, and intelligent test automation that can drastically reduce build times. It supports multiple deployment strategies OOTB, including blue/green and

canary. In addition, Harness can pinpoint code that's caused pipeline errors, helping speed remediation. Analytics was also top scoring and offered dev, process, and operational metrics. Customization of the UI and some ISDP components is possible and easy, but the limited scope of those changes resulted in a below-par score. Other areas for improvement include expanding the number of third-party integrations and its footprint in the user and open source community. Customers appreciate Harness' "willingness to engage directly with our development teams." Harness is great for development teams that want to spend less time managing tools and more time developing code.

- **AWS is strong in administration and management, but dev metrics are a miss.**

Amazon Web Services (AWS) pioneered infrastructure as a service (IaaS) and offers nearly every possible form of it. Its ISDP product includes CodeBuild, CodeStar, CloudFormation, and several other SaaS offerings that are integrated on demand when users create their account. The overall strategy score was solid. Innovation scored on par for current delivery and upcoming features for helping developers code faster. The vision was on par, focusing mainly on CodeCatalyst that will enable AWS to make its developer platform more accessible to a broader set of end users with different levels of technical ability. AWS actively supports its user community and open source projects, getting an on-par score for community. The roadmap is also on par and includes the aforementioned CodeCatalyst, a product currently in beta that offers a preassembled and more opinionated ISDP to help customers accelerate project setup and collaborate more fluidly.

AWS shines with excellent administration and management capabilities that enable user permissions to be managed down to the API level. AWS supports all major authentication protocols including emerging standards such as SCIM. AWS gets top scores for governance and compliance that include OOTB compliance templates. Platform cohesion and developer experience are on par. Build automation would have scored higher had it offered test execution optimization. AWS got low scores for analytics and reports; other vendors offered better dashboards with more dev-related metrics. Release readiness also scored low. Reference customer sentiment was mostly positive, saying, "Security is a first-class citizen." AWS is great for companies that want speed, scale, and security hosted in the same environment as their public cloud infrastructure.

- **CircleCI provides outstanding build automation, but its vision does not go beyond CI/CD.** CircleCI was one of the first vendors to offer CI as SaaS. Over time, it added CD and extensibility features to make a more well-rounded ISDP offering. CircleCI's on-par vision to speed time to value while simplifying the ISDP

experience is well articulated and strikes a pragmatic balance between individual developer concerns, the needs of platform teams, and support for enterprises in regulated industries. CircleCI's strategic vision is to remain a best-of-breed CI/CD tool vendor and not extend beyond that into other ISDP areas, resulting in a below-par score. The on-par roadmap aims at enterprise teams and includes features to help platform teams better manage an ISDP at scale with pervasive validation at each step of software delivery.

CircleCI has invested in optimized build processes, giving it a top score for build automation and CI. Its support for a wide range of platforms, including all public clouds and even mainframe, earns above-par scores. IaC capabilities were also top scoring. Product support leverages dedicated customer success engineers for each account, but average reference customer scores kept it on par. CircleCI user identities are derived from third-party VCS identities (e.g., those managed by GitHub or Bitbucket) and thus received a low score for administration and management. However, a new identity management system is on the roadmap to address this shortcoming. CircleCI has a good set of OOTB metrics and analytics, including DORA metrics, but the dashboards and capabilities were not as extensive as those scoring higher. One reference customer listed the following capabilities as most appreciated: "parallelization of workflows and rerunning only failed parts of workflows." CircleCI is a great fit for dev teams that value team autonomy and superfast build automation.

- **Codefresh has differentiating CI/CD but needs to improve governance and compliance.** Codefresh provides high-scoring CI/CD capabilities that emphasize the GitOps methodology, which requires all changes to infrastructure to be stored and managed as code in a Git repository prior to deployment. This methodology gained popularity from the Argo CD open source project, which powers Codefresh. Codefresh is also a registered project maintainer and contributor, which factored into its on-par community score. The Codefresh vision and roadmap focus on improvements to the developer experience and deployment capabilities such as Kubernetes context injection, improved disk utilization, and better support for teams. Innovation is confined to Argo and GitOps improvements, earning on-par scores. The one below-par score was for its smaller-than-average partner ecosystem.

Codefresh earned above-par scores for build automation, CI, and deployment automation featuring unique pipeline debugging capabilities that enable users to step in and out of containerized build steps, which no other vendor demonstrated.

Product support was strong, earning on-par scores. IaC provides strong support for Kubernetes configurations, but support for other legacy platforms lacks, keeping this score at par. Developer experience is on par; vendors scoring higher had integrated knowledge management. Other on-par scores include pipeline security, administration and management, and analytics that incorporate OOTB DORA metrics. Codefresh earned below-par scores for governance and compliance (providing limited support for ITSM integration) and support for functional test automation. When asked what was best about Codefresh, one customer said, “easy, standardized deployments for a hybrid universe.” Codefresh is a great fit for organizations that have adopted GitOps and place high value on developer autonomy.

Contenders

- **Flexagon has excellent customer support, but developer experience was below par.** Flexagon builds an end-to-end ISDP that supports traditional forms of software and infrastructure (e.g., Java and Linux) as well as business platforms such as Salesforce, Oracle, and SAP. The product vision is clear and focuses on delivering enterprise value through security and lowering risk. Planned updates include managed Oracle Cloud services, SAP blueprints, and a compliance and policy engine. Innovation is below par, investment in leading-edge areas is lacking, and the partner ecosystem was small, compared with others in this evaluation. Flexagon pricing features multiple tiers and full transparency of pricing options. Flexagon got above-par scores for test automation because it was one of the few vendors that featured native functional test automation capabilities and support.

Flexagon offers good platform cohesion with its offering that includes everything from agile planning to release automation in one package. It would have scored higher if not for some complexity in setting up and operating fleets of build runners. Flexagon got high scores for version control, with its support for the largest number of VCSes including Git and non-Git systems, and environment management that enabled teams to easily operate and manage any number of environments to support engineering teams. Deployment automation, release readiness, and platform support were also solid. Dev experience was not a focus area and resulted in a low score for that capability. Reference customers gave Flexagon perfect high scores for customer support, with one stating, “Flexagon continues to knock our socks off.” Flexagon is a great fit for enterprises looking for DevOps capabilities that span both open systems and business systems.

- **VMware delivers strong pipeline security but is light on data and analytics.**

VMware is best known for its infrastructure virtualization software. Tanzu Application Platform is VMware's ISDP and has a vision that is broad and incorporates elements of sociological factors to help developers be happier and more productive. VMware scored on par for community with strong support for open source tools and its user community. Innovation includes investments in autoremediation and AI/ML, earning an on-par score. The on-par roadmap will help users simplify the product lifecycle, improve developer collaboration, provide cost governance, and have more secure supply chains.

VMware pipeline security is one of the best in the evaluation with signed pipelines, signed commits, and additional security capabilities throughout the pipeline. Governance and compliance were on par, with OOTB compliance policies. Platform cohesion was also good with minimal setup required. Developer experience included a built-in knowledge management system to help onboard new developers and scored on par. Deployment automation included blue/green and canary deployment strategies, earning an on-par score. Administration and management were below par because they were missing features of those scoring higher. Data and analytics were few, and platform support seemed limited to Kubernetes workloads. One reference customer remarked, "The vendor is always willing to accept and act on feedback." VMware is a great fit for enterprises operating on Kubernetes and places high value on pipeline security.

- **OpenText has top-rated platform support, but DevSecOps capabilities score**

low. OpenText is best known for its portfolio of software development tools ranging from portfolio to test automation. OpenText's ValueEdge is an open platform that bundles these capabilities together in one ISDP. Its roadmap is on par, featuring elements that will better connect business and development. Innovation features AI/ML infused in risk management and testing, and OpenText boasts a healthy partner ecosystem. Both areas scored on par, with good evidence of progress from its AI/ML investments. But its vision focused more on business concerns than stronger CI/CD capabilities. In addition, pricing is opaque, compared with those scoring higher.

OpenText has one of the strongest governance and compliance offerings of this evaluation because nearly everything connects to change management systems. It has superior support for test automation capabilities that include autoclassification of test failures. In addition to providing its own VCS, users can connect with virtually any other VCS, earning above-par scores for that capability. Support for a

multitude of platforms including mainframe was a plus. Administration, management, and developer experience were on par, as were data and analytics that featured DORA and flow metrics. Deployment automation was also on par with blue/green and canary deployment support. Pipeline security lagged others evaluated. IaC, an area of growing importance to developers, also scored low. OpenText reference customers were positive about their experience, with one stating, “We don’t have any challenges at this stage.” OpenText is a great fit for enterprises with varied platforms in need of strong ties to test automation and compliance.

- **JFrog provides differentiated pipeline security capabilities, but the core ISDP falls short.** JFrog is best known for its flagship product, Artifactory, a binary artifact repository with a name that is so well known that it is synonymous with artifact repositories in general. JFrog’s strategy includes strong on-par scores for innovation based on investments in helping secure enterprise software supply chains. JFrog has a strong user community and actively participates in open source projects. Pricing flexibility was also on par, offering several payment tiers. JFrog scored below par for vision because it seemed overly focused on security at the expense of other ISDP capabilities. The roadmap was also below par due to its limited scope.

JFrog has superior version control due to the strength of Artifactory. Pipeline security is also top-notch with signed artifacts and signed pipelines that provide some of the strongest supply chain security measures available. The JFrog platform has good cohesion and requires few steps to set up and get started. Administration and management are on par and offer LDAP, OAuth, and support for SCIM to manage users at scale. Xray, a security product that scans artifacts, comes bundled and offers container and image scanning along with OSS supply chain management. To make this a more competitive ISDP offering, JFrog needs to invest in release readiness, data, and analytics and expand policy as code to all aspects of the ISDP while doing more to support test automation. Customers like how they are treated by JFrog, saying: It’s a “partner, not a vendor — they bring this mentality to everything.” JFrog is great for enterprises that place high value on supply chain security.

Challengers

- **IBM UrbanCode Deploy offers solid build and deployment capabilities but looks dated.** IBM UrbanCode offers UrbanCode Build and UrbanCode Deploy as two components that work together to deliver an enterprise-scale ISDP. IBM has a

solid development community because IBM Garage has thousands of contributors. IBM's partner ecosystem is also solidly on par with a number of top systems integrators listed as partners. The strategy has several weak elements including a roadmap that emphasizes maintenance versus new features, and innovation proof points are lacking.

IBM has solid build automation, deployment automation, release readiness, and support for a large variety of target platforms, which are core ISDP capabilities valued by enterprise dev teams. IBM's UrbanCode Velocity module offers strong analytics and graphics, earning it on-par scores for analytics, and there's a healthy number of third-party integrations for an on-par score in that category. But platform cohesion suffers from a complicated installation process, administration and management offer limited capabilities such as LDAP, and the developer experience is bare bones. Pipelines are present but don't offer the same level of capability as more modern tools. IBM is a good fit for enterprises that need to build an ISDP bridge between legacy and modern infrastructure. IBM declined to participate in the full Forrester Wave evaluation process.

Evaluation Overview

We grouped our evaluation criteria into three high-level categories:

- **Current offering.** Each vendor's position on the vertical axis of the Forrester Wave graphic indicates the strength of its current offering. Key criteria for these solutions include platform capabilities to facilitate management at scale, CI/CD to perform core ISDP functions, extensibility to enable platform operators to incorporate third-party tools as needed, and integrated systems to secure the software delivery pipeline.
- **Strategy.** Placement on the horizontal axis indicates the strength of the vendors' strategies. We evaluated vision, roadmap, and innovation to determine which vendor was placing the best bets to fulfill long-term product promises.
- **Market presence.** Represented by the size of the markers on the graphic, our market presence scores reflect each vendor's ISDP revenue and number of enterprise customers.

Vendor Inclusion Criteria

Each of the vendors we included in this assessment has:

- **Integrated CI/CD (e.g., deployment and release automation) capabilities.** The vendor provides integrated CI/CD capabilities OOTB that enable teams to rapidly establish build and deploy processes with minimal configuration.
- **Extensibility via APIs or plug-ins.** The vendor provides APIs to support third-party tool integration to extend the platform to incorporate performance, test automation, and security.
- **A system for centralized administration.** A single dashboard helps platform administrators operate and maintain an ISDP at enterprise scale.
- **Strong client interest.** Analyst inquiry logs indicate the vendor is top of mind among Forrester's end-user clientele with regard to ISDPs.
- **General-purpose programming languages and platforms.** The vendor supports build and deployment automation for open source programming languages like C# and Java as well as deployment to general-purpose runtime platforms such as Linux or Windows. It is also not exclusively tied to any particular business platform such as Salesforce or SAP.
- **High market performance.** The vendor's market performance demonstrates that it is a significant player in the ISDP market.

Supplemental Material

Online Resource

We publish all our Forrester Wave scores and weightings in an Excel file that provides detailed product evaluations and customizable rankings; download this tool by clicking the link at the beginning of this report on Forrester.com. We intend these scores and default weightings to serve only as a starting point and encourage readers to adapt the weightings to fit their individual needs.

The Forrester Wave Methodology

A Forrester Wave is a guide for buyers considering their purchasing options in a technology marketplace. To offer an equitable process for all participants, Forrester follows [The Forrester Wave™ Methodology](#) to evaluate participating vendors.

In our review, we conduct primary research to develop a list of vendors to consider for the evaluation. From that initial pool of vendors, we narrow our final list based on the inclusion criteria. We then gather details of product and strategy through a detailed questionnaire, demos/briefings, and customer reference surveys/interviews. We use those inputs, along with the analyst's experience and expertise in the marketplace, to

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score vendors, using a relative rating system that compares each vendor against the others in the evaluation.

We include the Forrester Wave publishing date (quarter and year) clearly in the title of each Forrester Wave report. We evaluated the vendors participating in this Forrester Wave using materials they provided to us by March 2023 and did not allow additional information after that point. We encourage readers to evaluate how the market and vendor offerings change over time.

In accordance with [our vendor review policy](#), Forrester asks vendors to review our findings prior to publishing to check for accuracy. Vendors marked as nonparticipating vendors in the Forrester Wave graphic met our defined inclusion criteria but declined to participate in or contributed only partially to the evaluation. We score these vendors in accordance with [our vendor participation policy](#) and publish their positioning along with those of the participating vendors.

Integrity Policy

We conduct all our research, including Forrester Wave evaluations, in accordance with the [integrity policy](#) posted on our website.



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WAVE REPORT

The Forrester WaveTM: Integrated Software Delivery Platforms, Q2 2023

The 13 Providers That Matter Most And How They
Stack Up

May 31, 2023

By Christopher Condo with Chris Gardner, Sara Sjoblom, Kara Hartig

FORRESTER®

Summary

In our 26-criterion evaluation of integrated software delivery platform (ISDP) providers, we identified the most significant ones and researched, analyzed, and scored them. This report shows how each provider measures up and helps modern application development leaders and DevOps platform team leaders select the right one for their needs.

Additional resources are available in the [online version](#) of this report.

ISDPs Provide A Strong Platform Foundation But Still Need Best-Of-Breed Tools

In our reports, [The Future Of DevOps](#) and [The Forrester Guide To DevOps](#), Forrester interviewed dozens of enterprise software development leaders from diverse companies ranging from Netflix to TransUnion to understand the current state of DevOps, its relevance, and next steps enterprise leaders were taking to further advance their practices. In addition, we asked which tools they were using, how effective the tools were, which were using [ISDPs](#), and how well those ISDPs were performing. We learned that many enterprises have shifted to platform teams to support their DevOps efforts and many platform leaders are evaluating and using ISDPs, but few are using all of their ISDP vendor's capabilities. They explained that while these ISDP tools hold great potential for a complete end-to-end toolchain, many individual capabilities fell short of enterprise needs, forcing these leaders to source those capabilities from separate best-of-breed vendors.

Our reference customer survey backs this up by indicating that enterprises are using (on average) only 53% of the capabilities they are paying for, with the remaining capabilities either being delivered by a separate vendor or simply not used. This means that ISDP vendors are not only competing with each other but also competing with the best-of-breeds, which is why the bar has risen so high for this evaluation. ISDP vendors recognize these challenges and are responding with a number of innovative solutions that will make them more attractive than a traditional DIY toolchain but open and extensible enough that end users can add the tools they want without sacrificing capability. The challenge is striking the right balance to create overall value.

As a result of these trends, ISDP customers should look for providers that:

- **Offer a strong core of CI/CD capabilities.** Continuous integration (CI) and continuous delivery (CD) are the core capabilities that the rest of the software delivery toolchain is built around. Several vendors offered build optimization algorithms that slash build and unit test times by intelligently building and testing only the code that changed. Deployment automation is also a differentiating capability, with only a few vendors offering advanced capabilities out of the box (OOTB) such as blue/green, canary, and feature management deployment strategies.
- **Provide connected platforms that best-of-breed tools can augment.** ISDPs provide engineering leverage in the form of preintegrated toolchains that enable work to easily flow from one process to the next, where administration occurs at

the platform level versus the tool level, and reduced context switching improves the developer experience. Often, platform leaders want to override the default capability with third-party tools that they believe offer superior capability, which can lead to a bit of a balancing act. That's why some ISDP vendors are heavily investing in improvements to their end-to-end platform, while others are banking on investments in third-party connectivity and partnerships.

- **Help enterprise leaders shift from DevOps to DevSecOps.** Forrester fields calls every day from enterprise development leaders that are transforming their DevOps practice into a development, security, and operations (DevSecOps) practice to make security a first-class concern for the entire development organization. ISDP vendors are responding with capabilities that better secure software delivery pipelines, including integrated secrets management, signed pipelines, and signed artifacts. Reference customers noted that not all of these tools are direct replacements for best-of-breed (best-of-breed security tools often having a significant advantage over bundled ones), but end users like them as forms of additional assurance and coverage.

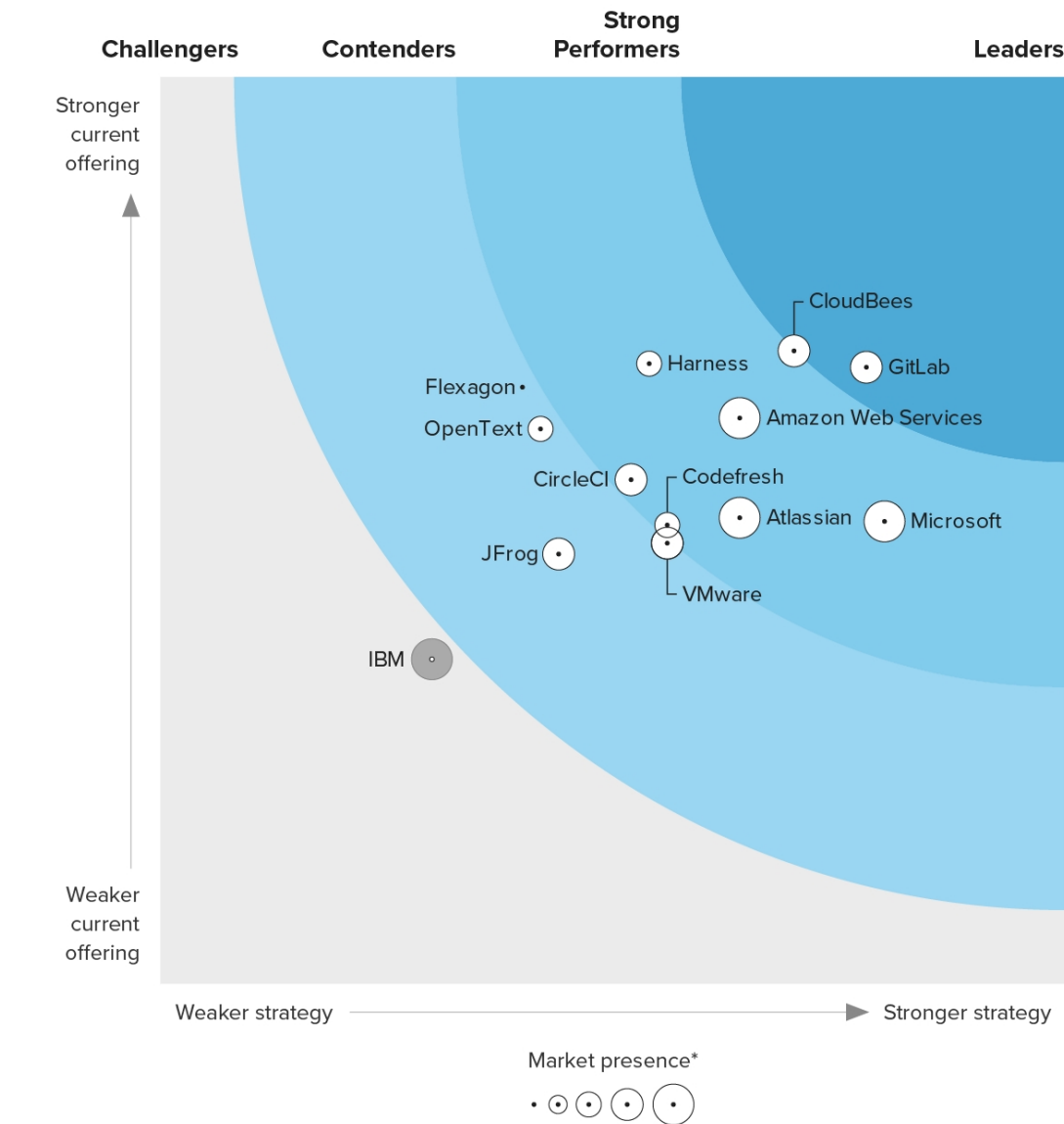
Evaluation Summary

The Forrester Wave™ evaluation highlights Leaders, Strong Performers, Contenders, and Challengers. It's an assessment of the top vendors in the market; it doesn't represent the entire vendor landscape. You'll find more information about this market in Forrester's [Now Tech: Integrated Software Delivery Platforms, Q2 2022](#).

We intend this evaluation to be a starting point only and encourage clients to view product evaluations and adapt criteria weightings using the Excel-based vendor comparison tool (see Figures 1 and 2). Click the link at the beginning of this report on [Forrester.com](#) to download the tool.

Figure 1
Forrester Wave™: Integrated Software Delivery Platforms, Q2 2023

THE FORRESTER WAVE™
Integrated Software Delivery Platforms
Q2 2023



*A gray bubble or open dot indicates a nonparticipating vendor.

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Figure 2**Forrester Wave™: Integrated Software Delivery Platforms Scorecard, Q2 2023**

		Forrester's weighting	Amazon Web Services	Atlassian	CircleCI	CloudBees	Codefresh	Flexagon	GitLab
Current offering		50%	2.57	3.12	2.78	3.49	2.53	3.29	3.40
Platform capabilities		20%	3.50	3.00	2.00	2.50	3.00	2.50	3.50
CI/CD capabilities		25%	2.00	2.60	3.00	4.00	4.00	3.20	3.40
Development, security, and operations		10%	1.00	3.00	2.00	2.00	2.00	3.00	4.00
Extensibility		10%	3.00	3.40	3.00	3.40	1.00	3.00	3.00
Governance and compliance		10%	3.00	5.00	3.00	5.00	1.00	3.00	3.00
Infrastructure management		10%	2.20	3.80	3.80	5.00	1.80	3.40	3.00
Product support		10%	3.00	3.00	3.00	3.00	3.00	5.00	3.00
Support for test automation		5%	3.00	1.00	3.00	3.00	1.00	5.00	5.00
Strategy		50%	3.20	3.20	2.60	3.50	2.80	2.00	3.90
Vision		20%	3.00	3.00	1.00	3.00	3.00	3.00	3.00
Roadmap		20%	3.00	3.00	3.00	3.00	3.00	3.00	5.00
Partner ecosystem		10%	5.00	5.00	1.00	5.00	1.00	1.00	3.00
Innovation		25%	3.00	3.00	3.00	3.00	3.00	1.00	3.00
Community		15%	3.00	3.00	3.00	5.00	3.00	1.00	5.00
Pricing flexibility and transparency		10%	3.00	3.00	5.00	3.00	3.00	3.00	5.00
Market presence		0%	5.00	4.50	3.50	3.50	2.50	1.00	3.50
ISDP revenue		50%	5.00	5.00	4.00	4.00	2.00	1.00	4.00
Number of enterprise customers		50%	5.00	4.00	3.00	3.00	3.00	1.00	3.00

All scores are based on a scale of 0 (weak) to 5 (strong).

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	Forrester's weighting	Harness	IBM*	JSFrog	Microsoft	OpenText	VMware
Current offering	50%	3.42	1.79	2.37	2.55	3.06	2.43
Platform capabilities	20%	4.00	1.50	2.00	3.00	2.50	2.00
CI/CD capabilities	25%	4.20	2.80	2.40	2.40	3.40	2.40
Development, security, and operations	10%	3.00	1.00	4.00	4.00	1.00	3.00
Extensibility	10%	1.00	1.40	4.20	4.20	3.40	3.00
Governance and compliance	10%	3.00	3.00	1.00	1.00	5.00	3.00
Infrastructure management	10%	4.20	1.00	1.00	1.80	2.20	1.80
Product support	10%	3.00	1.00	3.00	1.00	3.00	3.00
Support for test automation	5%	3.00	1.00	1.00	3.00	5.00	1.00
Strategy	50%	2.70	1.50	2.20	4.00	2.10	2.80
Vision	20%	3.00	1.00	1.00	3.00	1.00	3.00
Roadmap	20%	3.00	1.00	1.00	3.00	3.00	3.00
Partner ecosystem	10%	3.00	3.00	3.00	5.00	3.00	3.00
Innovation	25%	3.00	1.00	3.00	5.00	3.00	3.00
Community	15%	1.00	3.00	3.00	5.00	1.00	3.00
Pricing flexibility and transparency	10%	3.00	1.00	3.00	3.00	1.00	1.00
Market presence	0%	3.00	4.50	4.00	5.00	2.50	4.00
ISDP revenue	50%	3.00	5.00	4.00	5.00	4.00	4.00
Number of enterprise customers	50%	3.00	4.00	4.00	5.00	1.00	4.00

All scores are based on a scale of 0 (weak) to 5 (strong).

*Indicates a nonparticipating vendor

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Vendor Offerings

Forrester evaluated the offerings listed below (see Figure 3).

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Figure 3
Evaluated Vendors And Product Information

Vendor	Product evaluated
Amazon Web Services	AWS Code Services
Atlassian	Atlassian Open DevOps
CircleCI	CircleCI
CloudBees	CloudBees Platform
Codefresh	Codefresh
Flexagon	Flexagon FlexDeploy
GitLab	GitLab
Harness	Harness Software Delivery Platform
IBM	IBM UrbanCode Deploy
JFrog	JFrog Software Supply Chain Platform
Microsoft	Microsoft Azure DevOps and GitHub Enterprise
OpenText (formerly Micro Focus)	Micro Focus ValueEdge
VMware	VMware Tanzu Application Platform

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Vendor Profiles

Our analysis uncovered the following strengths and weaknesses of individual vendors.

Leaders

- **GitLab leads with a balance between solid platform capabilities and overall strategy.** GitLab is known for being the vendor that made end-to-end OOTB toolchains hip again and has led the industry toward consolidated ISDPs. GitLab’s strategy includes an on-par vision to deliver an excellent developer experience without sacrificing security or compliance. Its innovation is also good, going beyond traditional developers to include AI/ML engineering. GitLab is an open core product that not only invests heavily in the open source software (OSS) community but also enables its customers to contribute to the product, earning it

high scores for community. Pricing is completely transparent and includes a free tier. Its roadmap gets leading scores and includes enhanced supply chain security, enhanced UI, granular security and compliance controls, and pipeline security — all things enterprises need.

GitLab's platform cohesion received high scores for simplicity of setup and operation as both an on-premises product and a software-as-a-service (SaaS) product. Though GitLab comes with its own GitLab version control system (VCS), it supports complete integration with all major Git providers, earning it high scores for version control. GitLab gets high scores for DevSecOps by providing OOTB security tools for static application security testing (SAST), dynamic application security testing (DAST), and Secret Detection. GitLab is one of the few ISDP vendors to incorporate a test automation dashboard, earning a top score. Dev experience is on par, as are administration and management, deployment automation, and release readiness. Build automation, governance and compliance, and infrastructure as code (IaC) were on par with others. One reference customer commented, "The CI/CD experience using secrets, environments, runners, and SAST/DAST/license scans/etc. is unparalleled in my opinion." GitLab is great for enterprises wishing to consolidate their best-of-breed toolchain into one high-performing ISDP.

Strong Performers

- **CloudBees delivers on its core promise of CI/CD, but platform cohesion needs work.** CloudBees is primarily associated with Jenkins; this well-known open source CI tool that it donated to the Continuous Delivery Foundation is one of the original best-of-breed tools that accompanied the beginning of the DevOps era more than 10 years ago. CloudBees leverages open source Jenkins and augments that with enterprise-scale deployment automation, release readiness, and value stream management (VSM) analytics to provide an enterprise-class ISDP. It got high scores for community, with significant contributions to open source projects and cultivation of its own user community. Its strategy features a solid vision and one of the larger partner ecosystems in this market. Innovation was on par and includes efforts to improve codification of policies. A planned SaaS offering should address its platform cohesion weakness and will help CloudBees serve clients that want a more integrated and maintenance-free ISDP.

CloudBees deployment automation is above par, covering blue/green, canary, and powerful feature management capabilities that are third-party add-ons for most

other platforms. CloudBees release readiness, governance, and compliance are also top-notch with complete integration with popular IT service management (ITSM) tools such as ServiceNow that bring those UIs directly into the ISDP experience. Build automation requires a bit of effort to manage fleets of build runners, and pipeline security needs to go beyond branch security to earn higher scores. When asked what they liked about working with CloudBees, reference customers had good things to say: “The vendor relationship has been a true partnership. It’s listening to our internal customers.” CloudBees is a great fit for enterprise platform teams that value open source and open choice.

- **Microsoft differentiates with innovation, but governance and compliance need attention.** Microsoft acquired GitHub, the industry’s largest open source registry, in 2018. Since that time, Microsoft’s Azure DevOps and GitHub have slowly been moving toward a unified offering. However, both the product vision and the roadmap could have been clearer on how this will shape future offerings, resulting in an on-par score. Innovation strategy is tops because GitHub Copilot has set new standards for the development tools industry and planned investments will enable it to continue this leadership position. Its partner ecosystem also scored high, with Microsoft having one of the most extensive. GitHub also supports numerous open source projects, and its online community is one of the biggest, earning a top score. The roadmap heavily weighted developer productivity but was missing key enterprise needs for governance.

Microsoft’s developer experience gets top scores for GitHub Actions, which turned GitHub into a CI leader almost overnight; Codespaces, a virtual dev environment that requires no setup by developers to get started; and Copilot, the AI-powered coding assistant. It also received top scores for OOTB security tools such as Dependabot. Support for APIs and third-party tools was also top scoring, with numerous ways to incorporate not only software but also actions. Pipeline security and platform cohesion were on par, while administration and management were missing the granular permissions of those scoring higher. Microsoft data and analytics were mostly operational, resulting in a below-par score. Deployment automation was also below par — it lacked blue/green and canary deployment scenarios. Reference customers were mostly positive about Microsoft, noting, “Generally, adoption of standard features is quick and doesn’t require a great deal of time to become productive.” Microsoft is great for organizations looking to boost developer effectiveness with cutting-edge tools.

- **Atlassian gets top scores for cohesiveness, but pipeline security capabilities are lacking.** Atlassian is best known for Jira, one of the most popular agile planning

tools on the market serving 80% of the Fortune 500. Atlassian's product strategy is on par and strives to serve two audiences, teams of developers and platform teams, by building out a data layer to connect the tools together. Its partner ecosystem is top-notch. Innovation is on par and includes platform capabilities and a data lake component to capture and aggregate data for greater analytical insight. The roadmap reflects enterprise concerns with plans to aggregate third-party security data, automatically infuse issue tickets with context, provide greater VSM insights, and expand support plans for DevOps and platform teams.

Atlassian's Confluence, which enables teams to create custom developer onboarding content and a hub for team collaboration, enhances the developer experience. This plus integrated development environment plug-ins give Atlassian a solid on-par score for developer experience. The OOTB build and CI were on par, with reusable build templates and parallel build execution to reduce build times. Deployment automation was below par, as were capabilities to secure pipelines. OOTB security tools are few, earning a below-par score, because Atlassian prefers to leverage its partner ecosystem for those. A reference customer said, "We chose Atlassian for the product features and the maturity of the well-integrated suites of products and plug-ins, which help simplify using the well-designed solution." Atlassian is great fit for software teams and enterprise platform teams that are looking for a standard ISDP approach but want to leverage best-of-breed CD and security components.

- **Harness ISDP delivers on platform cohesion, but community contributions are nascent.** Harness has a SaaS ISDP with core CI/CD modules and cloud cost management capabilities. Its vision is to simplify software development with an end-to-end software delivery platform that offers a modern developer experience and automated governance and guardrails. The strategy has mostly on-par elements such as a growing partner ecosystem and investments in innovation. The only weak area was the smaller-than-average community program — those scoring higher had greater support for internal and external communities. The roadmap includes support for database development, test optimization, end-to-end visibility, and infusion of AI/ML throughout the platform.

Harness gets top scores for platform cohesion, delivering a great OOTB experience for first-time users. It also earned top scores in build automation that includes pipelines as code, reusable pipelines, resource management, support for test automation, and intelligent test automation that can drastically reduce build times. It supports multiple deployment strategies OOTB, including blue/green and

canary. In addition, Harness can pinpoint code that's caused pipeline errors, helping speed remediation. Analytics was also top scoring and offered dev, process, and operational metrics. Customization of the UI and some ISDP components is possible and easy, but the limited scope of those changes resulted in a below-par score. Other areas for improvement include expanding the number of third-party integrations and its footprint in the user and open source community. Customers appreciate Harness' "willingness to engage directly with our development teams." Harness is great for development teams that want to spend less time managing tools and more time developing code.

- **AWS is strong in administration and management, but dev metrics are a miss.**

Amazon Web Services (AWS) pioneered infrastructure as a service (IaaS) and offers nearly every possible form of it. Its ISDP product includes CodeBuild, CodeStar, CloudFormation, and several other SaaS offerings that are integrated on demand when users create their account. The overall strategy score was solid. Innovation scored on par for current delivery and upcoming features for helping developers code faster. The vision was on par, focusing mainly on CodeCatalyst that will enable AWS to make its developer platform more accessible to a broader set of end users with different levels of technical ability. AWS actively supports its user community and open source projects, getting an on-par score for community. The roadmap is also on par and includes the aforementioned CodeCatalyst, a product currently in beta that offers a preassembled and more opinionated ISDP to help customers accelerate project setup and collaborate more fluidly.

AWS shines with excellent administration and management capabilities that enable user permissions to be managed down to the API level. AWS supports all major authentication protocols including emerging standards such as SCIM. AWS gets top scores for governance and compliance that include OOTB compliance templates. Platform cohesion and developer experience are on par. Build automation would have scored higher had it offered test execution optimization. AWS got low scores for analytics and reports; other vendors offered better dashboards with more dev-related metrics. Release readiness also scored low. Reference customer sentiment was mostly positive, saying, "Security is a first-class citizen." AWS is great for companies that want speed, scale, and security hosted in the same environment as their public cloud infrastructure.

- **CircleCI provides outstanding build automation, but its vision does not go beyond CI/CD.** CircleCI was one of the first vendors to offer CI as SaaS. Over time, it added CD and extensibility features to make a more well-rounded ISDP offering. CircleCI's on-par vision to speed time to value while simplifying the ISDP

experience is well articulated and strikes a pragmatic balance between individual developer concerns, the needs of platform teams, and support for enterprises in regulated industries. CircleCI's strategic vision is to remain a best-of-breed CI/CD tool vendor and not extend beyond that into other ISDP areas, resulting in a below-par score. The on-par roadmap aims at enterprise teams and includes features to help platform teams better manage an ISDP at scale with pervasive validation at each step of software delivery.

CircleCI has invested in optimized build processes, giving it a top score for build automation and CI. Its support for a wide range of platforms, including all public clouds and even mainframe, earns above-par scores. IaC capabilities were also top scoring. Product support leverages dedicated customer success engineers for each account, but average reference customer scores kept it on par. CircleCI user identities are derived from third-party VCS identities (e.g., those managed by GitHub or Bitbucket) and thus received a low score for administration and management. However, a new identity management system is on the roadmap to address this shortcoming. CircleCI has a good set of OOTB metrics and analytics, including DORA metrics, but the dashboards and capabilities were not as extensive as those scoring higher. One reference customer listed the following capabilities as most appreciated: "parallelization of workflows and rerunning only failed parts of workflows." CircleCI is a great fit for dev teams that value team autonomy and superfast build automation.

- **Codefresh has differentiating CI/CD but needs to improve governance and compliance.** Codefresh provides high-scoring CI/CD capabilities that emphasize the GitOps methodology, which requires all changes to infrastructure to be stored and managed as code in a Git repository prior to deployment. This methodology gained popularity from the Argo CD open source project, which powers Codefresh. Codefresh is also a registered project maintainer and contributor, which factored into its on-par community score. The Codefresh vision and roadmap focus on improvements to the developer experience and deployment capabilities such as Kubernetes context injection, improved disk utilization, and better support for teams. Innovation is confined to Argo and GitOps improvements, earning on-par scores. The one below-par score was for its smaller-than-average partner ecosystem.

Codefresh earned above-par scores for build automation, CI, and deployment automation featuring unique pipeline debugging capabilities that enable users to step in and out of containerized build steps, which no other vendor demonstrated.

Product support was strong, earning on-par scores. IaC provides strong support for Kubernetes configurations, but support for other legacy platforms lacks, keeping this score at par. Developer experience is on par; vendors scoring higher had integrated knowledge management. Other on-par scores include pipeline security, administration and management, and analytics that incorporate OOTB DORA metrics. Codefresh earned below-par scores for governance and compliance (providing limited support for ITSM integration) and support for functional test automation. When asked what was best about Codefresh, one customer said, “easy, standardized deployments for a hybrid universe.” Codefresh is a great fit for organizations that have adopted GitOps and place high value on developer autonomy.

Contenders

- **Flexagon has excellent customer support, but developer experience was below par.** Flexagon builds an end-to-end ISDP that supports traditional forms of software and infrastructure (e.g., Java and Linux) as well as business platforms such as Salesforce, Oracle, and SAP. The product vision is clear and focuses on delivering enterprise value through security and lowering risk. Planned updates include managed Oracle Cloud services, SAP blueprints, and a compliance and policy engine. Innovation is below par, investment in leading-edge areas is lacking, and the partner ecosystem was small, compared with others in this evaluation. Flexagon pricing features multiple tiers and full transparency of pricing options. Flexagon got above-par scores for test automation because it was one of the few vendors that featured native functional test automation capabilities and support.

Flexagon offers good platform cohesion with its offering that includes everything from agile planning to release automation in one package. It would have scored higher if not for some complexity in setting up and operating fleets of build runners. Flexagon got high scores for version control, with its support for the largest number of VCSes including Git and non-Git systems, and environment management that enabled teams to easily operate and manage any number of environments to support engineering teams. Deployment automation, release readiness, and platform support were also solid. Dev experience was not a focus area and resulted in a low score for that capability. Reference customers gave Flexagon perfect high scores for customer support, with one stating, “Flexagon continues to knock our socks off.” Flexagon is a great fit for enterprises looking for DevOps capabilities that span both open systems and business systems.

- **VMware delivers strong pipeline security but is light on data and analytics.**

VMware is best known for its infrastructure virtualization software. Tanzu Application Platform is VMware's ISDP and has a vision that is broad and incorporates elements of sociological factors to help developers be happier and more productive. VMware scored on par for community with strong support for open source tools and its user community. Innovation includes investments in autoremediation and AI/ML, earning an on-par score. The on-par roadmap will help users simplify the product lifecycle, improve developer collaboration, provide cost governance, and have more secure supply chains.

VMware pipeline security is one of the best in the evaluation with signed pipelines, signed commits, and additional security capabilities throughout the pipeline. Governance and compliance were on par, with OOTB compliance policies. Platform cohesion was also good with minimal setup required. Developer experience included a built-in knowledge management system to help onboard new developers and scored on par. Deployment automation included blue/green and canary deployment strategies, earning an on-par score. Administration and management were below par because they were missing features of those scoring higher. Data and analytics were few, and platform support seemed limited to Kubernetes workloads. One reference customer remarked, "The vendor is always willing to accept and act on feedback." VMware is a great fit for enterprises operating on Kubernetes and places high value on pipeline security.

- **OpenText has top-rated platform support, but DevSecOps capabilities score**

low. OpenText is best known for its portfolio of software development tools ranging from portfolio to test automation. OpenText's ValueEdge is an open platform that bundles these capabilities together in one ISDP. Its roadmap is on par, featuring elements that will better connect business and development. Innovation features AI/ML infused in risk management and testing, and OpenText boasts a healthy partner ecosystem. Both areas scored on par, with good evidence of progress from its AI/ML investments. But its vision focused more on business concerns than stronger CI/CD capabilities. In addition, pricing is opaque, compared with those scoring higher.

OpenText has one of the strongest governance and compliance offerings of this evaluation because nearly everything connects to change management systems. It has superior support for test automation capabilities that include autoclassification of test failures. In addition to providing its own VCS, users can connect with virtually any other VCS, earning above-par scores for that capability. Support for a

multitude of platforms including mainframe was a plus. Administration, management, and developer experience were on par, as were data and analytics that featured DORA and flow metrics. Deployment automation was also on par with blue/green and canary deployment support. Pipeline security lagged others evaluated. IaC, an area of growing importance to developers, also scored low. OpenText reference customers were positive about their experience, with one stating, “We don’t have any challenges at this stage.” OpenText is a great fit for enterprises with varied platforms in need of strong ties to test automation and compliance.

- **JFrog provides differentiated pipeline security capabilities, but the core ISDP falls short.** JFrog is best known for its flagship product, Artifactory, a binary artifact repository with a name that is so well known that it is synonymous with artifact repositories in general. JFrog’s strategy includes strong on-par scores for innovation based on investments in helping secure enterprise software supply chains. JFrog has a strong user community and actively participates in open source projects. Pricing flexibility was also on par, offering several payment tiers. JFrog scored below par for vision because it seemed overly focused on security at the expense of other ISDP capabilities. The roadmap was also below par due to its limited scope.

JFrog has superior version control due to the strength of Artifactory. Pipeline security is also top-notch with signed artifacts and signed pipelines that provide some of the strongest supply chain security measures available. The JFrog platform has good cohesion and requires few steps to set up and get started. Administration and management are on par and offer LDAP, OAuth, and support for SCIM to manage users at scale. Xray, a security product that scans artifacts, comes bundled and offers container and image scanning along with OSS supply chain management. To make this a more competitive ISDP offering, JFrog needs to invest in release readiness, data, and analytics and expand policy as code to all aspects of the ISDP while doing more to support test automation. Customers like how they are treated by JFrog, saying: It’s a “partner, not a vendor — they bring this mentality to everything.” JFrog is great for enterprises that place high value on supply chain security.

Challengers

- **IBM UrbanCode Deploy offers solid build and deployment capabilities but looks dated.** IBM UrbanCode offers UrbanCode Build and UrbanCode Deploy as two components that work together to deliver an enterprise-scale ISDP. IBM has a

solid development community because IBM Garage has thousands of contributors. IBM's partner ecosystem is also solidly on par with a number of top systems integrators listed as partners. The strategy has several weak elements including a roadmap that emphasizes maintenance versus new features, and innovation proof points are lacking.

IBM has solid build automation, deployment automation, release readiness, and support for a large variety of target platforms, which are core ISDP capabilities valued by enterprise dev teams. IBM's UrbanCode Velocity module offers strong analytics and graphics, earning it on-par scores for analytics, and there's a healthy number of third-party integrations for an on-par score in that category. But platform cohesion suffers from a complicated installation process, administration and management offer limited capabilities such as LDAP, and the developer experience is bare bones. Pipelines are present but don't offer the same level of capability as more modern tools. IBM is a good fit for enterprises that need to build an ISDP bridge between legacy and modern infrastructure. IBM declined to participate in the full Forrester Wave evaluation process.

Evaluation Overview

We grouped our evaluation criteria into three high-level categories:

- **Current offering.** Each vendor's position on the vertical axis of the Forrester Wave graphic indicates the strength of its current offering. Key criteria for these solutions include platform capabilities to facilitate management at scale, CI/CD to perform core ISDP functions, extensibility to enable platform operators to incorporate third-party tools as needed, and integrated systems to secure the software delivery pipeline.
- **Strategy.** Placement on the horizontal axis indicates the strength of the vendors' strategies. We evaluated vision, roadmap, and innovation to determine which vendor was placing the best bets to fulfill long-term product promises.
- **Market presence.** Represented by the size of the markers on the graphic, our market presence scores reflect each vendor's ISDP revenue and number of enterprise customers.

Vendor Inclusion Criteria

Each of the vendors we included in this assessment has:

- **Integrated CI/CD (e.g., deployment and release automation) capabilities.** The vendor provides integrated CI/CD capabilities OOTB that enable teams to rapidly establish build and deploy processes with minimal configuration.
- **Extensibility via APIs or plug-ins.** The vendor provides APIs to support third-party tool integration to extend the platform to incorporate performance, test automation, and security.
- **A system for centralized administration.** A single dashboard helps platform administrators operate and maintain an ISDP at enterprise scale.
- **Strong client interest.** Analyst inquiry logs indicate the vendor is top of mind among Forrester's end-user clientele with regard to ISDPs.
- **General-purpose programming languages and platforms.** The vendor supports build and deployment automation for open source programming languages like C# and Java as well as deployment to general-purpose runtime platforms such as Linux or Windows. It is also not exclusively tied to any particular business platform such as Salesforce or SAP.
- **High market performance.** The vendor's market performance demonstrates that it is a significant player in the ISDP market.

Supplemental Material

Online Resource

We publish all our Forrester Wave scores and weightings in an Excel file that provides detailed product evaluations and customizable rankings; download this tool by clicking the link at the beginning of this report on Forrester.com. We intend these scores and default weightings to serve only as a starting point and encourage readers to adapt the weightings to fit their individual needs.

The Forrester Wave Methodology

A Forrester Wave is a guide for buyers considering their purchasing options in a technology marketplace. To offer an equitable process for all participants, Forrester follows [The Forrester Wave™ Methodology](#) to evaluate participating vendors.

In our review, we conduct primary research to develop a list of vendors to consider for the evaluation. From that initial pool of vendors, we narrow our final list based on the inclusion criteria. We then gather details of product and strategy through a detailed questionnaire, demos/briefings, and customer reference surveys/interviews. We use those inputs, along with the analyst's experience and expertise in the marketplace, to

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score vendors, using a relative rating system that compares each vendor against the others in the evaluation.

We include the Forrester Wave publishing date (quarter and year) clearly in the title of each Forrester Wave report. We evaluated the vendors participating in this Forrester Wave using materials they provided to us by March 2023 and did not allow additional information after that point. We encourage readers to evaluate how the market and vendor offerings change over time.

In accordance with [our vendor review policy](#), Forrester asks vendors to review our findings prior to publishing to check for accuracy. Vendors marked as nonparticipating vendors in the Forrester Wave graphic met our defined inclusion criteria but declined to participate in or contributed only partially to the evaluation. We score these vendors in accordance with [our vendor participation policy](#) and publish their positioning along with those of the participating vendors.

Integrity Policy

We conduct all our research, including Forrester Wave evaluations, in accordance with the [integrity policy](#) posted on our website.



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WAVE REPORT

The Forrester WaveTM: Integrated Software Delivery Platforms, Q2 2023

The 13 Providers That Matter Most And How They
Stack Up

May 31, 2023

By Christopher Condo with Chris Gardner, Sara Sjoblom, Kara Hartig

FORRESTER®

Summary

In our 26-criterion evaluation of integrated software delivery platform (ISDP) providers, we identified the most significant ones and researched, analyzed, and scored them. This report shows how each provider measures up and helps modern application development leaders and DevOps platform team leaders select the right one for their needs.

Additional resources are available in the [online version](#) of this report.

ISDPs Provide A Strong Platform Foundation But Still Need Best-Of-Breed Tools

In our reports, [The Future Of DevOps](#) and [The Forrester Guide To DevOps](#), Forrester interviewed dozens of enterprise software development leaders from diverse companies ranging from Netflix to TransUnion to understand the current state of DevOps, its relevance, and next steps enterprise leaders were taking to further advance their practices. In addition, we asked which tools they were using, how effective the tools were, which were using ISDPs, and how well those ISDPs were performing. We learned that many enterprises have shifted to platform teams to support their DevOps efforts and many platform leaders are evaluating and using ISDPs, but few are using all of their ISDP vendor's capabilities. They explained that while these ISDP tools hold great potential for a complete end-to-end toolchain, many individual capabilities fell short of enterprise needs, forcing these leaders to source those capabilities from separate best-of-breed vendors.

Our reference customer survey backs this up by indicating that enterprises are using (on average) only 53% of the capabilities they are paying for, with the remaining capabilities either being delivered by a separate vendor or simply not used. This means that ISDP vendors are not only competing with each other but also competing with the best-of-breeds, which is why the bar has risen so high for this evaluation. ISDP vendors recognize these challenges and are responding with a number of innovative solutions that will make them more attractive than a traditional DIY toolchain but open and extensible enough that end users can add the tools they want without sacrificing capability. The challenge is striking the right balance to create overall value.

As a result of these trends, ISDP customers should look for providers that:

- **Offer a strong core of CI/CD capabilities.** Continuous integration (CI) and continuous delivery (CD) are the core capabilities that the rest of the software delivery toolchain is built around. Several vendors offered build optimization algorithms that slash build and unit test times by intelligently building and testing only the code that changed. Deployment automation is also a differentiating capability, with only a few vendors offering advanced capabilities out of the box (OOTB) such as blue/green, canary, and feature management deployment strategies.
- **Provide connected platforms that best-of-breed tools can augment.** ISDPs provide engineering leverage in the form of preintegrated toolchains that enable work to easily flow from one process to the next, where administration occurs at

the platform level versus the tool level, and reduced context switching improves the developer experience. Often, platform leaders want to override the default capability with third-party tools that they believe offer superior capability, which can lead to a bit of a balancing act. That's why some ISDP vendors are heavily investing in improvements to their end-to-end platform, while others are banking on investments in third-party connectivity and partnerships.

- **Help enterprise leaders shift from DevOps to DevSecOps.** Forrester fields calls every day from enterprise development leaders that are transforming their DevOps practice into a development, security, and operations (DevSecOps) practice to make security a first-class concern for the entire development organization. ISDP vendors are responding with capabilities that better secure software delivery pipelines, including integrated secrets management, signed pipelines, and signed artifacts. Reference customers noted that not all of these tools are direct replacements for best-of-breed (best-of-breed security tools often having a significant advantage over bundled ones), but end users like them as forms of additional assurance and coverage.

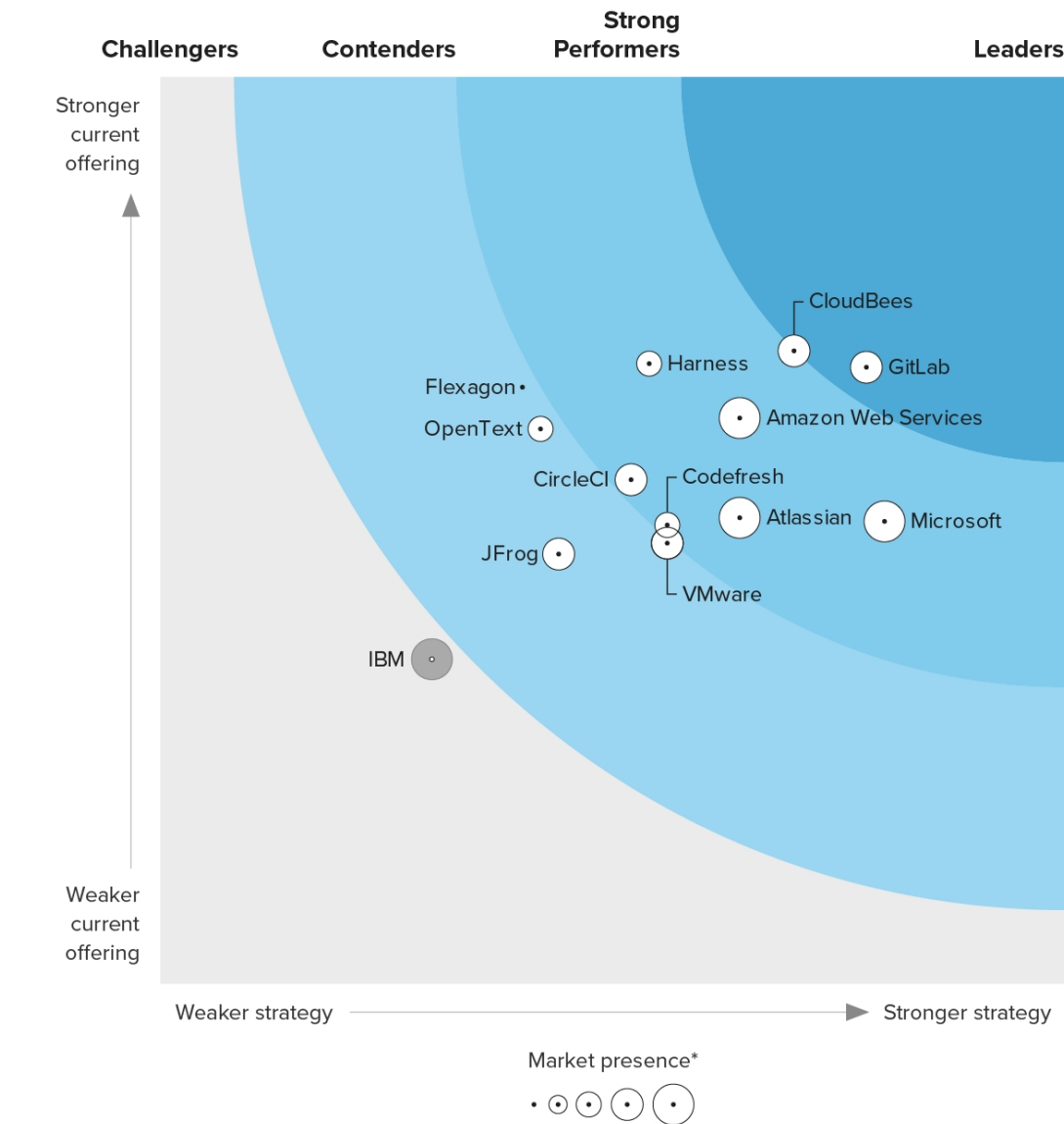
Evaluation Summary

The Forrester Wave™ evaluation highlights Leaders, Strong Performers, Contenders, and Challengers. It's an assessment of the top vendors in the market; it doesn't represent the entire vendor landscape. You'll find more information about this market in Forrester's [Now Tech: Integrated Software Delivery Platforms, Q2 2022](#).

We intend this evaluation to be a starting point only and encourage clients to view product evaluations and adapt criteria weightings using the Excel-based vendor comparison tool (see Figures 1 and 2). Click the link at the beginning of this report on [Forrester.com](#) to download the tool.

Figure 1
Forrester Wave™: Integrated Software Delivery Platforms, Q2 2023

THE FORRESTER WAVE™
Integrated Software Delivery Platforms
Q2 2023



*A gray bubble or open dot indicates a nonparticipating vendor.

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Figure 2**Forrester Wave™: Integrated Software Delivery Platforms Scorecard, Q2 2023**

	Forrester's weighting	Amazon Web Services	Atlassian	CircleCI	CloudBees	Codefresh	Flexagon	GitLab
Current offering	50%	2.57	3.12	2.78	3.49	2.53	3.29	3.40
Platform capabilities	20%	3.50	3.00	2.00	2.50	3.00	2.50	3.50
CI/CD capabilities	25%	2.00	2.60	3.00	4.00	4.00	3.20	3.40
Development, security, and operations	10%	1.00	3.00	2.00	2.00	2.00	3.00	4.00
Extensibility	10%	3.00	3.40	3.00	3.40	1.00	3.00	3.00
Governance and compliance	10%	3.00	5.00	3.00	5.00	1.00	3.00	3.00
Infrastructure management	10%	2.20	3.80	3.80	5.00	1.80	3.40	3.00
Product support	10%	3.00	3.00	3.00	3.00	3.00	5.00	3.00
Support for test automation	5%	3.00	1.00	3.00	3.00	1.00	5.00	5.00
Strategy	50%	3.20	3.20	2.60	3.50	2.80	2.00	3.90
Vision	20%	3.00	3.00	1.00	3.00	3.00	3.00	3.00
Roadmap	20%	3.00	3.00	3.00	3.00	3.00	3.00	5.00
Partner ecosystem	10%	5.00	5.00	1.00	5.00	1.00	1.00	3.00
Innovation	25%	3.00	3.00	3.00	3.00	3.00	1.00	3.00
Community	15%	3.00	3.00	3.00	5.00	3.00	1.00	5.00
Pricing flexibility and transparency	10%	3.00	3.00	5.00	3.00	3.00	3.00	5.00
Market presence	0%	5.00	4.50	3.50	3.50	2.50	1.00	3.50
ISDP revenue	50%	5.00	5.00	4.00	4.00	2.00	1.00	4.00
Number of enterprise customers	50%	5.00	4.00	3.00	3.00	3.00	1.00	3.00

All scores are based on a scale of 0 (weak) to 5 (strong).

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	Forrester's weighting	Harness	IBM*	JFrog	Microsoft	OpenText	VMware
Current offering	50%	3.42	1.79	2.37	2.55	3.06	2.43
Platform capabilities	20%	4.00	1.50	2.00	3.00	2.50	2.00
CI/CD capabilities	25%	4.20	2.80	2.40	2.40	3.40	2.40
Development, security, and operations	10%	3.00	1.00	4.00	4.00	1.00	3.00
Extensibility	10%	1.00	1.40	4.20	4.20	3.40	3.00
Governance and compliance	10%	3.00	3.00	1.00	1.00	5.00	3.00
Infrastructure management	10%	4.20	1.00	1.00	1.80	2.20	1.80
Product support	10%	3.00	1.00	3.00	1.00	3.00	3.00
Support for test automation	5%	3.00	1.00	1.00	3.00	5.00	1.00
Strategy	50%	2.70	1.50	2.20	4.00	2.10	2.80
Vision	20%	3.00	1.00	1.00	3.00	1.00	3.00
Roadmap	20%	3.00	1.00	1.00	3.00	3.00	3.00
Partner ecosystem	10%	3.00	3.00	3.00	5.00	3.00	3.00
Innovation	25%	3.00	1.00	3.00	5.00	3.00	3.00
Community	15%	1.00	3.00	3.00	5.00	1.00	3.00
Pricing flexibility and transparency	10%	3.00	1.00	3.00	3.00	1.00	1.00
Market presence	0%	3.00	4.50	4.00	5.00	2.50	4.00
ISDP revenue	50%	3.00	5.00	4.00	5.00	4.00	4.00
Number of enterprise customers	50%	3.00	4.00	4.00	5.00	1.00	4.00

All scores are based on a scale of 0 (weak) to 5 (strong).

*Indicates a nonparticipating vendor

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Vendor Offerings

Forrester evaluated the offerings listed below (see Figure 3).

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Figure 3
Evaluated Vendors And Product Information

Vendor	Product evaluated
Amazon Web Services	AWS Code Services
Atlassian	Atlassian Open DevOps
CircleCI	CircleCI
CloudBees	CloudBees Platform
Codefresh	Codefresh
Flexagon	Flexagon FlexDeploy
GitLab	GitLab
Harness	Harness Software Delivery Platform
IBM	IBM UrbanCode Deploy
JFrog	JFrog Software Supply Chain Platform
Microsoft	Microsoft Azure DevOps and GitHub Enterprise
OpenText (formerly Micro Focus)	Micro Focus ValueEdge
VMware	VMware Tanzu Application Platform

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Vendor Profiles

Our analysis uncovered the following strengths and weaknesses of individual vendors.

Leaders

- **GitLab leads with a balance between solid platform capabilities and overall strategy.** GitLab is known for being the vendor that made end-to-end OOTB toolchains hip again and has led the industry toward consolidated ISDPs. GitLab’s strategy includes an on-par vision to deliver an excellent developer experience without sacrificing security or compliance. Its innovation is also good, going beyond traditional developers to include AI/ML engineering. GitLab is an open core product that not only invests heavily in the open source software (OSS) community but also enables its customers to contribute to the product, earning it

high scores for community. Pricing is completely transparent and includes a free tier. Its roadmap gets leading scores and includes enhanced supply chain security, enhanced UI, granular security and compliance controls, and pipeline security — all things enterprises need.

GitLab's platform cohesion received high scores for simplicity of setup and operation as both an on-premises product and a software-as-a-service (SaaS) product. Though GitLab comes with its own GitLab version control system (VCS), it supports complete integration with all major Git providers, earning it high scores for version control. GitLab gets high scores for DevSecOps by providing OOTB security tools for static application security testing (SAST), dynamic application security testing (DAST), and Secret Detection. GitLab is one of the few ISDP vendors to incorporate a test automation dashboard, earning a top score. Dev experience is on par, as are administration and management, deployment automation, and release readiness. Build automation, governance and compliance, and infrastructure as code (IaC) were on par with others. One reference customer commented, "The CI/CD experience using secrets, environments, runners, and SAST/DAST/license scans/etc. is unparalleled in my opinion." GitLab is great for enterprises wishing to consolidate their best-of-breed toolchain into one high-performing ISDP.

Strong Performers

- **CloudBees delivers on its core promise of CI/CD, but platform cohesion needs work.** CloudBees is primarily associated with Jenkins; this well-known open source CI tool that it donated to the Continuous Delivery Foundation is one of the original best-of-breed tools that accompanied the beginning of the DevOps era more than 10 years ago. CloudBees leverages open source Jenkins and augments that with enterprise-scale deployment automation, release readiness, and value stream management (VSM) analytics to provide an enterprise-class ISDP. It got high scores for community, with significant contributions to open source projects and cultivation of its own user community. Its strategy features a solid vision and one of the larger partner ecosystems in this market. Innovation was on par and includes efforts to improve codification of policies. A planned SaaS offering should address its platform cohesion weakness and will help CloudBees serve clients that want a more integrated and maintenance-free ISDP.

CloudBees deployment automation is above par, covering blue/green, canary, and powerful feature management capabilities that are third-party add-ons for most

other platforms. CloudBees release readiness, governance, and compliance are also top-notch with complete integration with popular IT service management (ITSM) tools such as ServiceNow that bring those UIs directly into the ISDP experience. Build automation requires a bit of effort to manage fleets of build runners, and pipeline security needs to go beyond branch security to earn higher scores. When asked what they liked about working with CloudBees, reference customers had good things to say: “The vendor relationship has been a true partnership. It’s listening to our internal customers.” CloudBees is a great fit for enterprise platform teams that value open source and open choice.

- **Microsoft differentiates with innovation, but governance and compliance need attention.** Microsoft acquired GitHub, the industry’s largest open source registry, in 2018. Since that time, Microsoft’s Azure DevOps and GitHub have slowly been moving toward a unified offering. However, both the product vision and the roadmap could have been clearer on how this will shape future offerings, resulting in an on-par score. Innovation strategy is tops because GitHub Copilot has set new standards for the development tools industry and planned investments will enable it to continue this leadership position. Its partner ecosystem also scored high, with Microsoft having one of the most extensive. GitHub also supports numerous open source projects, and its online community is one of the biggest, earning a top score. The roadmap heavily weighted developer productivity but was missing key enterprise needs for governance.

Microsoft’s developer experience gets top scores for GitHub Actions, which turned GitHub into a CI leader almost overnight; Codespaces, a virtual dev environment that requires no setup by developers to get started; and Copilot, the AI-powered coding assistant. It also received top scores for OOTB security tools such as Dependabot. Support for APIs and third-party tools was also top scoring, with numerous ways to incorporate not only software but also actions. Pipeline security and platform cohesion were on par, while administration and management were missing the granular permissions of those scoring higher. Microsoft data and analytics were mostly operational, resulting in a below-par score. Deployment automation was also below par — it lacked blue/green and canary deployment scenarios. Reference customers were mostly positive about Microsoft, noting, “Generally, adoption of standard features is quick and doesn’t require a great deal of time to become productive.” Microsoft is great for organizations looking to boost developer effectiveness with cutting-edge tools.

- **Atlassian gets top scores for cohesiveness, but pipeline security capabilities are lacking.** Atlassian is best known for Jira, one of the most popular agile planning

tools on the market serving 80% of the Fortune 500. Atlassian's product strategy is on par and strives to serve two audiences, teams of developers and platform teams, by building out a data layer to connect the tools together. Its partner ecosystem is top-notch. Innovation is on par and includes platform capabilities and a data lake component to capture and aggregate data for greater analytical insight. The roadmap reflects enterprise concerns with plans to aggregate third-party security data, automatically infuse issue tickets with context, provide greater VSM insights, and expand support plans for DevOps and platform teams.

Atlassian's Confluence, which enables teams to create custom developer onboarding content and a hub for team collaboration, enhances the developer experience. This plus integrated development environment plug-ins give Atlassian a solid on-par score for developer experience. The OOTB build and CI were on par, with reusable build templates and parallel build execution to reduce build times. Deployment automation was below par, as were capabilities to secure pipelines. OOTB security tools are few, earning a below-par score, because Atlassian prefers to leverage its partner ecosystem for those. A reference customer said, "We chose Atlassian for the product features and the maturity of the well-integrated suites of products and plug-ins, which help simplify using the well-designed solution." Atlassian is great fit for software teams and enterprise platform teams that are looking for a standard ISDP approach but want to leverage best-of-breed CD and security components.

- **Harness ISDP delivers on platform cohesion, but community contributions are nascent.** Harness has a SaaS ISDP with core CI/CD modules and cloud cost management capabilities. Its vision is to simplify software development with an end-to-end software delivery platform that offers a modern developer experience and automated governance and guardrails. The strategy has mostly on-par elements such as a growing partner ecosystem and investments in innovation. The only weak area was the smaller-than-average community program — those scoring higher had greater support for internal and external communities. The roadmap includes support for database development, test optimization, end-to-end visibility, and infusion of AI/ML throughout the platform.

Harness gets top scores for platform cohesion, delivering a great OOTB experience for first-time users. It also earned top scores in build automation that includes pipelines as code, reusable pipelines, resource management, support for test automation, and intelligent test automation that can drastically reduce build times. It supports multiple deployment strategies OOTB, including blue/green and

canary. In addition, Harness can pinpoint code that's caused pipeline errors, helping speed remediation. Analytics was also top scoring and offered dev, process, and operational metrics. Customization of the UI and some ISDP components is possible and easy, but the limited scope of those changes resulted in a below-par score. Other areas for improvement include expanding the number of third-party integrations and its footprint in the user and open source community. Customers appreciate Harness' "willingness to engage directly with our development teams." Harness is great for development teams that want to spend less time managing tools and more time developing code.

- **AWS is strong in administration and management, but dev metrics are a miss.**

Amazon Web Services (AWS) pioneered infrastructure as a service (IaaS) and offers nearly every possible form of it. Its ISDP product includes CodeBuild, CodeStar, CloudFormation, and several other SaaS offerings that are integrated on demand when users create their account. The overall strategy score was solid. Innovation scored on par for current delivery and upcoming features for helping developers code faster. The vision was on par, focusing mainly on CodeCatalyst that will enable AWS to make its developer platform more accessible to a broader set of end users with different levels of technical ability. AWS actively supports its user community and open source projects, getting an on-par score for community. The roadmap is also on par and includes the aforementioned CodeCatalyst, a product currently in beta that offers a preassembled and more opinionated ISDP to help customers accelerate project setup and collaborate more fluidly.

AWS shines with excellent administration and management capabilities that enable user permissions to be managed down to the API level. AWS supports all major authentication protocols including emerging standards such as SCIM. AWS gets top scores for governance and compliance that include OOTB compliance templates. Platform cohesion and developer experience are on par. Build automation would have scored higher had it offered test execution optimization. AWS got low scores for analytics and reports; other vendors offered better dashboards with more dev-related metrics. Release readiness also scored low. Reference customer sentiment was mostly positive, saying, "Security is a first-class citizen." AWS is great for companies that want speed, scale, and security hosted in the same environment as their public cloud infrastructure.

- **CircleCI provides outstanding build automation, but its vision does not go beyond CI/CD.** CircleCI was one of the first vendors to offer CI as SaaS. Over time, it added CD and extensibility features to make a more well-rounded ISDP offering. CircleCI's on-par vision to speed time to value while simplifying the ISDP

experience is well articulated and strikes a pragmatic balance between individual developer concerns, the needs of platform teams, and support for enterprises in regulated industries. CircleCI's strategic vision is to remain a best-of-breed CI/CD tool vendor and not extend beyond that into other ISDP areas, resulting in a below-par score. The on-par roadmap aims at enterprise teams and includes features to help platform teams better manage an ISDP at scale with pervasive validation at each step of software delivery.

CircleCI has invested in optimized build processes, giving it a top score for build automation and CI. Its support for a wide range of platforms, including all public clouds and even mainframe, earns above-par scores. IaC capabilities were also top scoring. Product support leverages dedicated customer success engineers for each account, but average reference customer scores kept it on par. CircleCI user identities are derived from third-party VCS identities (e.g., those managed by GitHub or Bitbucket) and thus received a low score for administration and management. However, a new identity management system is on the roadmap to address this shortcoming. CircleCI has a good set of OOTB metrics and analytics, including DORA metrics, but the dashboards and capabilities were not as extensive as those scoring higher. One reference customer listed the following capabilities as most appreciated: "parallelization of workflows and rerunning only failed parts of workflows." CircleCI is a great fit for dev teams that value team autonomy and superfast build automation.

- **Codefresh has differentiating CI/CD but needs to improve governance and compliance.** Codefresh provides high-scoring CI/CD capabilities that emphasize the GitOps methodology, which requires all changes to infrastructure to be stored and managed as code in a Git repository prior to deployment. This methodology gained popularity from the Argo CD open source project, which powers Codefresh. Codefresh is also a registered project maintainer and contributor, which factored into its on-par community score. The Codefresh vision and roadmap focus on improvements to the developer experience and deployment capabilities such as Kubernetes context injection, improved disk utilization, and better support for teams. Innovation is confined to Argo and GitOps improvements, earning on-par scores. The one below-par score was for its smaller-than-average partner ecosystem.

Codefresh earned above-par scores for build automation, CI, and deployment automation featuring unique pipeline debugging capabilities that enable users to step in and out of containerized build steps, which no other vendor demonstrated.

Product support was strong, earning on-par scores. IaC provides strong support for Kubernetes configurations, but support for other legacy platforms lacks, keeping this score at par. Developer experience is on par; vendors scoring higher had integrated knowledge management. Other on-par scores include pipeline security, administration and management, and analytics that incorporate OOTB DORA metrics. Codefresh earned below-par scores for governance and compliance (providing limited support for ITSM integration) and support for functional test automation. When asked what was best about Codefresh, one customer said, “easy, standardized deployments for a hybrid universe.” Codefresh is a great fit for organizations that have adopted GitOps and place high value on developer autonomy.

Contenders

- **Flexagon has excellent customer support, but developer experience was below par.** Flexagon builds an end-to-end ISDP that supports traditional forms of software and infrastructure (e.g., Java and Linux) as well as business platforms such as Salesforce, Oracle, and SAP. The product vision is clear and focuses on delivering enterprise value through security and lowering risk. Planned updates include managed Oracle Cloud services, SAP blueprints, and a compliance and policy engine. Innovation is below par, investment in leading-edge areas is lacking, and the partner ecosystem was small, compared with others in this evaluation. Flexagon pricing features multiple tiers and full transparency of pricing options. Flexagon got above-par scores for test automation because it was one of the few vendors that featured native functional test automation capabilities and support.

Flexagon offers good platform cohesion with its offering that includes everything from agile planning to release automation in one package. It would have scored higher if not for some complexity in setting up and operating fleets of build runners. Flexagon got high scores for version control, with its support for the largest number of VCSes including Git and non-Git systems, and environment management that enabled teams to easily operate and manage any number of environments to support engineering teams. Deployment automation, release readiness, and platform support were also solid. Dev experience was not a focus area and resulted in a low score for that capability. Reference customers gave Flexagon perfect high scores for customer support, with one stating, “Flexagon continues to knock our socks off.” Flexagon is a great fit for enterprises looking for DevOps capabilities that span both open systems and business systems.

- **VMware delivers strong pipeline security but is light on data and analytics.**

VMware is best known for its infrastructure virtualization software. Tanzu Application Platform is VMware's ISDP and has a vision that is broad and incorporates elements of sociological factors to help developers be happier and more productive. VMware scored on par for community with strong support for open source tools and its user community. Innovation includes investments in autoremediation and AI/ML, earning an on-par score. The on-par roadmap will help users simplify the product lifecycle, improve developer collaboration, provide cost governance, and have more secure supply chains.

VMware pipeline security is one of the best in the evaluation with signed pipelines, signed commits, and additional security capabilities throughout the pipeline. Governance and compliance were on par, with OOTB compliance policies. Platform cohesion was also good with minimal setup required. Developer experience included a built-in knowledge management system to help onboard new developers and scored on par. Deployment automation included blue/green and canary deployment strategies, earning an on-par score. Administration and management were below par because they were missing features of those scoring higher. Data and analytics were few, and platform support seemed limited to Kubernetes workloads. One reference customer remarked, "The vendor is always willing to accept and act on feedback." VMware is a great fit for enterprises operating on Kubernetes and places high value on pipeline security.

- **OpenText has top-rated platform support, but DevSecOps capabilities score**

low. OpenText is best known for its portfolio of software development tools ranging from portfolio to test automation. OpenText's ValueEdge is an open platform that bundles these capabilities together in one ISDP. Its roadmap is on par, featuring elements that will better connect business and development. Innovation features AI/ML infused in risk management and testing, and OpenText boasts a healthy partner ecosystem. Both areas scored on par, with good evidence of progress from its AI/ML investments. But its vision focused more on business concerns than stronger CI/CD capabilities. In addition, pricing is opaque, compared with those scoring higher.

OpenText has one of the strongest governance and compliance offerings of this evaluation because nearly everything connects to change management systems. It has superior support for test automation capabilities that include autoclassification of test failures. In addition to providing its own VCS, users can connect with virtually any other VCS, earning above-par scores for that capability. Support for a

multitude of platforms including mainframe was a plus. Administration, management, and developer experience were on par, as were data and analytics that featured DORA and flow metrics. Deployment automation was also on par with blue/green and canary deployment support. Pipeline security lagged others evaluated. IaC, an area of growing importance to developers, also scored low. OpenText reference customers were positive about their experience, with one stating, “We don’t have any challenges at this stage.” OpenText is a great fit for enterprises with varied platforms in need of strong ties to test automation and compliance.

- **JFrog provides differentiated pipeline security capabilities, but the core ISDP falls short.** JFrog is best known for its flagship product, Artifactory, a binary artifact repository with a name that is so well known that it is synonymous with artifact repositories in general. JFrog’s strategy includes strong on-par scores for innovation based on investments in helping secure enterprise software supply chains. JFrog has a strong user community and actively participates in open source projects. Pricing flexibility was also on par, offering several payment tiers. JFrog scored below par for vision because it seemed overly focused on security at the expense of other ISDP capabilities. The roadmap was also below par due to its limited scope.

JFrog has superior version control due to the strength of Artifactory. Pipeline security is also top-notch with signed artifacts and signed pipelines that provide some of the strongest supply chain security measures available. The JFrog platform has good cohesion and requires few steps to set up and get started. Administration and management are on par and offer LDAP, OAuth, and support for SCIM to manage users at scale. Xray, a security product that scans artifacts, comes bundled and offers container and image scanning along with OSS supply chain management. To make this a more competitive ISDP offering, JFrog needs to invest in release readiness, data, and analytics and expand policy as code to all aspects of the ISDP while doing more to support test automation. Customers like how they are treated by JFrog, saying: It’s a “partner, not a vendor — they bring this mentality to everything.” JFrog is great for enterprises that place high value on supply chain security.

Challengers

- **IBM UrbanCode Deploy offers solid build and deployment capabilities but looks dated.** IBM UrbanCode offers UrbanCode Build and UrbanCode Deploy as two components that work together to deliver an enterprise-scale ISDP. IBM has a

solid development community because IBM Garage has thousands of contributors. IBM's partner ecosystem is also solidly on par with a number of top systems integrators listed as partners. The strategy has several weak elements including a roadmap that emphasizes maintenance versus new features, and innovation proof points are lacking.

IBM has solid build automation, deployment automation, release readiness, and support for a large variety of target platforms, which are core ISDP capabilities valued by enterprise dev teams. IBM's UrbanCode Velocity module offers strong analytics and graphics, earning it on-par scores for analytics, and there's a healthy number of third-party integrations for an on-par score in that category. But platform cohesion suffers from a complicated installation process, administration and management offer limited capabilities such as LDAP, and the developer experience is bare bones. Pipelines are present but don't offer the same level of capability as more modern tools. IBM is a good fit for enterprises that need to build an ISDP bridge between legacy and modern infrastructure. IBM declined to participate in the full Forrester Wave evaluation process.

Evaluation Overview

We grouped our evaluation criteria into three high-level categories:

- **Current offering.** Each vendor's position on the vertical axis of the Forrester Wave graphic indicates the strength of its current offering. Key criteria for these solutions include platform capabilities to facilitate management at scale, CI/CD to perform core ISDP functions, extensibility to enable platform operators to incorporate third-party tools as needed, and integrated systems to secure the software delivery pipeline.
- **Strategy.** Placement on the horizontal axis indicates the strength of the vendors' strategies. We evaluated vision, roadmap, and innovation to determine which vendor was placing the best bets to fulfill long-term product promises.
- **Market presence.** Represented by the size of the markers on the graphic, our market presence scores reflect each vendor's ISDP revenue and number of enterprise customers.

Vendor Inclusion Criteria

Each of the vendors we included in this assessment has:

- **Integrated CI/CD (e.g., deployment and release automation) capabilities.** The vendor provides integrated CI/CD capabilities OOTB that enable teams to rapidly establish build and deploy processes with minimal configuration.
- **Extensibility via APIs or plug-ins.** The vendor provides APIs to support third-party tool integration to extend the platform to incorporate performance, test automation, and security.
- **A system for centralized administration.** A single dashboard helps platform administrators operate and maintain an ISDP at enterprise scale.
- **Strong client interest.** Analyst inquiry logs indicate the vendor is top of mind among Forrester's end-user clientele with regard to ISDPs.
- **General-purpose programming languages and platforms.** The vendor supports build and deployment automation for open source programming languages like C# and Java as well as deployment to general-purpose runtime platforms such as Linux or Windows. It is also not exclusively tied to any particular business platform such as Salesforce or SAP.
- **High market performance.** The vendor's market performance demonstrates that it is a significant player in the ISDP market.

Supplemental Material

Online Resource

We publish all our Forrester Wave scores and weightings in an Excel file that provides detailed product evaluations and customizable rankings; download this tool by clicking the link at the beginning of this report on Forrester.com. We intend these scores and default weightings to serve only as a starting point and encourage readers to adapt the weightings to fit their individual needs.

The Forrester Wave Methodology

A Forrester Wave is a guide for buyers considering their purchasing options in a technology marketplace. To offer an equitable process for all participants, Forrester follows [The Forrester Wave™ Methodology](#) to evaluate participating vendors.

In our review, we conduct primary research to develop a list of vendors to consider for the evaluation. From that initial pool of vendors, we narrow our final list based on the inclusion criteria. We then gather details of product and strategy through a detailed questionnaire, demos/briefings, and customer reference surveys/interviews. We use those inputs, along with the analyst's experience and expertise in the marketplace, to

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score vendors, using a relative rating system that compares each vendor against the others in the evaluation.

We include the Forrester Wave publishing date (quarter and year) clearly in the title of each Forrester Wave report. We evaluated the vendors participating in this Forrester Wave using materials they provided to us by March 2023 and did not allow additional information after that point. We encourage readers to evaluate how the market and vendor offerings change over time.

In accordance with [our vendor review policy](#), Forrester asks vendors to review our findings prior to publishing to check for accuracy. Vendors marked as nonparticipating vendors in the Forrester Wave graphic met our defined inclusion criteria but declined to participate in or contributed only partially to the evaluation. We score these vendors in accordance with [our vendor participation policy](#) and publish their positioning along with those of the participating vendors.

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WAVE REPORT

The Forrester WaveTM: Integrated Software Delivery Platforms, Q2 2023

The 13 Providers That Matter Most And How They
Stack Up

May 31, 2023

By Christopher Condo with Chris Gardner, Sara Sjoblom, Kara Hartig

FORRESTER®

Summary

In our 26-criterion evaluation of integrated software delivery platform (ISDP) providers, we identified the most significant ones and researched, analyzed, and scored them. This report shows how each provider measures up and helps modern application development leaders and DevOps platform team leaders select the right one for their needs.

Additional resources are available in the [online version](#) of this report.

ISDPs Provide A Strong Platform Foundation But Still Need Best-Of-Breed Tools

In our reports, [The Future Of DevOps](#) and [The Forrester Guide To DevOps](#), Forrester interviewed dozens of enterprise software development leaders from diverse companies ranging from Netflix to TransUnion to understand the current state of DevOps, its relevance, and next steps enterprise leaders were taking to further advance their practices. In addition, we asked which tools they were using, how effective the tools were, which were using ISDPs, and how well those ISDPs were performing. We learned that many enterprises have shifted to platform teams to support their DevOps efforts and many platform leaders are evaluating and using ISDPs, but few are using all of their ISDP vendor's capabilities. They explained that while these ISDP tools hold great potential for a complete end-to-end toolchain, many individual capabilities fell short of enterprise needs, forcing these leaders to source those capabilities from separate best-of-breed vendors.

Our reference customer survey backs this up by indicating that enterprises are using (on average) only 53% of the capabilities they are paying for, with the remaining capabilities either being delivered by a separate vendor or simply not used. This means that ISDP vendors are not only competing with each other but also competing with the best-of-breeds, which is why the bar has risen so high for this evaluation. ISDP vendors recognize these challenges and are responding with a number of innovative solutions that will make them more attractive than a traditional DIY toolchain but open and extensible enough that end users can add the tools they want without sacrificing capability. The challenge is striking the right balance to create overall value.

As a result of these trends, ISDP customers should look for providers that:

- **Offer a strong core of CI/CD capabilities.** Continuous integration (CI) and continuous delivery (CD) are the core capabilities that the rest of the software delivery toolchain is built around. Several vendors offered build optimization algorithms that slash build and unit test times by intelligently building and testing only the code that changed. Deployment automation is also a differentiating capability, with only a few vendors offering advanced capabilities out of the box (OOTB) such as blue/green, canary, and feature management deployment strategies.
- **Provide connected platforms that best-of-breed tools can augment.** ISDPs provide engineering leverage in the form of preintegrated toolchains that enable work to easily flow from one process to the next, where administration occurs at

the platform level versus the tool level, and reduced context switching improves the developer experience. Often, platform leaders want to override the default capability with third-party tools that they believe offer superior capability, which can lead to a bit of a balancing act. That's why some ISDP vendors are heavily investing in improvements to their end-to-end platform, while others are banking on investments in third-party connectivity and partnerships.

- **Help enterprise leaders shift from DevOps to DevSecOps.** Forrester fields calls every day from enterprise development leaders that are transforming their DevOps practice into a development, security, and operations (DevSecOps) practice to make security a first-class concern for the entire development organization. ISDP vendors are responding with capabilities that better secure software delivery pipelines, including integrated secrets management, signed pipelines, and signed artifacts. Reference customers noted that not all of these tools are direct replacements for best-of-breed (best-of-breed security tools often having a significant advantage over bundled ones), but end users like them as forms of additional assurance and coverage.

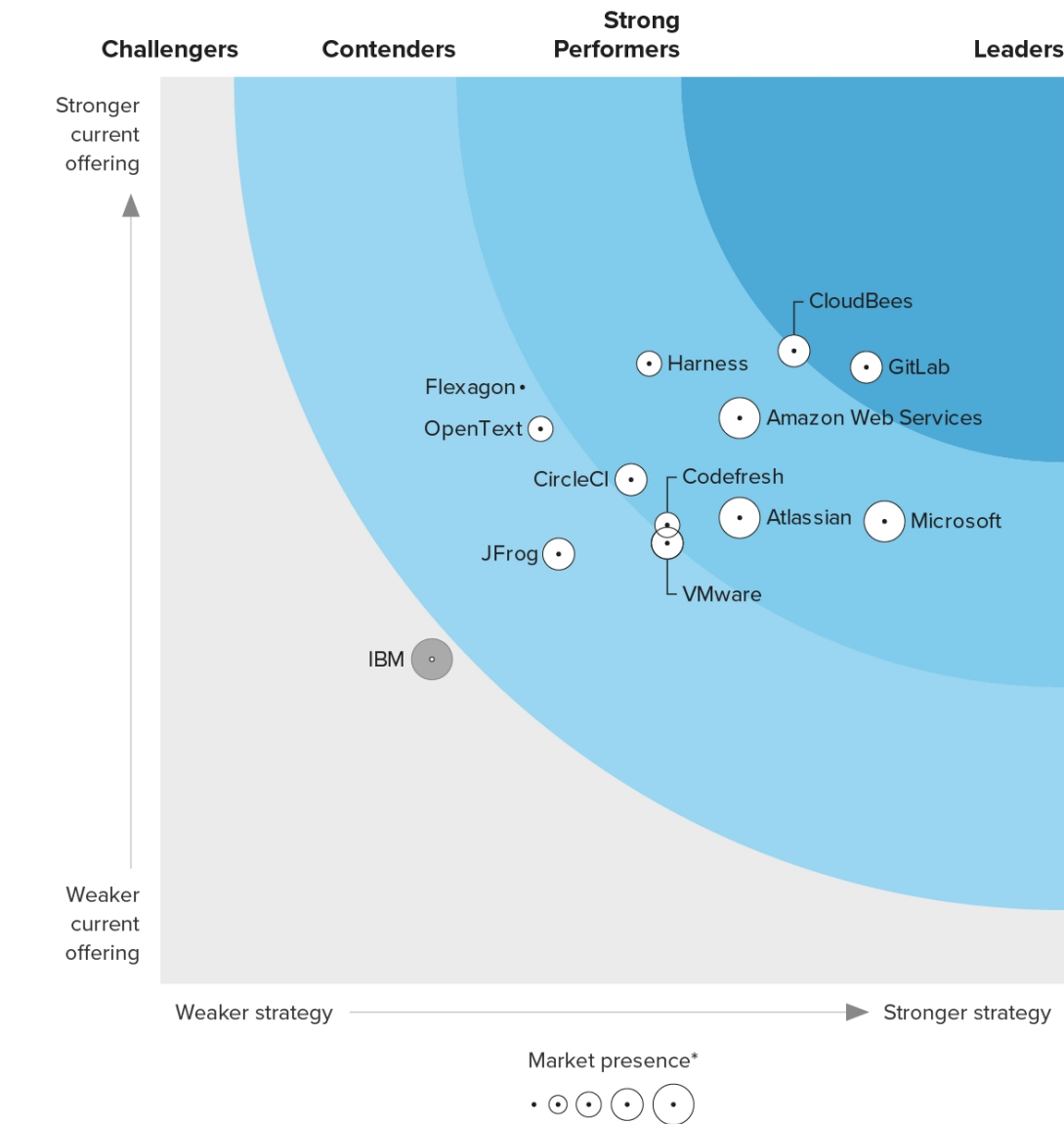
Evaluation Summary

The Forrester Wave™ evaluation highlights Leaders, Strong Performers, Contenders, and Challengers. It's an assessment of the top vendors in the market; it doesn't represent the entire vendor landscape. You'll find more information about this market in Forrester's [Now Tech: Integrated Software Delivery Platforms, Q2 2022](#).

We intend this evaluation to be a starting point only and encourage clients to view product evaluations and adapt criteria weightings using the Excel-based vendor comparison tool (see Figures 1 and 2). Click the link at the beginning of this report on [Forrester.com](#) to download the tool.

Figure 1
Forrester Wave™: Integrated Software Delivery Platforms, Q2 2023

THE FORRESTER WAVE™
Integrated Software Delivery Platforms
Q2 2023



*A gray bubble or open dot indicates a nonparticipating vendor.

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Figure 2**Forrester Wave™: Integrated Software Delivery Platforms Scorecard, Q2 2023**

	Forrester's weighting	Amazon Web Services	Atlassian	CircleCI	CloudBees	Codefresh	Flexagon	GitLab
Current offering	50%	2.57	3.12	2.78	3.49	2.53	3.29	3.40
Platform capabilities	20%	3.50	3.00	2.00	2.50	3.00	2.50	3.50
CI/CD capabilities	25%	2.00	2.60	3.00	4.00	4.00	3.20	3.40
Development, security, and operations	10%	1.00	3.00	2.00	2.00	2.00	3.00	4.00
Extensibility	10%	3.00	3.40	3.00	3.40	1.00	3.00	3.00
Governance and compliance	10%	3.00	5.00	3.00	5.00	1.00	3.00	3.00
Infrastructure management	10%	2.20	3.80	3.80	5.00	1.80	3.40	3.00
Product support	10%	3.00	3.00	3.00	3.00	3.00	5.00	3.00
Support for test automation	5%	3.00	1.00	3.00	3.00	1.00	5.00	5.00
Strategy	50%	3.20	3.20	2.60	3.50	2.80	2.00	3.90
Vision	20%	3.00	3.00	1.00	3.00	3.00	3.00	3.00
Roadmap	20%	3.00	3.00	3.00	3.00	3.00	3.00	5.00
Partner ecosystem	10%	5.00	5.00	1.00	5.00	1.00	1.00	3.00
Innovation	25%	3.00	3.00	3.00	3.00	3.00	1.00	3.00
Community	15%	3.00	3.00	3.00	5.00	3.00	1.00	5.00
Pricing flexibility and transparency	10%	3.00	3.00	5.00	3.00	3.00	3.00	5.00
Market presence	0%	5.00	4.50	3.50	3.50	2.50	1.00	3.50
ISDP revenue	50%	5.00	5.00	4.00	4.00	2.00	1.00	4.00
Number of enterprise customers	50%	5.00	4.00	3.00	3.00	3.00	1.00	3.00

All scores are based on a scale of 0 (weak) to 5 (strong).

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	Forrester's weighting	Harness	IBM*	JSFrog	Microsoft	OpenText	VMware
Current offering	50%	3.42	1.79	2.37	2.55	3.06	2.43
Platform capabilities	20%	4.00	1.50	2.00	3.00	2.50	2.00
CI/CD capabilities	25%	4.20	2.80	2.40	2.40	3.40	2.40
Development, security, and operations	10%	3.00	1.00	4.00	4.00	1.00	3.00
Extensibility	10%	1.00	1.40	4.20	4.20	3.40	3.00
Governance and compliance	10%	3.00	3.00	1.00	1.00	5.00	3.00
Infrastructure management	10%	4.20	1.00	1.00	1.80	2.20	1.80
Product support	10%	3.00	1.00	3.00	1.00	3.00	3.00
Support for test automation	5%	3.00	1.00	1.00	3.00	5.00	1.00
Strategy	50%	2.70	1.50	2.20	4.00	2.10	2.80
Vision	20%	3.00	1.00	1.00	3.00	1.00	3.00
Roadmap	20%	3.00	1.00	1.00	3.00	3.00	3.00
Partner ecosystem	10%	3.00	3.00	3.00	5.00	3.00	3.00
Innovation	25%	3.00	1.00	3.00	5.00	3.00	3.00
Community	15%	1.00	3.00	3.00	5.00	1.00	3.00
Pricing flexibility and transparency	10%	3.00	1.00	3.00	3.00	1.00	1.00
Market presence	0%	3.00	4.50	4.00	5.00	2.50	4.00
ISDP revenue	50%	3.00	5.00	4.00	5.00	4.00	4.00
Number of enterprise customers	50%	3.00	4.00	4.00	5.00	1.00	4.00

All scores are based on a scale of 0 (weak) to 5 (strong).

*Indicates a nonparticipating vendor

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Vendor Offerings

Forrester evaluated the offerings listed below (see Figure 3).

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Figure 3
Evaluated Vendors And Product Information

Vendor	Product evaluated
Amazon Web Services	AWS Code Services
Atlassian	Atlassian Open DevOps
CircleCI	CircleCI
CloudBees	CloudBees Platform
Codefresh	Codefresh
Flexagon	Flexagon FlexDeploy
GitLab	GitLab
Harness	Harness Software Delivery Platform
IBM	IBM UrbanCode Deploy
JFrog	JFrog Software Supply Chain Platform
Microsoft	Microsoft Azure DevOps and GitHub Enterprise
OpenText (formerly Micro Focus)	Micro Focus ValueEdge
VMware	VMware Tanzu Application Platform

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Vendor Profiles

Our analysis uncovered the following strengths and weaknesses of individual vendors.

Leaders

- **GitLab leads with a balance between solid platform capabilities and overall strategy.** GitLab is known for being the vendor that made end-to-end OOTB toolchains hip again and has led the industry toward consolidated ISDPs. GitLab’s strategy includes an on-par vision to deliver an excellent developer experience without sacrificing security or compliance. Its innovation is also good, going beyond traditional developers to include AI/ML engineering. GitLab is an open core product that not only invests heavily in the open source software (OSS) community but also enables its customers to contribute to the product, earning it

high scores for community. Pricing is completely transparent and includes a free tier. Its roadmap gets leading scores and includes enhanced supply chain security, enhanced UI, granular security and compliance controls, and pipeline security — all things enterprises need.

GitLab's platform cohesion received high scores for simplicity of setup and operation as both an on-premises product and a software-as-a-service (SaaS) product. Though GitLab comes with its own GitLab version control system (VCS), it supports complete integration with all major Git providers, earning it high scores for version control. GitLab gets high scores for DevSecOps by providing OOTB security tools for static application security testing (SAST), dynamic application security testing (DAST), and Secret Detection. GitLab is one of the few ISDP vendors to incorporate a test automation dashboard, earning a top score. Dev experience is on par, as are administration and management, deployment automation, and release readiness. Build automation, governance and compliance, and infrastructure as code (IaC) were on par with others. One reference customer commented, "The CI/CD experience using secrets, environments, runners, and SAST/DAST/license scans/etc. is unparalleled in my opinion." GitLab is great for enterprises wishing to consolidate their best-of-breed toolchain into one high-performing ISDP.

Strong Performers

- **CloudBees delivers on its core promise of CI/CD, but platform cohesion needs work.** CloudBees is primarily associated with Jenkins; this well-known open source CI tool that it donated to the Continuous Delivery Foundation is one of the original best-of-breed tools that accompanied the beginning of the DevOps era more than 10 years ago. CloudBees leverages open source Jenkins and augments that with enterprise-scale deployment automation, release readiness, and value stream management (VSM) analytics to provide an enterprise-class ISDP. It got high scores for community, with significant contributions to open source projects and cultivation of its own user community. Its strategy features a solid vision and one of the larger partner ecosystems in this market. Innovation was on par and includes efforts to improve codification of policies. A planned SaaS offering should address its platform cohesion weakness and will help CloudBees serve clients that want a more integrated and maintenance-free ISDP.

CloudBees deployment automation is above par, covering blue/green, canary, and powerful feature management capabilities that are third-party add-ons for most

other platforms. CloudBees release readiness, governance, and compliance are also top-notch with complete integration with popular IT service management (ITSM) tools such as ServiceNow that bring those UIs directly into the ISDP experience. Build automation requires a bit of effort to manage fleets of build runners, and pipeline security needs to go beyond branch security to earn higher scores. When asked what they liked about working with CloudBees, reference customers had good things to say: “The vendor relationship has been a true partnership. It’s listening to our internal customers.” CloudBees is a great fit for enterprise platform teams that value open source and open choice.

- **Microsoft differentiates with innovation, but governance and compliance need attention.** Microsoft acquired GitHub, the industry’s largest open source registry, in 2018. Since that time, Microsoft’s Azure DevOps and GitHub have slowly been moving toward a unified offering. However, both the product vision and the roadmap could have been clearer on how this will shape future offerings, resulting in an on-par score. Innovation strategy is tops because GitHub Copilot has set new standards for the development tools industry and planned investments will enable it to continue this leadership position. Its partner ecosystem also scored high, with Microsoft having one of the most extensive. GitHub also supports numerous open source projects, and its online community is one of the biggest, earning a top score. The roadmap heavily weighted developer productivity but was missing key enterprise needs for governance.

Microsoft’s developer experience gets top scores for GitHub Actions, which turned GitHub into a CI leader almost overnight; Codespaces, a virtual dev environment that requires no setup by developers to get started; and Copilot, the AI-powered coding assistant. It also received top scores for OOTB security tools such as Dependabot. Support for APIs and third-party tools was also top scoring, with numerous ways to incorporate not only software but also actions. Pipeline security and platform cohesion were on par, while administration and management were missing the granular permissions of those scoring higher. Microsoft data and analytics were mostly operational, resulting in a below-par score. Deployment automation was also below par — it lacked blue/green and canary deployment scenarios. Reference customers were mostly positive about Microsoft, noting, “Generally, adoption of standard features is quick and doesn’t require a great deal of time to become productive.” Microsoft is great for organizations looking to boost developer effectiveness with cutting-edge tools.

- **Atlassian gets top scores for cohesiveness, but pipeline security capabilities are lacking.** Atlassian is best known for Jira, one of the most popular agile planning

tools on the market serving 80% of the Fortune 500. Atlassian's product strategy is on par and strives to serve two audiences, teams of developers and platform teams, by building out a data layer to connect the tools together. Its partner ecosystem is top-notch. Innovation is on par and includes platform capabilities and a data lake component to capture and aggregate data for greater analytical insight. The roadmap reflects enterprise concerns with plans to aggregate third-party security data, automatically infuse issue tickets with context, provide greater VSM insights, and expand support plans for DevOps and platform teams.

Atlassian's Confluence, which enables teams to create custom developer onboarding content and a hub for team collaboration, enhances the developer experience. This plus integrated development environment plug-ins give Atlassian a solid on-par score for developer experience. The OOTB build and CI were on par, with reusable build templates and parallel build execution to reduce build times. Deployment automation was below par, as were capabilities to secure pipelines. OOTB security tools are few, earning a below-par score, because Atlassian prefers to leverage its partner ecosystem for those. A reference customer said, "We chose Atlassian for the product features and the maturity of the well-integrated suites of products and plug-ins, which help simplify using the well-designed solution." Atlassian is great fit for software teams and enterprise platform teams that are looking for a standard ISDP approach but want to leverage best-of-breed CD and security components.

- **Harness ISDP delivers on platform cohesion, but community contributions are nascent.** Harness has a SaaS ISDP with core CI/CD modules and cloud cost management capabilities. Its vision is to simplify software development with an end-to-end software delivery platform that offers a modern developer experience and automated governance and guardrails. The strategy has mostly on-par elements such as a growing partner ecosystem and investments in innovation. The only weak area was the smaller-than-average community program — those scoring higher had greater support for internal and external communities. The roadmap includes support for database development, test optimization, end-to-end visibility, and infusion of AI/ML throughout the platform.

Harness gets top scores for platform cohesion, delivering a great OOTB experience for first-time users. It also earned top scores in build automation that includes pipelines as code, reusable pipelines, resource management, support for test automation, and intelligent test automation that can drastically reduce build times. It supports multiple deployment strategies OOTB, including blue/green and

canary. In addition, Harness can pinpoint code that's caused pipeline errors, helping speed remediation. Analytics was also top scoring and offered dev, process, and operational metrics. Customization of the UI and some ISDP components is possible and easy, but the limited scope of those changes resulted in a below-par score. Other areas for improvement include expanding the number of third-party integrations and its footprint in the user and open source community. Customers appreciate Harness' "willingness to engage directly with our development teams." Harness is great for development teams that want to spend less time managing tools and more time developing code.

- **AWS is strong in administration and management, but dev metrics are a miss.**

Amazon Web Services (AWS) pioneered infrastructure as a service (IaaS) and offers nearly every possible form of it. Its ISDP product includes CodeBuild, CodeStar, CloudFormation, and several other SaaS offerings that are integrated on demand when users create their account. The overall strategy score was solid. Innovation scored on par for current delivery and upcoming features for helping developers code faster. The vision was on par, focusing mainly on CodeCatalyst that will enable AWS to make its developer platform more accessible to a broader set of end users with different levels of technical ability. AWS actively supports its user community and open source projects, getting an on-par score for community. The roadmap is also on par and includes the aforementioned CodeCatalyst, a product currently in beta that offers a preassembled and more opinionated ISDP to help customers accelerate project setup and collaborate more fluidly.

AWS shines with excellent administration and management capabilities that enable user permissions to be managed down to the API level. AWS supports all major authentication protocols including emerging standards such as SCIM. AWS gets top scores for governance and compliance that include OOTB compliance templates. Platform cohesion and developer experience are on par. Build automation would have scored higher had it offered test execution optimization. AWS got low scores for analytics and reports; other vendors offered better dashboards with more dev-related metrics. Release readiness also scored low. Reference customer sentiment was mostly positive, saying, "Security is a first-class citizen." AWS is great for companies that want speed, scale, and security hosted in the same environment as their public cloud infrastructure.

- **CircleCI provides outstanding build automation, but its vision does not go beyond CI/CD.** CircleCI was one of the first vendors to offer CI as SaaS. Over time, it added CD and extensibility features to make a more well-rounded ISDP offering. CircleCI's on-par vision to speed time to value while simplifying the ISDP

experience is well articulated and strikes a pragmatic balance between individual developer concerns, the needs of platform teams, and support for enterprises in regulated industries. CircleCI's strategic vision is to remain a best-of-breed CI/CD tool vendor and not extend beyond that into other ISDP areas, resulting in a below-par score. The on-par roadmap aims at enterprise teams and includes features to help platform teams better manage an ISDP at scale with pervasive validation at each step of software delivery.

CircleCI has invested in optimized build processes, giving it a top score for build automation and CI. Its support for a wide range of platforms, including all public clouds and even mainframe, earns above-par scores. IaC capabilities were also top scoring. Product support leverages dedicated customer success engineers for each account, but average reference customer scores kept it on par. CircleCI user identities are derived from third-party VCS identities (e.g., those managed by GitHub or Bitbucket) and thus received a low score for administration and management. However, a new identity management system is on the roadmap to address this shortcoming. CircleCI has a good set of OOTB metrics and analytics, including DORA metrics, but the dashboards and capabilities were not as extensive as those scoring higher. One reference customer listed the following capabilities as most appreciated: "parallelization of workflows and rerunning only failed parts of workflows." CircleCI is a great fit for dev teams that value team autonomy and superfast build automation.

- **Codefresh has differentiating CI/CD but needs to improve governance and compliance.** Codefresh provides high-scoring CI/CD capabilities that emphasize the GitOps methodology, which requires all changes to infrastructure to be stored and managed as code in a Git repository prior to deployment. This methodology gained popularity from the Argo CD open source project, which powers Codefresh. Codefresh is also a registered project maintainer and contributor, which factored into its on-par community score. The Codefresh vision and roadmap focus on improvements to the developer experience and deployment capabilities such as Kubernetes context injection, improved disk utilization, and better support for teams. Innovation is confined to Argo and GitOps improvements, earning on-par scores. The one below-par score was for its smaller-than-average partner ecosystem.

Codefresh earned above-par scores for build automation, CI, and deployment automation featuring unique pipeline debugging capabilities that enable users to step in and out of containerized build steps, which no other vendor demonstrated.

Product support was strong, earning on-par scores. IaC provides strong support for Kubernetes configurations, but support for other legacy platforms lacks, keeping this score at par. Developer experience is on par; vendors scoring higher had integrated knowledge management. Other on-par scores include pipeline security, administration and management, and analytics that incorporate OOTB DORA metrics. Codefresh earned below-par scores for governance and compliance (providing limited support for ITSM integration) and support for functional test automation. When asked what was best about Codefresh, one customer said, “easy, standardized deployments for a hybrid universe.” Codefresh is a great fit for organizations that have adopted GitOps and place high value on developer autonomy.

Contenders

- **Flexagon has excellent customer support, but developer experience was below par.** Flexagon builds an end-to-end ISDP that supports traditional forms of software and infrastructure (e.g., Java and Linux) as well as business platforms such as Salesforce, Oracle, and SAP. The product vision is clear and focuses on delivering enterprise value through security and lowering risk. Planned updates include managed Oracle Cloud services, SAP blueprints, and a compliance and policy engine. Innovation is below par, investment in leading-edge areas is lacking, and the partner ecosystem was small, compared with others in this evaluation. Flexagon pricing features multiple tiers and full transparency of pricing options. Flexagon got above-par scores for test automation because it was one of the few vendors that featured native functional test automation capabilities and support.

Flexagon offers good platform cohesion with its offering that includes everything from agile planning to release automation in one package. It would have scored higher if not for some complexity in setting up and operating fleets of build runners. Flexagon got high scores for version control, with its support for the largest number of VCSes including Git and non-Git systems, and environment management that enabled teams to easily operate and manage any number of environments to support engineering teams. Deployment automation, release readiness, and platform support were also solid. Dev experience was not a focus area and resulted in a low score for that capability. Reference customers gave Flexagon perfect high scores for customer support, with one stating, “Flexagon continues to knock our socks off.” Flexagon is a great fit for enterprises looking for DevOps capabilities that span both open systems and business systems.

- **VMware delivers strong pipeline security but is light on data and analytics.**

VMware is best known for its infrastructure virtualization software. Tanzu Application Platform is VMware's ISDP and has a vision that is broad and incorporates elements of sociological factors to help developers be happier and more productive. VMware scored on par for community with strong support for open source tools and its user community. Innovation includes investments in autoremediation and AI/ML, earning an on-par score. The on-par roadmap will help users simplify the product lifecycle, improve developer collaboration, provide cost governance, and have more secure supply chains.

VMware pipeline security is one of the best in the evaluation with signed pipelines, signed commits, and additional security capabilities throughout the pipeline. Governance and compliance were on par, with OOTB compliance policies. Platform cohesion was also good with minimal setup required. Developer experience included a built-in knowledge management system to help onboard new developers and scored on par. Deployment automation included blue/green and canary deployment strategies, earning an on-par score. Administration and management were below par because they were missing features of those scoring higher. Data and analytics were few, and platform support seemed limited to Kubernetes workloads. One reference customer remarked, "The vendor is always willing to accept and act on feedback." VMware is a great fit for enterprises operating on Kubernetes and places high value on pipeline security.

- **OpenText has top-rated platform support, but DevSecOps capabilities score**

low. OpenText is best known for its portfolio of software development tools ranging from portfolio to test automation. OpenText's ValueEdge is an open platform that bundles these capabilities together in one ISDP. Its roadmap is on par, featuring elements that will better connect business and development. Innovation features AI/ML infused in risk management and testing, and OpenText boasts a healthy partner ecosystem. Both areas scored on par, with good evidence of progress from its AI/ML investments. But its vision focused more on business concerns than stronger CI/CD capabilities. In addition, pricing is opaque, compared with those scoring higher.

OpenText has one of the strongest governance and compliance offerings of this evaluation because nearly everything connects to change management systems. It has superior support for test automation capabilities that include autoclassification of test failures. In addition to providing its own VCS, users can connect with virtually any other VCS, earning above-par scores for that capability. Support for a

multitude of platforms including mainframe was a plus. Administration, management, and developer experience were on par, as were data and analytics that featured DORA and flow metrics. Deployment automation was also on par with blue/green and canary deployment support. Pipeline security lagged others evaluated. IaC, an area of growing importance to developers, also scored low. OpenText reference customers were positive about their experience, with one stating, “We don’t have any challenges at this stage.” OpenText is a great fit for enterprises with varied platforms in need of strong ties to test automation and compliance.

- **JFrog provides differentiated pipeline security capabilities, but the core ISDP falls short.** JFrog is best known for its flagship product, Artifactory, a binary artifact repository with a name that is so well known that it is synonymous with artifact repositories in general. JFrog’s strategy includes strong on-par scores for innovation based on investments in helping secure enterprise software supply chains. JFrog has a strong user community and actively participates in open source projects. Pricing flexibility was also on par, offering several payment tiers. JFrog scored below par for vision because it seemed overly focused on security at the expense of other ISDP capabilities. The roadmap was also below par due to its limited scope.

JFrog has superior version control due to the strength of Artifactory. Pipeline security is also top-notch with signed artifacts and signed pipelines that provide some of the strongest supply chain security measures available. The JFrog platform has good cohesion and requires few steps to set up and get started. Administration and management are on par and offer LDAP, OAuth, and support for SCIM to manage users at scale. Xray, a security product that scans artifacts, comes bundled and offers container and image scanning along with OSS supply chain management. To make this a more competitive ISDP offering, JFrog needs to invest in release readiness, data, and analytics and expand policy as code to all aspects of the ISDP while doing more to support test automation. Customers like how they are treated by JFrog, saying: It’s a “partner, not a vendor — they bring this mentality to everything.” JFrog is great for enterprises that place high value on supply chain security.

Challengers

- **IBM UrbanCode Deploy offers solid build and deployment capabilities but looks dated.** IBM UrbanCode offers UrbanCode Build and UrbanCode Deploy as two components that work together to deliver an enterprise-scale ISDP. IBM has a

solid development community because IBM Garage has thousands of contributors. IBM's partner ecosystem is also solidly on par with a number of top systems integrators listed as partners. The strategy has several weak elements including a roadmap that emphasizes maintenance versus new features, and innovation proof points are lacking.

IBM has solid build automation, deployment automation, release readiness, and support for a large variety of target platforms, which are core ISDP capabilities valued by enterprise dev teams. IBM's UrbanCode Velocity module offers strong analytics and graphics, earning it on-par scores for analytics, and there's a healthy number of third-party integrations for an on-par score in that category. But platform cohesion suffers from a complicated installation process, administration and management offer limited capabilities such as LDAP, and the developer experience is bare bones. Pipelines are present but don't offer the same level of capability as more modern tools. IBM is a good fit for enterprises that need to build an ISDP bridge between legacy and modern infrastructure. IBM declined to participate in the full Forrester Wave evaluation process.

Evaluation Overview

We grouped our evaluation criteria into three high-level categories:

- **Current offering.** Each vendor's position on the vertical axis of the Forrester Wave graphic indicates the strength of its current offering. Key criteria for these solutions include platform capabilities to facilitate management at scale, CI/CD to perform core ISDP functions, extensibility to enable platform operators to incorporate third-party tools as needed, and integrated systems to secure the software delivery pipeline.
- **Strategy.** Placement on the horizontal axis indicates the strength of the vendors' strategies. We evaluated vision, roadmap, and innovation to determine which vendor was placing the best bets to fulfill long-term product promises.
- **Market presence.** Represented by the size of the markers on the graphic, our market presence scores reflect each vendor's ISDP revenue and number of enterprise customers.

Vendor Inclusion Criteria

Each of the vendors we included in this assessment has:

- **Integrated CI/CD (e.g., deployment and release automation) capabilities.** The vendor provides integrated CI/CD capabilities OOTB that enable teams to rapidly establish build and deploy processes with minimal configuration.
- **Extensibility via APIs or plug-ins.** The vendor provides APIs to support third-party tool integration to extend the platform to incorporate performance, test automation, and security.
- **A system for centralized administration.** A single dashboard helps platform administrators operate and maintain an ISDP at enterprise scale.
- **Strong client interest.** Analyst inquiry logs indicate the vendor is top of mind among Forrester's end-user clientele with regard to ISDPs.
- **General-purpose programming languages and platforms.** The vendor supports build and deployment automation for open source programming languages like C# and Java as well as deployment to general-purpose runtime platforms such as Linux or Windows. It is also not exclusively tied to any particular business platform such as Salesforce or SAP.
- **High market performance.** The vendor's market performance demonstrates that it is a significant player in the ISDP market.

Supplemental Material

Online Resource

We publish all our Forrester Wave scores and weightings in an Excel file that provides detailed product evaluations and customizable rankings; download this tool by clicking the link at the beginning of this report on Forrester.com. We intend these scores and default weightings to serve only as a starting point and encourage readers to adapt the weightings to fit their individual needs.

The Forrester Wave Methodology

A Forrester Wave is a guide for buyers considering their purchasing options in a technology marketplace. To offer an equitable process for all participants, Forrester follows [The Forrester Wave™ Methodology](#) to evaluate participating vendors.

In our review, we conduct primary research to develop a list of vendors to consider for the evaluation. From that initial pool of vendors, we narrow our final list based on the inclusion criteria. We then gather details of product and strategy through a detailed questionnaire, demos/briefings, and customer reference surveys/interviews. We use those inputs, along with the analyst's experience and expertise in the marketplace, to

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score vendors, using a relative rating system that compares each vendor against the others in the evaluation.

We include the Forrester Wave publishing date (quarter and year) clearly in the title of each Forrester Wave report. We evaluated the vendors participating in this Forrester Wave using materials they provided to us by March 2023 and did not allow additional information after that point. We encourage readers to evaluate how the market and vendor offerings change over time.

In accordance with [our vendor review policy](#), Forrester asks vendors to review our findings prior to publishing to check for accuracy. Vendors marked as nonparticipating vendors in the Forrester Wave graphic met our defined inclusion criteria but declined to participate in or contributed only partially to the evaluation. We score these vendors in accordance with [our vendor participation policy](#) and publish their positioning along with those of the participating vendors.

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WAVE REPORT

The Forrester WaveTM: Integrated Software Delivery Platforms, Q2 2023

The 13 Providers That Matter Most And How They
Stack Up

May 31, 2023

By Christopher Condo with Chris Gardner, Sara Sjoblom, Kara Hartig

FORRESTER®

Summary

In our 26-criterion evaluation of integrated software delivery platform (ISDP) providers, we identified the most significant ones and researched, analyzed, and scored them. This report shows how each provider measures up and helps modern application development leaders and DevOps platform team leaders select the right one for their needs.

Additional resources are available in the [online version](#) of this report.

ISDPs Provide A Strong Platform Foundation But Still Need Best-Of-Breed Tools

In our reports, [The Future Of DevOps](#) and [The Forrester Guide To DevOps](#), Forrester interviewed dozens of enterprise software development leaders from diverse companies ranging from Netflix to TransUnion to understand the current state of DevOps, its relevance, and next steps enterprise leaders were taking to further advance their practices. In addition, we asked which tools they were using, how effective the tools were, which were using [ISDPs](#), and how well those ISDPs were performing. We learned that many enterprises have shifted to platform teams to support their DevOps efforts and many platform leaders are evaluating and using ISDPs, but few are using all of their ISDP vendor's capabilities. They explained that while these ISDP tools hold great potential for a complete end-to-end toolchain, many individual capabilities fell short of enterprise needs, forcing these leaders to source those capabilities from separate best-of-breed vendors.

Our reference customer survey backs this up by indicating that enterprises are using (on average) only 53% of the capabilities they are paying for, with the remaining capabilities either being delivered by a separate vendor or simply not used. This means that ISDP vendors are not only competing with each other but also competing with the best-of-breeds, which is why the bar has risen so high for this evaluation. ISDP vendors recognize these challenges and are responding with a number of innovative solutions that will make them more attractive than a traditional DIY toolchain but open and extensible enough that end users can add the tools they want without sacrificing capability. The challenge is striking the right balance to create overall value.

As a result of these trends, ISDP customers should look for providers that:

- **Offer a strong core of CI/CD capabilities.** Continuous integration (CI) and continuous delivery (CD) are the core capabilities that the rest of the software delivery toolchain is built around. Several vendors offered build optimization algorithms that slash build and unit test times by intelligently building and testing only the code that changed. Deployment automation is also a differentiating capability, with only a few vendors offering advanced capabilities out of the box (OOTB) such as blue/green, canary, and feature management deployment strategies.
- **Provide connected platforms that best-of-breed tools can augment.** ISDPs provide engineering leverage in the form of preintegrated toolchains that enable work to easily flow from one process to the next, where administration occurs at

the platform level versus the tool level, and reduced context switching improves the developer experience. Often, platform leaders want to override the default capability with third-party tools that they believe offer superior capability, which can lead to a bit of a balancing act. That's why some ISDP vendors are heavily investing in improvements to their end-to-end platform, while others are banking on investments in third-party connectivity and partnerships.

- **Help enterprise leaders shift from DevOps to DevSecOps.** Forrester fields calls every day from enterprise development leaders that are transforming their DevOps practice into a development, security, and operations (DevSecOps) practice to make security a first-class concern for the entire development organization. ISDP vendors are responding with capabilities that better secure software delivery pipelines, including integrated secrets management, signed pipelines, and signed artifacts. Reference customers noted that not all of these tools are direct replacements for best-of-breed (best-of-breed security tools often having a significant advantage over bundled ones), but end users like them as forms of additional assurance and coverage.

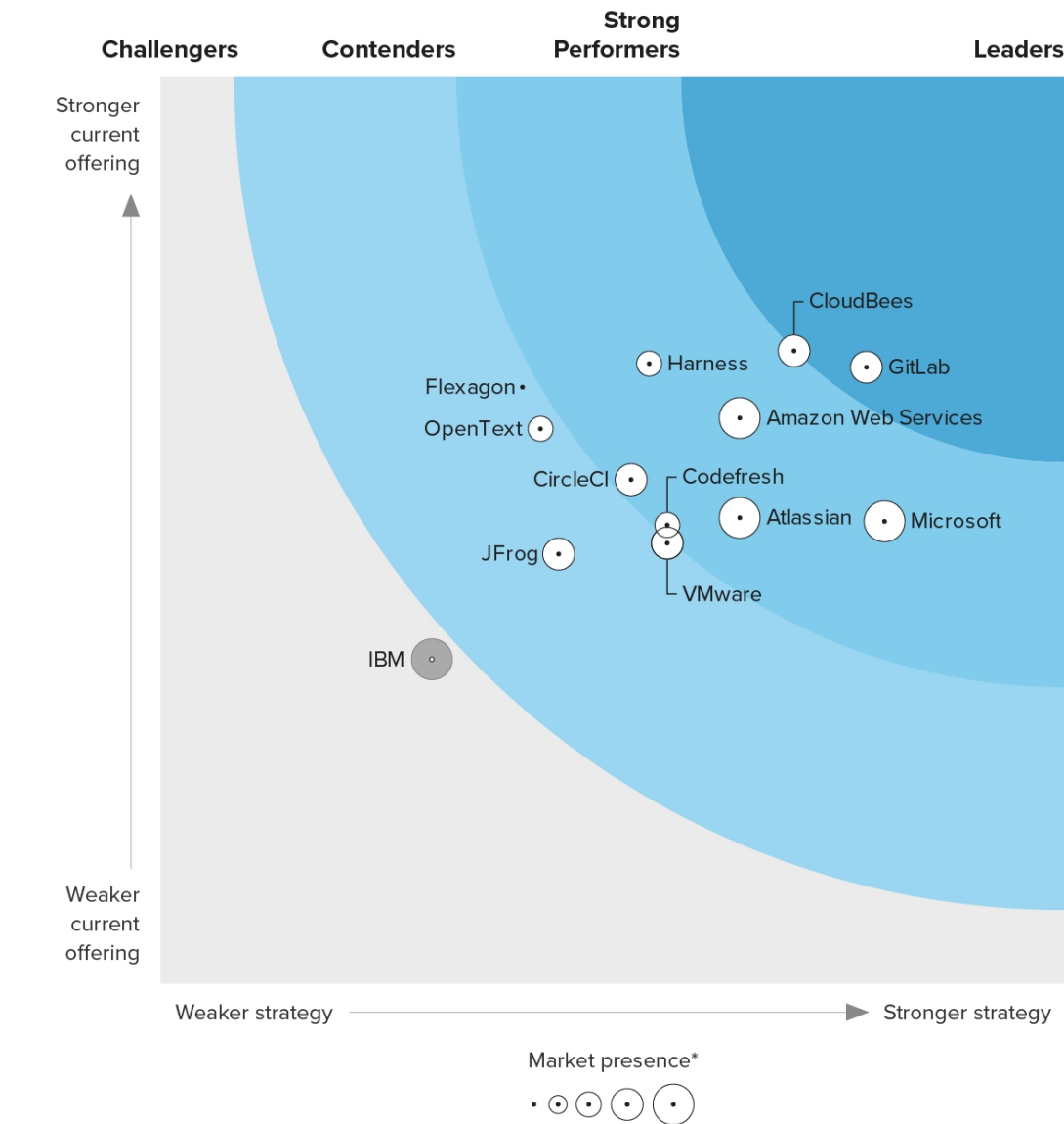
Evaluation Summary

The Forrester Wave™ evaluation highlights Leaders, Strong Performers, Contenders, and Challengers. It's an assessment of the top vendors in the market; it doesn't represent the entire vendor landscape. You'll find more information about this market in Forrester's [Now Tech: Integrated Software Delivery Platforms, Q2 2022](#).

We intend this evaluation to be a starting point only and encourage clients to view product evaluations and adapt criteria weightings using the Excel-based vendor comparison tool (see Figures 1 and 2). Click the link at the beginning of this report on [Forrester.com](#) to download the tool.

Figure 1
Forrester Wave™: Integrated Software Delivery Platforms, Q2 2023

THE FORRESTER WAVE™
Integrated Software Delivery Platforms
Q2 2023



*A gray bubble or open dot indicates a nonparticipating vendor.

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Figure 2**Forrester Wave™: Integrated Software Delivery Platforms Scorecard, Q2 2023**

	Forrester's weighting	Amazon Web Services	Atlassian	CircleCI	CloudBees	Codefresh	Flexagon	GitLab
Current offering	50%	2.57	3.12	2.78	3.49	2.53	3.29	3.40
Platform capabilities	20%	3.50	3.00	2.00	2.50	3.00	2.50	3.50
CI/CD capabilities	25%	2.00	2.60	3.00	4.00	4.00	3.20	3.40
Development, security, and operations	10%	1.00	3.00	2.00	2.00	2.00	3.00	4.00
Extensibility	10%	3.00	3.40	3.00	3.40	1.00	3.00	3.00
Governance and compliance	10%	3.00	5.00	3.00	5.00	1.00	3.00	3.00
Infrastructure management	10%	2.20	3.80	3.80	5.00	1.80	3.40	3.00
Product support	10%	3.00	3.00	3.00	3.00	3.00	5.00	3.00
Support for test automation	5%	3.00	1.00	3.00	3.00	1.00	5.00	5.00
Strategy	50%	3.20	3.20	2.60	3.50	2.80	2.00	3.90
Vision	20%	3.00	3.00	1.00	3.00	3.00	3.00	3.00
Roadmap	20%	3.00	3.00	3.00	3.00	3.00	3.00	5.00
Partner ecosystem	10%	5.00	5.00	1.00	5.00	1.00	1.00	3.00
Innovation	25%	3.00	3.00	3.00	3.00	3.00	1.00	3.00
Community	15%	3.00	3.00	3.00	5.00	3.00	1.00	5.00
Pricing flexibility and transparency	10%	3.00	3.00	5.00	3.00	3.00	3.00	5.00
Market presence	0%	5.00	4.50	3.50	3.50	2.50	1.00	3.50
ISDP revenue	50%	5.00	5.00	4.00	4.00	2.00	1.00	4.00
Number of enterprise customers	50%	5.00	4.00	3.00	3.00	3.00	1.00	3.00

All scores are based on a scale of 0 (weak) to 5 (strong).

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	Forrester's weighting	Harness	IBM*	JFrog	Microsoft	OpenText	VMware
Current offering	50%	3.42	1.79	2.37	2.55	3.06	2.43
Platform capabilities	20%	4.00	1.50	2.00	3.00	2.50	2.00
CI/CD capabilities	25%	4.20	2.80	2.40	2.40	3.40	2.40
Development, security, and operations	10%	3.00	1.00	4.00	4.00	1.00	3.00
Extensibility	10%	1.00	1.40	4.20	4.20	3.40	3.00
Governance and compliance	10%	3.00	3.00	1.00	1.00	5.00	3.00
Infrastructure management	10%	4.20	1.00	1.00	1.80	2.20	1.80
Product support	10%	3.00	1.00	3.00	1.00	3.00	3.00
Support for test automation	5%	3.00	1.00	1.00	3.00	5.00	1.00
Strategy	50%	2.70	1.50	2.20	4.00	2.10	2.80
Vision	20%	3.00	1.00	1.00	3.00	1.00	3.00
Roadmap	20%	3.00	1.00	1.00	3.00	3.00	3.00
Partner ecosystem	10%	3.00	3.00	3.00	5.00	3.00	3.00
Innovation	25%	3.00	1.00	3.00	5.00	3.00	3.00
Community	15%	1.00	3.00	3.00	5.00	1.00	3.00
Pricing flexibility and transparency	10%	3.00	1.00	3.00	3.00	1.00	1.00
Market presence	0%	3.00	4.50	4.00	5.00	2.50	4.00
ISDP revenue	50%	3.00	5.00	4.00	5.00	4.00	4.00
Number of enterprise customers	50%	3.00	4.00	4.00	5.00	1.00	4.00

All scores are based on a scale of 0 (weak) to 5 (strong).

*Indicates a nonparticipating vendor

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Vendor Offerings

Forrester evaluated the offerings listed below (see Figure 3).

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Figure 3
Evaluated Vendors And Product Information

Vendor	Product evaluated
Amazon Web Services	AWS Code Services
Atlassian	Atlassian Open DevOps
CircleCI	CircleCI
CloudBees	CloudBees Platform
Codefresh	Codefresh
Flexagon	Flexagon FlexDeploy
GitLab	GitLab
Harness	Harness Software Delivery Platform
IBM	IBM UrbanCode Deploy
JFrog	JFrog Software Supply Chain Platform
Microsoft	Microsoft Azure DevOps and GitHub Enterprise
OpenText (formerly Micro Focus)	Micro Focus ValueEdge
VMware	VMware Tanzu Application Platform

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Vendor Profiles

Our analysis uncovered the following strengths and weaknesses of individual vendors.

Leaders

- **GitLab leads with a balance between solid platform capabilities and overall strategy.** GitLab is known for being the vendor that made end-to-end OOTB toolchains hip again and has led the industry toward consolidated ISDPs. GitLab’s strategy includes an on-par vision to deliver an excellent developer experience without sacrificing security or compliance. Its innovation is also good, going beyond traditional developers to include AI/ML engineering. GitLab is an open core product that not only invests heavily in the open source software (OSS) community but also enables its customers to contribute to the product, earning it

high scores for community. Pricing is completely transparent and includes a free tier. Its roadmap gets leading scores and includes enhanced supply chain security, enhanced UI, granular security and compliance controls, and pipeline security — all things enterprises need.

GitLab's platform cohesion received high scores for simplicity of setup and operation as both an on-premises product and a software-as-a-service (SaaS) product. Though GitLab comes with its own GitLab version control system (VCS), it supports complete integration with all major Git providers, earning it high scores for version control. GitLab gets high scores for DevSecOps by providing OOTB security tools for static application security testing (SAST), dynamic application security testing (DAST), and Secret Detection. GitLab is one of the few ISDP vendors to incorporate a test automation dashboard, earning a top score. Dev experience is on par, as are administration and management, deployment automation, and release readiness. Build automation, governance and compliance, and infrastructure as code (IaC) were on par with others. One reference customer commented, "The CI/CD experience using secrets, environments, runners, and SAST/DAST/license scans/etc. is unparalleled in my opinion." GitLab is great for enterprises wishing to consolidate their best-of-breed toolchain into one high-performing ISDP.

Strong Performers

- **CloudBees delivers on its core promise of CI/CD, but platform cohesion needs work.** CloudBees is primarily associated with Jenkins; this well-known open source CI tool that it donated to the Continuous Delivery Foundation is one of the original best-of-breed tools that accompanied the beginning of the DevOps era more than 10 years ago. CloudBees leverages open source Jenkins and augments that with enterprise-scale deployment automation, release readiness, and value stream management (VSM) analytics to provide an enterprise-class ISDP. It got high scores for community, with significant contributions to open source projects and cultivation of its own user community. Its strategy features a solid vision and one of the larger partner ecosystems in this market. Innovation was on par and includes efforts to improve codification of policies. A planned SaaS offering should address its platform cohesion weakness and will help CloudBees serve clients that want a more integrated and maintenance-free ISDP.

CloudBees deployment automation is above par, covering blue/green, canary, and powerful feature management capabilities that are third-party add-ons for most

other platforms. CloudBees release readiness, governance, and compliance are also top-notch with complete integration with popular IT service management (ITSM) tools such as ServiceNow that bring those UIs directly into the ISDP experience. Build automation requires a bit of effort to manage fleets of build runners, and pipeline security needs to go beyond branch security to earn higher scores. When asked what they liked about working with CloudBees, reference customers had good things to say: “The vendor relationship has been a true partnership. It’s listening to our internal customers.” CloudBees is a great fit for enterprise platform teams that value open source and open choice.

- **Microsoft differentiates with innovation, but governance and compliance need attention.** Microsoft acquired GitHub, the industry’s largest open source registry, in 2018. Since that time, Microsoft’s Azure DevOps and GitHub have slowly been moving toward a unified offering. However, both the product vision and the roadmap could have been clearer on how this will shape future offerings, resulting in an on-par score. Innovation strategy is tops because GitHub Copilot has set new standards for the development tools industry and planned investments will enable it to continue this leadership position. Its partner ecosystem also scored high, with Microsoft having one of the most extensive. GitHub also supports numerous open source projects, and its online community is one of the biggest, earning a top score. The roadmap heavily weighted developer productivity but was missing key enterprise needs for governance.

Microsoft’s developer experience gets top scores for GitHub Actions, which turned GitHub into a CI leader almost overnight; Codespaces, a virtual dev environment that requires no setup by developers to get started; and Copilot, the AI-powered coding assistant. It also received top scores for OOTB security tools such as Dependabot. Support for APIs and third-party tools was also top scoring, with numerous ways to incorporate not only software but also actions. Pipeline security and platform cohesion were on par, while administration and management were missing the granular permissions of those scoring higher. Microsoft data and analytics were mostly operational, resulting in a below-par score. Deployment automation was also below par — it lacked blue/green and canary deployment scenarios. Reference customers were mostly positive about Microsoft, noting, “Generally, adoption of standard features is quick and doesn’t require a great deal of time to become productive.” Microsoft is great for organizations looking to boost developer effectiveness with cutting-edge tools.

- **Atlassian gets top scores for cohesiveness, but pipeline security capabilities are lacking.** Atlassian is best known for Jira, one of the most popular agile planning

tools on the market serving 80% of the Fortune 500. Atlassian's product strategy is on par and strives to serve two audiences, teams of developers and platform teams, by building out a data layer to connect the tools together. Its partner ecosystem is top-notch. Innovation is on par and includes platform capabilities and a data lake component to capture and aggregate data for greater analytical insight. The roadmap reflects enterprise concerns with plans to aggregate third-party security data, automatically infuse issue tickets with context, provide greater VSM insights, and expand support plans for DevOps and platform teams.

Atlassian's Confluence, which enables teams to create custom developer onboarding content and a hub for team collaboration, enhances the developer experience. This plus integrated development environment plug-ins give Atlassian a solid on-par score for developer experience. The OOTB build and CI were on par, with reusable build templates and parallel build execution to reduce build times. Deployment automation was below par, as were capabilities to secure pipelines. OOTB security tools are few, earning a below-par score, because Atlassian prefers to leverage its partner ecosystem for those. A reference customer said, "We chose Atlassian for the product features and the maturity of the well-integrated suites of products and plug-ins, which help simplify using the well-designed solution." Atlassian is great fit for software teams and enterprise platform teams that are looking for a standard ISDP approach but want to leverage best-of-breed CD and security components.

- **Harness ISDP delivers on platform cohesion, but community contributions are nascent.** Harness has a SaaS ISDP with core CI/CD modules and cloud cost management capabilities. Its vision is to simplify software development with an end-to-end software delivery platform that offers a modern developer experience and automated governance and guardrails. The strategy has mostly on-par elements such as a growing partner ecosystem and investments in innovation. The only weak area was the smaller-than-average community program — those scoring higher had greater support for internal and external communities. The roadmap includes support for database development, test optimization, end-to-end visibility, and infusion of AI/ML throughout the platform.

Harness gets top scores for platform cohesion, delivering a great OOTB experience for first-time users. It also earned top scores in build automation that includes pipelines as code, reusable pipelines, resource management, support for test automation, and intelligent test automation that can drastically reduce build times. It supports multiple deployment strategies OOTB, including blue/green and

canary. In addition, Harness can pinpoint code that's caused pipeline errors, helping speed remediation. Analytics was also top scoring and offered dev, process, and operational metrics. Customization of the UI and some ISDP components is possible and easy, but the limited scope of those changes resulted in a below-par score. Other areas for improvement include expanding the number of third-party integrations and its footprint in the user and open source community. Customers appreciate Harness' "willingness to engage directly with our development teams." Harness is great for development teams that want to spend less time managing tools and more time developing code.

- **AWS is strong in administration and management, but dev metrics are a miss.**

Amazon Web Services (AWS) pioneered infrastructure as a service (IaaS) and offers nearly every possible form of it. Its ISDP product includes CodeBuild, CodeStar, CloudFormation, and several other SaaS offerings that are integrated on demand when users create their account. The overall strategy score was solid. Innovation scored on par for current delivery and upcoming features for helping developers code faster. The vision was on par, focusing mainly on CodeCatalyst that will enable AWS to make its developer platform more accessible to a broader set of end users with different levels of technical ability. AWS actively supports its user community and open source projects, getting an on-par score for community. The roadmap is also on par and includes the aforementioned CodeCatalyst, a product currently in beta that offers a preassembled and more opinionated ISDP to help customers accelerate project setup and collaborate more fluidly.

AWS shines with excellent administration and management capabilities that enable user permissions to be managed down to the API level. AWS supports all major authentication protocols including emerging standards such as SCIM. AWS gets top scores for governance and compliance that include OOTB compliance templates. Platform cohesion and developer experience are on par. Build automation would have scored higher had it offered test execution optimization. AWS got low scores for analytics and reports; other vendors offered better dashboards with more dev-related metrics. Release readiness also scored low. Reference customer sentiment was mostly positive, saying, "Security is a first-class citizen." AWS is great for companies that want speed, scale, and security hosted in the same environment as their public cloud infrastructure.

- **CircleCI provides outstanding build automation, but its vision does not go beyond CI/CD.** CircleCI was one of the first vendors to offer CI as SaaS. Over time, it added CD and extensibility features to make a more well-rounded ISDP offering. CircleCI's on-par vision to speed time to value while simplifying the ISDP

experience is well articulated and strikes a pragmatic balance between individual developer concerns, the needs of platform teams, and support for enterprises in regulated industries. CircleCI's strategic vision is to remain a best-of-breed CI/CD tool vendor and not extend beyond that into other ISDP areas, resulting in a below-par score. The on-par roadmap aims at enterprise teams and includes features to help platform teams better manage an ISDP at scale with pervasive validation at each step of software delivery.

CircleCI has invested in optimized build processes, giving it a top score for build automation and CI. Its support for a wide range of platforms, including all public clouds and even mainframe, earns above-par scores. IaC capabilities were also top scoring. Product support leverages dedicated customer success engineers for each account, but average reference customer scores kept it on par. CircleCI user identities are derived from third-party VCS identities (e.g., those managed by GitHub or Bitbucket) and thus received a low score for administration and management. However, a new identity management system is on the roadmap to address this shortcoming. CircleCI has a good set of OOTB metrics and analytics, including DORA metrics, but the dashboards and capabilities were not as extensive as those scoring higher. One reference customer listed the following capabilities as most appreciated: "parallelization of workflows and rerunning only failed parts of workflows." CircleCI is a great fit for dev teams that value team autonomy and superfast build automation.

- **Codefresh has differentiating CI/CD but needs to improve governance and compliance.** Codefresh provides high-scoring CI/CD capabilities that emphasize the GitOps methodology, which requires all changes to infrastructure to be stored and managed as code in a Git repository prior to deployment. This methodology gained popularity from the Argo CD open source project, which powers Codefresh. Codefresh is also a registered project maintainer and contributor, which factored into its on-par community score. The Codefresh vision and roadmap focus on improvements to the developer experience and deployment capabilities such as Kubernetes context injection, improved disk utilization, and better support for teams. Innovation is confined to Argo and GitOps improvements, earning on-par scores. The one below-par score was for its smaller-than-average partner ecosystem.

Codefresh earned above-par scores for build automation, CI, and deployment automation featuring unique pipeline debugging capabilities that enable users to step in and out of containerized build steps, which no other vendor demonstrated.

Product support was strong, earning on-par scores. IaC provides strong support for Kubernetes configurations, but support for other legacy platforms lacks, keeping this score at par. Developer experience is on par; vendors scoring higher had integrated knowledge management. Other on-par scores include pipeline security, administration and management, and analytics that incorporate OOTB DORA metrics. Codefresh earned below-par scores for governance and compliance (providing limited support for ITSM integration) and support for functional test automation. When asked what was best about Codefresh, one customer said, “easy, standardized deployments for a hybrid universe.” Codefresh is a great fit for organizations that have adopted GitOps and place high value on developer autonomy.

Contenders

- **Flexagon has excellent customer support, but developer experience was below par.** Flexagon builds an end-to-end ISDP that supports traditional forms of software and infrastructure (e.g., Java and Linux) as well as business platforms such as Salesforce, Oracle, and SAP. The product vision is clear and focuses on delivering enterprise value through security and lowering risk. Planned updates include managed Oracle Cloud services, SAP blueprints, and a compliance and policy engine. Innovation is below par, investment in leading-edge areas is lacking, and the partner ecosystem was small, compared with others in this evaluation. Flexagon pricing features multiple tiers and full transparency of pricing options. Flexagon got above-par scores for test automation because it was one of the few vendors that featured native functional test automation capabilities and support.

Flexagon offers good platform cohesion with its offering that includes everything from agile planning to release automation in one package. It would have scored higher if not for some complexity in setting up and operating fleets of build runners. Flexagon got high scores for version control, with its support for the largest number of VCSes including Git and non-Git systems, and environment management that enabled teams to easily operate and manage any number of environments to support engineering teams. Deployment automation, release readiness, and platform support were also solid. Dev experience was not a focus area and resulted in a low score for that capability. Reference customers gave Flexagon perfect high scores for customer support, with one stating, “Flexagon continues to knock our socks off.” Flexagon is a great fit for enterprises looking for DevOps capabilities that span both open systems and business systems.

- **VMware delivers strong pipeline security but is light on data and analytics.**

VMware is best known for its infrastructure virtualization software. Tanzu Application Platform is VMware's ISDP and has a vision that is broad and incorporates elements of sociological factors to help developers be happier and more productive. VMware scored on par for community with strong support for open source tools and its user community. Innovation includes investments in autoremediation and AI/ML, earning an on-par score. The on-par roadmap will help users simplify the product lifecycle, improve developer collaboration, provide cost governance, and have more secure supply chains.

VMware pipeline security is one of the best in the evaluation with signed pipelines, signed commits, and additional security capabilities throughout the pipeline. Governance and compliance were on par, with OOTB compliance policies. Platform cohesion was also good with minimal setup required. Developer experience included a built-in knowledge management system to help onboard new developers and scored on par. Deployment automation included blue/green and canary deployment strategies, earning an on-par score. Administration and management were below par because they were missing features of those scoring higher. Data and analytics were few, and platform support seemed limited to Kubernetes workloads. One reference customer remarked, "The vendor is always willing to accept and act on feedback." VMware is a great fit for enterprises operating on Kubernetes and places high value on pipeline security.

- **OpenText has top-rated platform support, but DevSecOps capabilities score**

low. OpenText is best known for its portfolio of software development tools ranging from portfolio to test automation. OpenText's ValueEdge is an open platform that bundles these capabilities together in one ISDP. Its roadmap is on par, featuring elements that will better connect business and development. Innovation features AI/ML infused in risk management and testing, and OpenText boasts a healthy partner ecosystem. Both areas scored on par, with good evidence of progress from its AI/ML investments. But its vision focused more on business concerns than stronger CI/CD capabilities. In addition, pricing is opaque, compared with those scoring higher.

OpenText has one of the strongest governance and compliance offerings of this evaluation because nearly everything connects to change management systems. It has superior support for test automation capabilities that include autoclassification of test failures. In addition to providing its own VCS, users can connect with virtually any other VCS, earning above-par scores for that capability. Support for a

multitude of platforms including mainframe was a plus. Administration, management, and developer experience were on par, as were data and analytics that featured DORA and flow metrics. Deployment automation was also on par with blue/green and canary deployment support. Pipeline security lagged others evaluated. IaC, an area of growing importance to developers, also scored low. OpenText reference customers were positive about their experience, with one stating, “We don’t have any challenges at this stage.” OpenText is a great fit for enterprises with varied platforms in need of strong ties to test automation and compliance.

- **JFrog provides differentiated pipeline security capabilities, but the core ISDP falls short.** JFrog is best known for its flagship product, Artifactory, a binary artifact repository with a name that is so well known that it is synonymous with artifact repositories in general. JFrog’s strategy includes strong on-par scores for innovation based on investments in helping secure enterprise software supply chains. JFrog has a strong user community and actively participates in open source projects. Pricing flexibility was also on par, offering several payment tiers. JFrog scored below par for vision because it seemed overly focused on security at the expense of other ISDP capabilities. The roadmap was also below par due to its limited scope.

JFrog has superior version control due to the strength of Artifactory. Pipeline security is also top-notch with signed artifacts and signed pipelines that provide some of the strongest supply chain security measures available. The JFrog platform has good cohesion and requires few steps to set up and get started. Administration and management are on par and offer LDAP, OAuth, and support for SCIM to manage users at scale. Xray, a security product that scans artifacts, comes bundled and offers container and image scanning along with OSS supply chain management. To make this a more competitive ISDP offering, JFrog needs to invest in release readiness, data, and analytics and expand policy as code to all aspects of the ISDP while doing more to support test automation. Customers like how they are treated by JFrog, saying: It’s a “partner, not a vendor — they bring this mentality to everything.” JFrog is great for enterprises that place high value on supply chain security.

Challengers

- **IBM UrbanCode Deploy offers solid build and deployment capabilities but looks dated.** IBM UrbanCode offers UrbanCode Build and UrbanCode Deploy as two components that work together to deliver an enterprise-scale ISDP. IBM has a

solid development community because IBM Garage has thousands of contributors. IBM's partner ecosystem is also solidly on par with a number of top systems integrators listed as partners. The strategy has several weak elements including a roadmap that emphasizes maintenance versus new features, and innovation proof points are lacking.

IBM has solid build automation, deployment automation, release readiness, and support for a large variety of target platforms, which are core ISDP capabilities valued by enterprise dev teams. IBM's UrbanCode Velocity module offers strong analytics and graphics, earning it on-par scores for analytics, and there's a healthy number of third-party integrations for an on-par score in that category. But platform cohesion suffers from a complicated installation process, administration and management offer limited capabilities such as LDAP, and the developer experience is bare bones. Pipelines are present but don't offer the same level of capability as more modern tools. IBM is a good fit for enterprises that need to build an ISDP bridge between legacy and modern infrastructure. IBM declined to participate in the full Forrester Wave evaluation process.

Evaluation Overview

We grouped our evaluation criteria into three high-level categories:

- **Current offering.** Each vendor's position on the vertical axis of the Forrester Wave graphic indicates the strength of its current offering. Key criteria for these solutions include platform capabilities to facilitate management at scale, CI/CD to perform core ISDP functions, extensibility to enable platform operators to incorporate third-party tools as needed, and integrated systems to secure the software delivery pipeline.
- **Strategy.** Placement on the horizontal axis indicates the strength of the vendors' strategies. We evaluated vision, roadmap, and innovation to determine which vendor was placing the best bets to fulfill long-term product promises.
- **Market presence.** Represented by the size of the markers on the graphic, our market presence scores reflect each vendor's ISDP revenue and number of enterprise customers.

Vendor Inclusion Criteria

Each of the vendors we included in this assessment has:

- **Integrated CI/CD (e.g., deployment and release automation) capabilities.** The vendor provides integrated CI/CD capabilities OOTB that enable teams to rapidly establish build and deploy processes with minimal configuration.
- **Extensibility via APIs or plug-ins.** The vendor provides APIs to support third-party tool integration to extend the platform to incorporate performance, test automation, and security.
- **A system for centralized administration.** A single dashboard helps platform administrators operate and maintain an ISDP at enterprise scale.
- **Strong client interest.** Analyst inquiry logs indicate the vendor is top of mind among Forrester's end-user clientele with regard to ISDPs.
- **General-purpose programming languages and platforms.** The vendor supports build and deployment automation for open source programming languages like C# and Java as well as deployment to general-purpose runtime platforms such as Linux or Windows. It is also not exclusively tied to any particular business platform such as Salesforce or SAP.
- **High market performance.** The vendor's market performance demonstrates that it is a significant player in the ISDP market.

Supplemental Material

Online Resource

We publish all our Forrester Wave scores and weightings in an Excel file that provides detailed product evaluations and customizable rankings; download this tool by clicking the link at the beginning of this report on Forrester.com. We intend these scores and default weightings to serve only as a starting point and encourage readers to adapt the weightings to fit their individual needs.

The Forrester Wave Methodology

A Forrester Wave is a guide for buyers considering their purchasing options in a technology marketplace. To offer an equitable process for all participants, Forrester follows [The Forrester Wave™ Methodology](#) to evaluate participating vendors.

In our review, we conduct primary research to develop a list of vendors to consider for the evaluation. From that initial pool of vendors, we narrow our final list based on the inclusion criteria. We then gather details of product and strategy through a detailed questionnaire, demos/briefings, and customer reference surveys/interviews. We use those inputs, along with the analyst's experience and expertise in the marketplace, to

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score vendors, using a relative rating system that compares each vendor against the others in the evaluation.

We include the Forrester Wave publishing date (quarter and year) clearly in the title of each Forrester Wave report. We evaluated the vendors participating in this Forrester Wave using materials they provided to us by March 2023 and did not allow additional information after that point. We encourage readers to evaluate how the market and vendor offerings change over time.

In accordance with [our vendor review policy](#), Forrester asks vendors to review our findings prior to publishing to check for accuracy. Vendors marked as nonparticipating vendors in the Forrester Wave graphic met our defined inclusion criteria but declined to participate in or contributed only partially to the evaluation. We score these vendors in accordance with [our vendor participation policy](#) and publish their positioning along with those of the participating vendors.

Integrity Policy

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WAVE REPORT

The Forrester WaveTM: Integrated Software Delivery Platforms, Q2 2023

The 13 Providers That Matter Most And How They
Stack Up

May 31, 2023

By Christopher Condo with Chris Gardner, Sara Sjoblom, Kara Hartig

FORRESTER®

Summary

In our 26-criterion evaluation of integrated software delivery platform (ISDP) providers, we identified the most significant ones and researched, analyzed, and scored them. This report shows how each provider measures up and helps modern application development leaders and DevOps platform team leaders select the right one for their needs.

Additional resources are available in the [online version](#) of this report.

ISDPs Provide A Strong Platform Foundation But Still Need Best-Of-Breed Tools

In our reports, [The Future Of DevOps](#) and [The Forrester Guide To DevOps](#), Forrester interviewed dozens of enterprise software development leaders from diverse companies ranging from Netflix to TransUnion to understand the current state of DevOps, its relevance, and next steps enterprise leaders were taking to further advance their practices. In addition, we asked which tools they were using, how effective the tools were, which were using ISDPs, and how well those ISDPs were performing. We learned that many enterprises have shifted to platform teams to support their DevOps efforts and many platform leaders are evaluating and using ISDPs, but few are using all of their ISDP vendor's capabilities. They explained that while these ISDP tools hold great potential for a complete end-to-end toolchain, many individual capabilities fell short of enterprise needs, forcing these leaders to source those capabilities from separate best-of-breed vendors.

Our reference customer survey backs this up by indicating that enterprises are using (on average) only 53% of the capabilities they are paying for, with the remaining capabilities either being delivered by a separate vendor or simply not used. This means that ISDP vendors are not only competing with each other but also competing with the best-of-breeds, which is why the bar has risen so high for this evaluation. ISDP vendors recognize these challenges and are responding with a number of innovative solutions that will make them more attractive than a traditional DIY toolchain but open and extensible enough that end users can add the tools they want without sacrificing capability. The challenge is striking the right balance to create overall value.

As a result of these trends, ISDP customers should look for providers that:

- **Offer a strong core of CI/CD capabilities.** Continuous integration (CI) and continuous delivery (CD) are the core capabilities that the rest of the software delivery toolchain is built around. Several vendors offered build optimization algorithms that slash build and unit test times by intelligently building and testing only the code that changed. Deployment automation is also a differentiating capability, with only a few vendors offering advanced capabilities out of the box (OOTB) such as blue/green, canary, and feature management deployment strategies.
- **Provide connected platforms that best-of-breed tools can augment.** ISDPs provide engineering leverage in the form of preintegrated toolchains that enable work to easily flow from one process to the next, where administration occurs at

the platform level versus the tool level, and reduced context switching improves the developer experience. Often, platform leaders want to override the default capability with third-party tools that they believe offer superior capability, which can lead to a bit of a balancing act. That's why some ISDP vendors are heavily investing in improvements to their end-to-end platform, while others are banking on investments in third-party connectivity and partnerships.

- **Help enterprise leaders shift from DevOps to DevSecOps.** Forrester fields calls every day from enterprise development leaders that are transforming their DevOps practice into a development, security, and operations (DevSecOps) practice to make security a first-class concern for the entire development organization. ISDP vendors are responding with capabilities that better secure software delivery pipelines, including integrated secrets management, signed pipelines, and signed artifacts. Reference customers noted that not all of these tools are direct replacements for best-of-breed (best-of-breed security tools often having a significant advantage over bundled ones), but end users like them as forms of additional assurance and coverage.

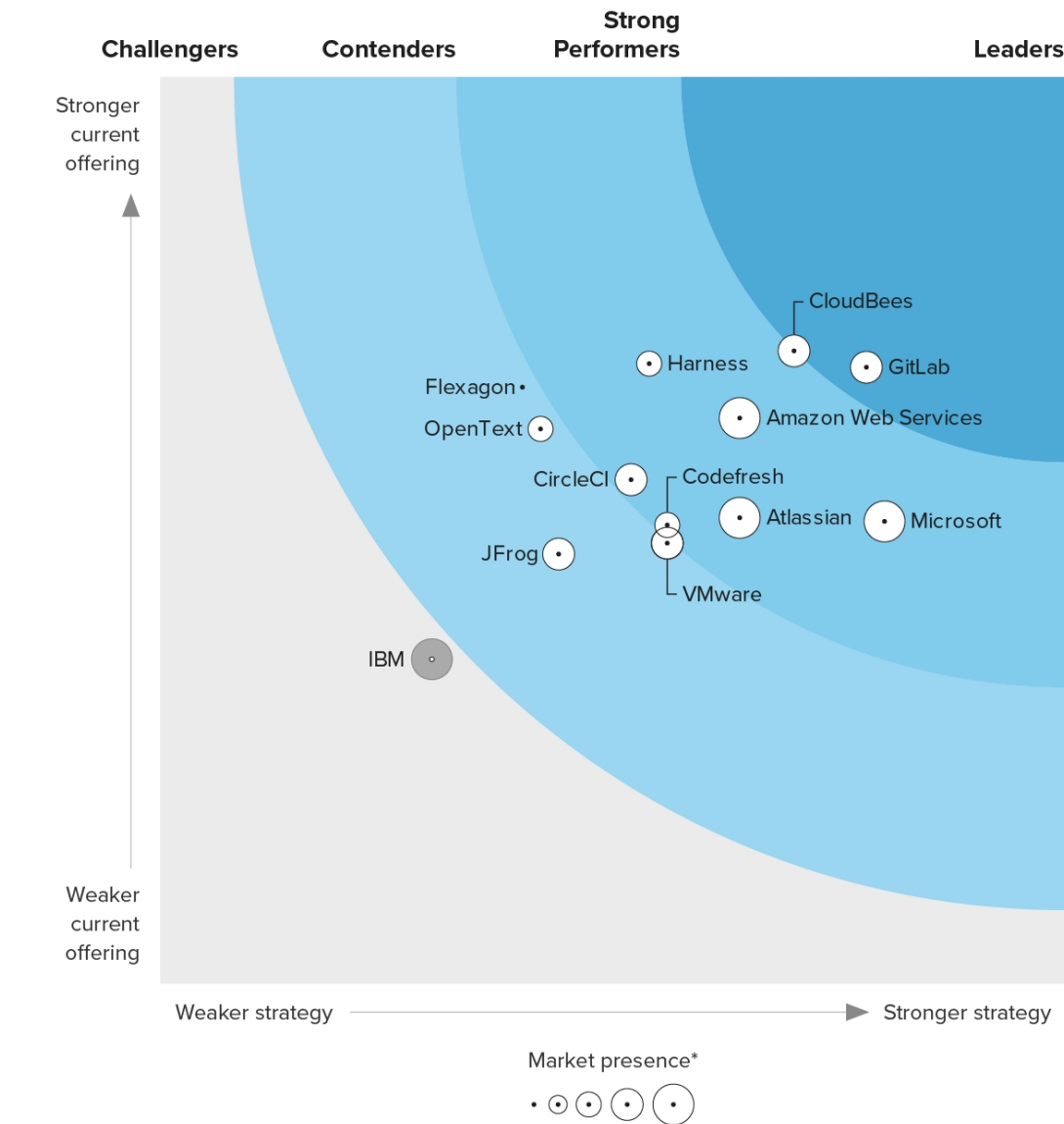
Evaluation Summary

The Forrester Wave™ evaluation highlights Leaders, Strong Performers, Contenders, and Challengers. It's an assessment of the top vendors in the market; it doesn't represent the entire vendor landscape. You'll find more information about this market in Forrester's [Now Tech: Integrated Software Delivery Platforms, Q2 2022](#).

We intend this evaluation to be a starting point only and encourage clients to view product evaluations and adapt criteria weightings using the Excel-based vendor comparison tool (see Figures 1 and 2). Click the link at the beginning of this report on [Forrester.com](#) to download the tool.

Figure 1
Forrester Wave™: Integrated Software Delivery Platforms, Q2 2023

THE FORRESTER WAVE™
Integrated Software Delivery Platforms
Q2 2023



*A gray bubble or open dot indicates a nonparticipating vendor.

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Figure 2**Forrester Wave™: Integrated Software Delivery Platforms Scorecard, Q2 2023**

	Forrester's weighting	Amazon Web Services	Atlassian	CircleCI	CloudBees	Codefresh	Flexagon	GitLab
Current offering	50%	2.57	3.12	2.78	3.49	2.53	3.29	3.40
Platform capabilities	20%	3.50	3.00	2.00	2.50	3.00	2.50	3.50
CI/CD capabilities	25%	2.00	2.60	3.00	4.00	4.00	3.20	3.40
Development, security, and operations	10%	1.00	3.00	2.00	2.00	2.00	3.00	4.00
Extensibility	10%	3.00	3.40	3.00	3.40	1.00	3.00	3.00
Governance and compliance	10%	3.00	5.00	3.00	5.00	1.00	3.00	3.00
Infrastructure management	10%	2.20	3.80	3.80	5.00	1.80	3.40	3.00
Product support	10%	3.00	3.00	3.00	3.00	3.00	5.00	3.00
Support for test automation	5%	3.00	1.00	3.00	3.00	1.00	5.00	5.00
Strategy	50%	3.20	3.20	2.60	3.50	2.80	2.00	3.90
Vision	20%	3.00	3.00	1.00	3.00	3.00	3.00	3.00
Roadmap	20%	3.00	3.00	3.00	3.00	3.00	3.00	5.00
Partner ecosystem	10%	5.00	5.00	1.00	5.00	1.00	1.00	3.00
Innovation	25%	3.00	3.00	3.00	3.00	3.00	1.00	3.00
Community	15%	3.00	3.00	3.00	5.00	3.00	1.00	5.00
Pricing flexibility and transparency	10%	3.00	3.00	5.00	3.00	3.00	3.00	5.00
Market presence	0%	5.00	4.50	3.50	3.50	2.50	1.00	3.50
ISDP revenue	50%	5.00	5.00	4.00	4.00	2.00	1.00	4.00
Number of enterprise customers	50%	5.00	4.00	3.00	3.00	3.00	1.00	3.00

All scores are based on a scale of 0 (weak) to 5 (strong).

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	Forrester's weighting	Harness	IBM*	JFrog	Microsoft	OpenText	VMware
Current offering	50%	3.42	1.79	2.37	2.55	3.06	2.43
Platform capabilities	20%	4.00	1.50	2.00	3.00	2.50	2.00
CI/CD capabilities	25%	4.20	2.80	2.40	2.40	3.40	2.40
Development, security, and operations	10%	3.00	1.00	4.00	4.00	1.00	3.00
Extensibility	10%	1.00	1.40	4.20	4.20	3.40	3.00
Governance and compliance	10%	3.00	3.00	1.00	1.00	5.00	3.00
Infrastructure management	10%	4.20	1.00	1.00	1.80	2.20	1.80
Product support	10%	3.00	1.00	3.00	1.00	3.00	3.00
Support for test automation	5%	3.00	1.00	1.00	3.00	5.00	1.00
Strategy	50%	2.70	1.50	2.20	4.00	2.10	2.80
Vision	20%	3.00	1.00	1.00	3.00	1.00	3.00
Roadmap	20%	3.00	1.00	1.00	3.00	3.00	3.00
Partner ecosystem	10%	3.00	3.00	3.00	5.00	3.00	3.00
Innovation	25%	3.00	1.00	3.00	5.00	3.00	3.00
Community	15%	1.00	3.00	3.00	5.00	1.00	3.00
Pricing flexibility and transparency	10%	3.00	1.00	3.00	3.00	1.00	1.00
Market presence	0%	3.00	4.50	4.00	5.00	2.50	4.00
ISDP revenue	50%	3.00	5.00	4.00	5.00	4.00	4.00
Number of enterprise customers	50%	3.00	4.00	4.00	5.00	1.00	4.00

All scores are based on a scale of 0 (weak) to 5 (strong).

*Indicates a nonparticipating vendor

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Vendor Offerings

Forrester evaluated the offerings listed below (see Figure 3).

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Figure 3
Evaluated Vendors And Product Information

Vendor	Product evaluated
Amazon Web Services	AWS Code Services
Atlassian	Atlassian Open DevOps
CircleCI	CircleCI
CloudBees	CloudBees Platform
Codefresh	Codefresh
Flexagon	Flexagon FlexDeploy
GitLab	GitLab
Harness	Harness Software Delivery Platform
IBM	IBM UrbanCode Deploy
JFrog	JFrog Software Supply Chain Platform
Microsoft	Microsoft Azure DevOps and GitHub Enterprise
OpenText (formerly Micro Focus)	Micro Focus ValueEdge
VMware	VMware Tanzu Application Platform

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Vendor Profiles

Our analysis uncovered the following strengths and weaknesses of individual vendors.

Leaders

- **GitLab leads with a balance between solid platform capabilities and overall strategy.** GitLab is known for being the vendor that made end-to-end OOTB toolchains hip again and has led the industry toward consolidated ISDPs. GitLab’s strategy includes an on-par vision to deliver an excellent developer experience without sacrificing security or compliance. Its innovation is also good, going beyond traditional developers to include AI/ML engineering. GitLab is an open core product that not only invests heavily in the open source software (OSS) community but also enables its customers to contribute to the product, earning it

high scores for community. Pricing is completely transparent and includes a free tier. Its roadmap gets leading scores and includes enhanced supply chain security, enhanced UI, granular security and compliance controls, and pipeline security — all things enterprises need.

GitLab's platform cohesion received high scores for simplicity of setup and operation as both an on-premises product and a software-as-a-service (SaaS) product. Though GitLab comes with its own GitLab version control system (VCS), it supports complete integration with all major Git providers, earning it high scores for version control. GitLab gets high scores for DevSecOps by providing OOTB security tools for static application security testing (SAST), dynamic application security testing (DAST), and Secret Detection. GitLab is one of the few ISDP vendors to incorporate a test automation dashboard, earning a top score. Dev experience is on par, as are administration and management, deployment automation, and release readiness. Build automation, governance and compliance, and infrastructure as code (IaC) were on par with others. One reference customer commented, "The CI/CD experience using secrets, environments, runners, and SAST/DAST/license scans/etc. is unparalleled in my opinion." GitLab is great for enterprises wishing to consolidate their best-of-breed toolchain into one high-performing ISDP.

Strong Performers

- **CloudBees delivers on its core promise of CI/CD, but platform cohesion needs work.** CloudBees is primarily associated with Jenkins; this well-known open source CI tool that it donated to the Continuous Delivery Foundation is one of the original best-of-breed tools that accompanied the beginning of the DevOps era more than 10 years ago. CloudBees leverages open source Jenkins and augments that with enterprise-scale deployment automation, release readiness, and value stream management (VSM) analytics to provide an enterprise-class ISDP. It got high scores for community, with significant contributions to open source projects and cultivation of its own user community. Its strategy features a solid vision and one of the larger partner ecosystems in this market. Innovation was on par and includes efforts to improve codification of policies. A planned SaaS offering should address its platform cohesion weakness and will help CloudBees serve clients that want a more integrated and maintenance-free ISDP.

CloudBees deployment automation is above par, covering blue/green, canary, and powerful feature management capabilities that are third-party add-ons for most

other platforms. CloudBees release readiness, governance, and compliance are also top-notch with complete integration with popular IT service management (ITSM) tools such as ServiceNow that bring those UIs directly into the ISDP experience. Build automation requires a bit of effort to manage fleets of build runners, and pipeline security needs to go beyond branch security to earn higher scores. When asked what they liked about working with CloudBees, reference customers had good things to say: “The vendor relationship has been a true partnership. It’s listening to our internal customers.” CloudBees is a great fit for enterprise platform teams that value open source and open choice.

- **Microsoft differentiates with innovation, but governance and compliance need attention.** Microsoft acquired GitHub, the industry’s largest open source registry, in 2018. Since that time, Microsoft’s Azure DevOps and GitHub have slowly been moving toward a unified offering. However, both the product vision and the roadmap could have been clearer on how this will shape future offerings, resulting in an on-par score. Innovation strategy is tops because GitHub Copilot has set new standards for the development tools industry and planned investments will enable it to continue this leadership position. Its partner ecosystem also scored high, with Microsoft having one of the most extensive. GitHub also supports numerous open source projects, and its online community is one of the biggest, earning a top score. The roadmap heavily weighted developer productivity but was missing key enterprise needs for governance.

Microsoft’s developer experience gets top scores for GitHub Actions, which turned GitHub into a CI leader almost overnight; Codespaces, a virtual dev environment that requires no setup by developers to get started; and Copilot, the AI-powered coding assistant. It also received top scores for OOTB security tools such as Dependabot. Support for APIs and third-party tools was also top scoring, with numerous ways to incorporate not only software but also actions. Pipeline security and platform cohesion were on par, while administration and management were missing the granular permissions of those scoring higher. Microsoft data and analytics were mostly operational, resulting in a below-par score. Deployment automation was also below par — it lacked blue/green and canary deployment scenarios. Reference customers were mostly positive about Microsoft, noting, “Generally, adoption of standard features is quick and doesn’t require a great deal of time to become productive.” Microsoft is great for organizations looking to boost developer effectiveness with cutting-edge tools.

- **Atlassian gets top scores for cohesiveness, but pipeline security capabilities are lacking.** Atlassian is best known for Jira, one of the most popular agile planning

tools on the market serving 80% of the Fortune 500. Atlassian's product strategy is on par and strives to serve two audiences, teams of developers and platform teams, by building out a data layer to connect the tools together. Its partner ecosystem is top-notch. Innovation is on par and includes platform capabilities and a data lake component to capture and aggregate data for greater analytical insight. The roadmap reflects enterprise concerns with plans to aggregate third-party security data, automatically infuse issue tickets with context, provide greater VSM insights, and expand support plans for DevOps and platform teams.

Atlassian's Confluence, which enables teams to create custom developer onboarding content and a hub for team collaboration, enhances the developer experience. This plus integrated development environment plug-ins give Atlassian a solid on-par score for developer experience. The OOTB build and CI were on par, with reusable build templates and parallel build execution to reduce build times. Deployment automation was below par, as were capabilities to secure pipelines. OOTB security tools are few, earning a below-par score, because Atlassian prefers to leverage its partner ecosystem for those. A reference customer said, "We chose Atlassian for the product features and the maturity of the well-integrated suites of products and plug-ins, which help simplify using the well-designed solution." Atlassian is great fit for software teams and enterprise platform teams that are looking for a standard ISDP approach but want to leverage best-of-breed CD and security components.

- **Harness ISDP delivers on platform cohesion, but community contributions are nascent.** Harness has a SaaS ISDP with core CI/CD modules and cloud cost management capabilities. Its vision is to simplify software development with an end-to-end software delivery platform that offers a modern developer experience and automated governance and guardrails. The strategy has mostly on-par elements such as a growing partner ecosystem and investments in innovation. The only weak area was the smaller-than-average community program — those scoring higher had greater support for internal and external communities. The roadmap includes support for database development, test optimization, end-to-end visibility, and infusion of AI/ML throughout the platform.

Harness gets top scores for platform cohesion, delivering a great OOTB experience for first-time users. It also earned top scores in build automation that includes pipelines as code, reusable pipelines, resource management, support for test automation, and intelligent test automation that can drastically reduce build times. It supports multiple deployment strategies OOTB, including blue/green and

canary. In addition, Harness can pinpoint code that's caused pipeline errors, helping speed remediation. Analytics was also top scoring and offered dev, process, and operational metrics. Customization of the UI and some ISDP components is possible and easy, but the limited scope of those changes resulted in a below-par score. Other areas for improvement include expanding the number of third-party integrations and its footprint in the user and open source community. Customers appreciate Harness' "willingness to engage directly with our development teams." Harness is great for development teams that want to spend less time managing tools and more time developing code.

- **AWS is strong in administration and management, but dev metrics are a miss.**

Amazon Web Services (AWS) pioneered infrastructure as a service (IaaS) and offers nearly every possible form of it. Its ISDP product includes CodeBuild, CodeStar, CloudFormation, and several other SaaS offerings that are integrated on demand when users create their account. The overall strategy score was solid. Innovation scored on par for current delivery and upcoming features for helping developers code faster. The vision was on par, focusing mainly on CodeCatalyst that will enable AWS to make its developer platform more accessible to a broader set of end users with different levels of technical ability. AWS actively supports its user community and open source projects, getting an on-par score for community. The roadmap is also on par and includes the aforementioned CodeCatalyst, a product currently in beta that offers a preassembled and more opinionated ISDP to help customers accelerate project setup and collaborate more fluidly.

AWS shines with excellent administration and management capabilities that enable user permissions to be managed down to the API level. AWS supports all major authentication protocols including emerging standards such as SCIM. AWS gets top scores for governance and compliance that include OOTB compliance templates. Platform cohesion and developer experience are on par. Build automation would have scored higher had it offered test execution optimization. AWS got low scores for analytics and reports; other vendors offered better dashboards with more dev-related metrics. Release readiness also scored low. Reference customer sentiment was mostly positive, saying, "Security is a first-class citizen." AWS is great for companies that want speed, scale, and security hosted in the same environment as their public cloud infrastructure.

- **CircleCI provides outstanding build automation, but its vision does not go beyond CI/CD.** CircleCI was one of the first vendors to offer CI as SaaS. Over time, it added CD and extensibility features to make a more well-rounded ISDP offering. CircleCI's on-par vision to speed time to value while simplifying the ISDP

experience is well articulated and strikes a pragmatic balance between individual developer concerns, the needs of platform teams, and support for enterprises in regulated industries. CircleCI's strategic vision is to remain a best-of-breed CI/CD tool vendor and not extend beyond that into other ISDP areas, resulting in a below-par score. The on-par roadmap aims at enterprise teams and includes features to help platform teams better manage an ISDP at scale with pervasive validation at each step of software delivery.

CircleCI has invested in optimized build processes, giving it a top score for build automation and CI. Its support for a wide range of platforms, including all public clouds and even mainframe, earns above-par scores. IaC capabilities were also top scoring. Product support leverages dedicated customer success engineers for each account, but average reference customer scores kept it on par. CircleCI user identities are derived from third-party VCS identities (e.g., those managed by GitHub or Bitbucket) and thus received a low score for administration and management. However, a new identity management system is on the roadmap to address this shortcoming. CircleCI has a good set of OOTB metrics and analytics, including DORA metrics, but the dashboards and capabilities were not as extensive as those scoring higher. One reference customer listed the following capabilities as most appreciated: "parallelization of workflows and rerunning only failed parts of workflows." CircleCI is a great fit for dev teams that value team autonomy and superfast build automation.

- **Codefresh has differentiating CI/CD but needs to improve governance and compliance.** Codefresh provides high-scoring CI/CD capabilities that emphasize the GitOps methodology, which requires all changes to infrastructure to be stored and managed as code in a Git repository prior to deployment. This methodology gained popularity from the Argo CD open source project, which powers Codefresh. Codefresh is also a registered project maintainer and contributor, which factored into its on-par community score. The Codefresh vision and roadmap focus on improvements to the developer experience and deployment capabilities such as Kubernetes context injection, improved disk utilization, and better support for teams. Innovation is confined to Argo and GitOps improvements, earning on-par scores. The one below-par score was for its smaller-than-average partner ecosystem.

Codefresh earned above-par scores for build automation, CI, and deployment automation featuring unique pipeline debugging capabilities that enable users to step in and out of containerized build steps, which no other vendor demonstrated.

Product support was strong, earning on-par scores. IaC provides strong support for Kubernetes configurations, but support for other legacy platforms lacks, keeping this score at par. Developer experience is on par; vendors scoring higher had integrated knowledge management. Other on-par scores include pipeline security, administration and management, and analytics that incorporate OOTB DORA metrics. Codefresh earned below-par scores for governance and compliance (providing limited support for ITSM integration) and support for functional test automation. When asked what was best about Codefresh, one customer said, “easy, standardized deployments for a hybrid universe.” Codefresh is a great fit for organizations that have adopted GitOps and place high value on developer autonomy.

Contenders

- **Flexagon has excellent customer support, but developer experience was below par.** Flexagon builds an end-to-end ISDP that supports traditional forms of software and infrastructure (e.g., Java and Linux) as well as business platforms such as Salesforce, Oracle, and SAP. The product vision is clear and focuses on delivering enterprise value through security and lowering risk. Planned updates include managed Oracle Cloud services, SAP blueprints, and a compliance and policy engine. Innovation is below par, investment in leading-edge areas is lacking, and the partner ecosystem was small, compared with others in this evaluation. Flexagon pricing features multiple tiers and full transparency of pricing options. Flexagon got above-par scores for test automation because it was one of the few vendors that featured native functional test automation capabilities and support.

Flexagon offers good platform cohesion with its offering that includes everything from agile planning to release automation in one package. It would have scored higher if not for some complexity in setting up and operating fleets of build runners. Flexagon got high scores for version control, with its support for the largest number of VCSes including Git and non-Git systems, and environment management that enabled teams to easily operate and manage any number of environments to support engineering teams. Deployment automation, release readiness, and platform support were also solid. Dev experience was not a focus area and resulted in a low score for that capability. Reference customers gave Flexagon perfect high scores for customer support, with one stating, “Flexagon continues to knock our socks off.” Flexagon is a great fit for enterprises looking for DevOps capabilities that span both open systems and business systems.

- **VMware delivers strong pipeline security but is light on data and analytics.**

VMware is best known for its infrastructure virtualization software. Tanzu Application Platform is VMware's ISDP and has a vision that is broad and incorporates elements of sociological factors to help developers be happier and more productive. VMware scored on par for community with strong support for open source tools and its user community. Innovation includes investments in autoremediation and AI/ML, earning an on-par score. The on-par roadmap will help users simplify the product lifecycle, improve developer collaboration, provide cost governance, and have more secure supply chains.

VMware pipeline security is one of the best in the evaluation with signed pipelines, signed commits, and additional security capabilities throughout the pipeline. Governance and compliance were on par, with OOTB compliance policies. Platform cohesion was also good with minimal setup required. Developer experience included a built-in knowledge management system to help onboard new developers and scored on par. Deployment automation included blue/green and canary deployment strategies, earning an on-par score. Administration and management were below par because they were missing features of those scoring higher. Data and analytics were few, and platform support seemed limited to Kubernetes workloads. One reference customer remarked, "The vendor is always willing to accept and act on feedback." VMware is a great fit for enterprises operating on Kubernetes and places high value on pipeline security.

- **OpenText has top-rated platform support, but DevSecOps capabilities score**

low. OpenText is best known for its portfolio of software development tools ranging from portfolio to test automation. OpenText's ValueEdge is an open platform that bundles these capabilities together in one ISDP. Its roadmap is on par, featuring elements that will better connect business and development. Innovation features AI/ML infused in risk management and testing, and OpenText boasts a healthy partner ecosystem. Both areas scored on par, with good evidence of progress from its AI/ML investments. But its vision focused more on business concerns than stronger CI/CD capabilities. In addition, pricing is opaque, compared with those scoring higher.

OpenText has one of the strongest governance and compliance offerings of this evaluation because nearly everything connects to change management systems. It has superior support for test automation capabilities that include autotest classification of test failures. In addition to providing its own VCS, users can connect with virtually any other VCS, earning above-par scores for that capability. Support for a

multitude of platforms including mainframe was a plus. Administration, management, and developer experience were on par, as were data and analytics that featured DORA and flow metrics. Deployment automation was also on par with blue/green and canary deployment support. Pipeline security lagged others evaluated. IaC, an area of growing importance to developers, also scored low. OpenText reference customers were positive about their experience, with one stating, “We don’t have any challenges at this stage.” OpenText is a great fit for enterprises with varied platforms in need of strong ties to test automation and compliance.

- **JFrog provides differentiated pipeline security capabilities, but the core ISDP falls short.** JFrog is best known for its flagship product, Artifactory, a binary artifact repository with a name that is so well known that it is synonymous with artifact repositories in general. JFrog’s strategy includes strong on-par scores for innovation based on investments in helping secure enterprise software supply chains. JFrog has a strong user community and actively participates in open source projects. Pricing flexibility was also on par, offering several payment tiers. JFrog scored below par for vision because it seemed overly focused on security at the expense of other ISDP capabilities. The roadmap was also below par due to its limited scope.

JFrog has superior version control due to the strength of Artifactory. Pipeline security is also top-notch with signed artifacts and signed pipelines that provide some of the strongest supply chain security measures available. The JFrog platform has good cohesion and requires few steps to set up and get started. Administration and management are on par and offer LDAP, OAuth, and support for SCIM to manage users at scale. Xray, a security product that scans artifacts, comes bundled and offers container and image scanning along with OSS supply chain management. To make this a more competitive ISDP offering, JFrog needs to invest in release readiness, data, and analytics and expand policy as code to all aspects of the ISDP while doing more to support test automation. Customers like how they are treated by JFrog, saying: It’s a “partner, not a vendor — they bring this mentality to everything.” JFrog is great for enterprises that place high value on supply chain security.

Challengers

- **IBM UrbanCode Deploy offers solid build and deployment capabilities but looks dated.** IBM UrbanCode offers UrbanCode Build and UrbanCode Deploy as two components that work together to deliver an enterprise-scale ISDP. IBM has a

solid development community because IBM Garage has thousands of contributors. IBM's partner ecosystem is also solidly on par with a number of top systems integrators listed as partners. The strategy has several weak elements including a roadmap that emphasizes maintenance versus new features, and innovation proof points are lacking.

IBM has solid build automation, deployment automation, release readiness, and support for a large variety of target platforms, which are core ISDP capabilities valued by enterprise dev teams. IBM's UrbanCode Velocity module offers strong analytics and graphics, earning it on-par scores for analytics, and there's a healthy number of third-party integrations for an on-par score in that category. But platform cohesion suffers from a complicated installation process, administration and management offer limited capabilities such as LDAP, and the developer experience is bare bones. Pipelines are present but don't offer the same level of capability as more modern tools. IBM is a good fit for enterprises that need to build an ISDP bridge between legacy and modern infrastructure. IBM declined to participate in the full Forrester Wave evaluation process.

Evaluation Overview

We grouped our evaluation criteria into three high-level categories:

- **Current offering.** Each vendor's position on the vertical axis of the Forrester Wave graphic indicates the strength of its current offering. Key criteria for these solutions include platform capabilities to facilitate management at scale, CI/CD to perform core ISDP functions, extensibility to enable platform operators to incorporate third-party tools as needed, and integrated systems to secure the software delivery pipeline.
- **Strategy.** Placement on the horizontal axis indicates the strength of the vendors' strategies. We evaluated vision, roadmap, and innovation to determine which vendor was placing the best bets to fulfill long-term product promises.
- **Market presence.** Represented by the size of the markers on the graphic, our market presence scores reflect each vendor's ISDP revenue and number of enterprise customers.

Vendor Inclusion Criteria

Each of the vendors we included in this assessment has:

- **Integrated CI/CD (e.g., deployment and release automation) capabilities.** The vendor provides integrated CI/CD capabilities OOTB that enable teams to rapidly establish build and deploy processes with minimal configuration.
- **Extensibility via APIs or plug-ins.** The vendor provides APIs to support third-party tool integration to extend the platform to incorporate performance, test automation, and security.
- **A system for centralized administration.** A single dashboard helps platform administrators operate and maintain an ISDP at enterprise scale.
- **Strong client interest.** Analyst inquiry logs indicate the vendor is top of mind among Forrester's end-user clientele with regard to ISDPs.
- **General-purpose programming languages and platforms.** The vendor supports build and deployment automation for open source programming languages like C# and Java as well as deployment to general-purpose runtime platforms such as Linux or Windows. It is also not exclusively tied to any particular business platform such as Salesforce or SAP.
- **High market performance.** The vendor's market performance demonstrates that it is a significant player in the ISDP market.

Supplemental Material

Online Resource

We publish all our Forrester Wave scores and weightings in an Excel file that provides detailed product evaluations and customizable rankings; download this tool by clicking the link at the beginning of this report on Forrester.com. We intend these scores and default weightings to serve only as a starting point and encourage readers to adapt the weightings to fit their individual needs.

The Forrester Wave Methodology

A Forrester Wave is a guide for buyers considering their purchasing options in a technology marketplace. To offer an equitable process for all participants, Forrester follows [The Forrester Wave™ Methodology](#) to evaluate participating vendors.

In our review, we conduct primary research to develop a list of vendors to consider for the evaluation. From that initial pool of vendors, we narrow our final list based on the inclusion criteria. We then gather details of product and strategy through a detailed questionnaire, demos/briefings, and customer reference surveys/interviews. We use those inputs, along with the analyst's experience and expertise in the marketplace, to

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score vendors, using a relative rating system that compares each vendor against the others in the evaluation.

We include the Forrester Wave publishing date (quarter and year) clearly in the title of each Forrester Wave report. We evaluated the vendors participating in this Forrester Wave using materials they provided to us by March 2023 and did not allow additional information after that point. We encourage readers to evaluate how the market and vendor offerings change over time.

In accordance with [our vendor review policy](#), Forrester asks vendors to review our findings prior to publishing to check for accuracy. Vendors marked as nonparticipating vendors in the Forrester Wave graphic met our defined inclusion criteria but declined to participate in or contributed only partially to the evaluation. We score these vendors in accordance with [our vendor participation policy](#) and publish their positioning along with those of the participating vendors.

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WAVE REPORT

The Forrester WaveTM: Integrated Software Delivery Platforms, Q2 2023

The 13 Providers That Matter Most And How They
Stack Up

May 31, 2023

By Christopher Condo with Chris Gardner, Sara Sjoblom, Kara Hartig

FORRESTER®

Summary

In our 26-criterion evaluation of integrated software delivery platform (ISDP) providers, we identified the most significant ones and researched, analyzed, and scored them. This report shows how each provider measures up and helps modern application development leaders and DevOps platform team leaders select the right one for their needs.

Additional resources are available in the [online version](#) of this report.

ISDPs Provide A Strong Platform Foundation But Still Need Best-Of-Breed Tools

In our reports, [The Future Of DevOps](#) and [The Forrester Guide To DevOps](#), Forrester interviewed dozens of enterprise software development leaders from diverse companies ranging from Netflix to TransUnion to understand the current state of DevOps, its relevance, and next steps enterprise leaders were taking to further advance their practices. In addition, we asked which tools they were using, how effective the tools were, which were using [ISDPs](#), and how well those ISDPs were performing. We learned that many enterprises have shifted to platform teams to support their DevOps efforts and many platform leaders are evaluating and using ISDPs, but few are using all of their ISDP vendor's capabilities. They explained that while these ISDP tools hold great potential for a complete end-to-end toolchain, many individual capabilities fell short of enterprise needs, forcing these leaders to source those capabilities from separate best-of-breed vendors.

Our reference customer survey backs this up by indicating that enterprises are using (on average) only 53% of the capabilities they are paying for, with the remaining capabilities either being delivered by a separate vendor or simply not used. This means that ISDP vendors are not only competing with each other but also competing with the best-of-breeds, which is why the bar has risen so high for this evaluation. ISDP vendors recognize these challenges and are responding with a number of innovative solutions that will make them more attractive than a traditional DIY toolchain but open and extensible enough that end users can add the tools they want without sacrificing capability. The challenge is striking the right balance to create overall value.

As a result of these trends, ISDP customers should look for providers that:

- **Offer a strong core of CI/CD capabilities.** Continuous integration (CI) and continuous delivery (CD) are the core capabilities that the rest of the software delivery toolchain is built around. Several vendors offered build optimization algorithms that slash build and unit test times by intelligently building and testing only the code that changed. Deployment automation is also a differentiating capability, with only a few vendors offering advanced capabilities out of the box (OOTB) such as blue/green, canary, and feature management deployment strategies.
- **Provide connected platforms that best-of-breed tools can augment.** ISDPs provide engineering leverage in the form of preintegrated toolchains that enable work to easily flow from one process to the next, where administration occurs at

the platform level versus the tool level, and reduced context switching improves the developer experience. Often, platform leaders want to override the default capability with third-party tools that they believe offer superior capability, which can lead to a bit of a balancing act. That's why some ISDP vendors are heavily investing in improvements to their end-to-end platform, while others are banking on investments in third-party connectivity and partnerships.

- **Help enterprise leaders shift from DevOps to DevSecOps.** Forrester fields calls every day from enterprise development leaders that are transforming their DevOps practice into a development, security, and operations (DevSecOps) practice to make security a first-class concern for the entire development organization. ISDP vendors are responding with capabilities that better secure software delivery pipelines, including integrated secrets management, signed pipelines, and signed artifacts. Reference customers noted that not all of these tools are direct replacements for best-of-breed (best-of-breed security tools often having a significant advantage over bundled ones), but end users like them as forms of additional assurance and coverage.

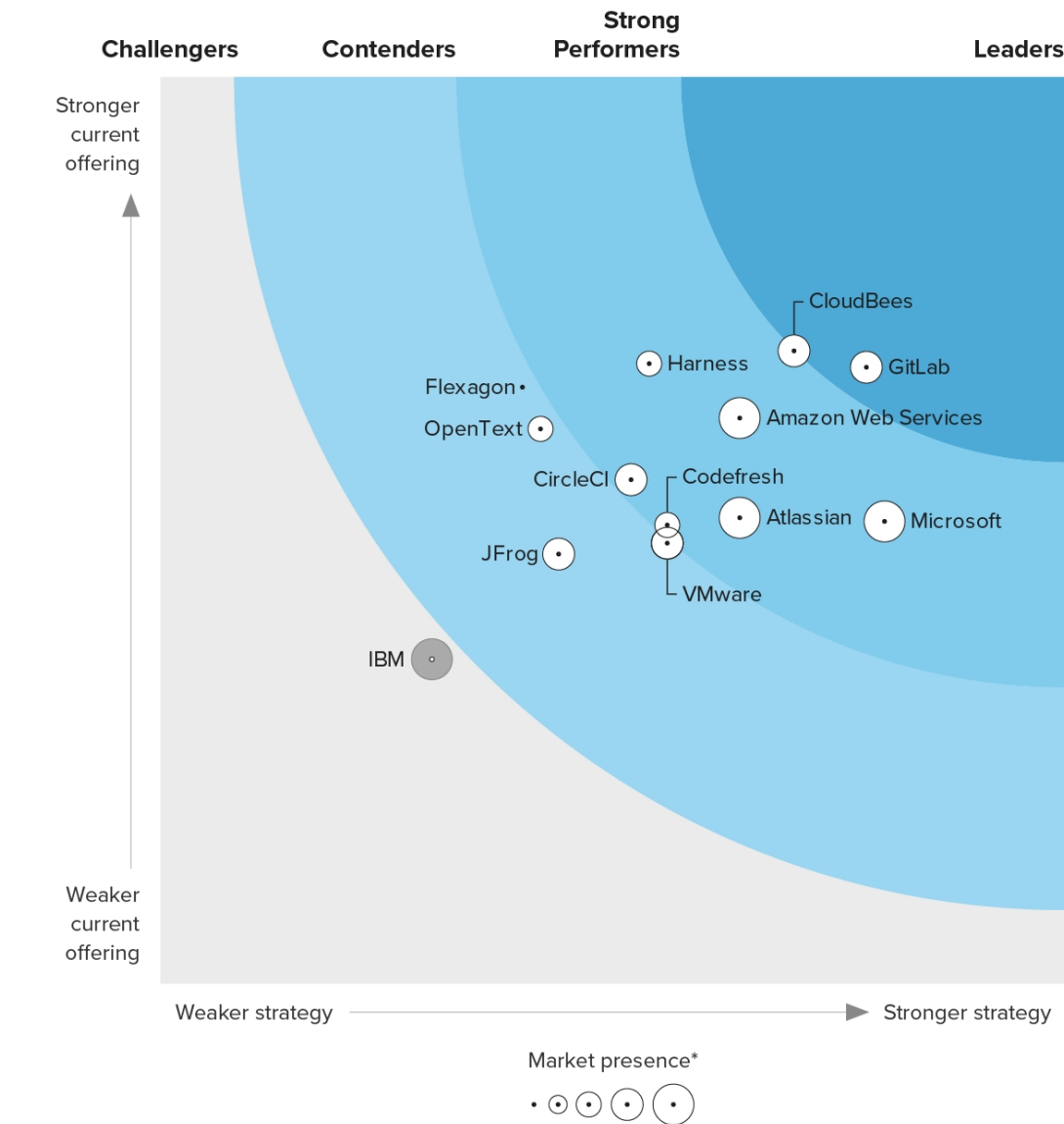
Evaluation Summary

The Forrester Wave™ evaluation highlights Leaders, Strong Performers, Contenders, and Challengers. It's an assessment of the top vendors in the market; it doesn't represent the entire vendor landscape. You'll find more information about this market in Forrester's [Now Tech: Integrated Software Delivery Platforms, Q2 2022](#).

We intend this evaluation to be a starting point only and encourage clients to view product evaluations and adapt criteria weightings using the Excel-based vendor comparison tool (see Figures 1 and 2). Click the link at the beginning of this report on [Forrester.com](#) to download the tool.

Figure 1
Forrester Wave™: Integrated Software Delivery Platforms, Q2 2023

THE FORRESTER WAVE™
Integrated Software Delivery Platforms
Q2 2023



*A gray bubble or open dot indicates a nonparticipating vendor.

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Figure 2**Forrester Wave™: Integrated Software Delivery Platforms Scorecard, Q2 2023**

	Forrester's weighting	Amazon Web Services	Atlassian	CircleCI	CloudBees	Codefresh	Flexagon	GitLab
Current offering	50%	2.57	3.12	2.78	3.49	2.53	3.29	3.40
Platform capabilities	20%	3.50	3.00	2.00	2.50	3.00	2.50	3.50
CI/CD capabilities	25%	2.00	2.60	3.00	4.00	4.00	3.20	3.40
Development, security, and operations	10%	1.00	3.00	2.00	2.00	2.00	3.00	4.00
Extensibility	10%	3.00	3.40	3.00	3.40	1.00	3.00	3.00
Governance and compliance	10%	3.00	5.00	3.00	5.00	1.00	3.00	3.00
Infrastructure management	10%	2.20	3.80	3.80	5.00	1.80	3.40	3.00
Product support	10%	3.00	3.00	3.00	3.00	3.00	5.00	3.00
Support for test automation	5%	3.00	1.00	3.00	3.00	1.00	5.00	5.00
Strategy	50%	3.20	3.20	2.60	3.50	2.80	2.00	3.90
Vision	20%	3.00	3.00	1.00	3.00	3.00	3.00	3.00
Roadmap	20%	3.00	3.00	3.00	3.00	3.00	3.00	5.00
Partner ecosystem	10%	5.00	5.00	1.00	5.00	1.00	1.00	3.00
Innovation	25%	3.00	3.00	3.00	3.00	3.00	1.00	3.00
Community	15%	3.00	3.00	3.00	5.00	3.00	1.00	5.00
Pricing flexibility and transparency	10%	3.00	3.00	5.00	3.00	3.00	3.00	5.00
Market presence	0%	5.00	4.50	3.50	3.50	2.50	1.00	3.50
ISDP revenue	50%	5.00	5.00	4.00	4.00	2.00	1.00	4.00
Number of enterprise customers	50%	5.00	4.00	3.00	3.00	3.00	1.00	3.00

All scores are based on a scale of 0 (weak) to 5 (strong).

Source: Forrester Research, Inc. Unauthorized reproduction, citation, or distribution prohibited.

	Forrester's weighting	Harness	IBM*	JFrog	Microsoft	OpenText	VMware
Current offering	50%	3.42	1.79	2.37	2.55	3.06	2.43
Platform capabilities	20%	4.00	1.50	2.00	3.00	2.50	2.00
CI/CD capabilities	25%	4.20	2.80	2.40	2.40	3.40	2.40
Development, security, and operations	10%	3.00	1.00	4.00	4.00	1.00	3.00
Extensibility	10%	1.00	1.40	4.20	4.20	3.40	3.00
Governance and compliance	10%	3.00	3.00	1.00	1.00	5.00	3.00
Infrastructure management	10%	4.20	1.00	1.00	1.80	2.20	1.80
Product support	10%	3.00	1.00	3.00	1.00	3.00	3.00
Support for test automation	5%	3.00	1.00	1.00	3.00	5.00	1.00
Strategy	50%	2.70	1.50	2.20	4.00	2.10	2.80
Vision	20%	3.00	1.00	1.00	3.00	1.00	3.00
Roadmap	20%	3.00	1.00	1.00	3.00	3.00	3.00
Partner ecosystem	10%	3.00	3.00	3.00	5.00	3.00	3.00
Innovation	25%	3.00	1.00	3.00	5.00	3.00	3.00
Community	15%	1.00	3.00	3.00	5.00	1.00	3.00
Pricing flexibility and transparency	10%	3.00	1.00	3.00	3.00	1.00	1.00
Market presence	0%	3.00	4.50	4.00	5.00	2.50	4.00
ISDP revenue	50%	3.00	5.00	4.00	5.00	4.00	4.00
Number of enterprise customers	50%	3.00	4.00	4.00	5.00	1.00	4.00

All scores are based on a scale of 0 (weak) to 5 (strong).

*Indicates a nonparticipating vendor

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Vendor Offerings

Forrester evaluated the offerings listed below (see Figure 3).

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Figure 3
Evaluated Vendors And Product Information

Vendor	Product evaluated
Amazon Web Services	AWS Code Services
Atlassian	Atlassian Open DevOps
CircleCI	CircleCI
CloudBees	CloudBees Platform
Codefresh	Codefresh
Flexagon	Flexagon FlexDeploy
GitLab	GitLab
Harness	Harness Software Delivery Platform
IBM	IBM UrbanCode Deploy
JFrog	JFrog Software Supply Chain Platform
Microsoft	Microsoft Azure DevOps and GitHub Enterprise
OpenText (formerly Micro Focus)	Micro Focus ValueEdge
VMware	VMware Tanzu Application Platform

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Vendor Profiles

Our analysis uncovered the following strengths and weaknesses of individual vendors.

Leaders

- **GitLab leads with a balance between solid platform capabilities and overall strategy.** GitLab is known for being the vendor that made end-to-end OOTB toolchains hip again and has led the industry toward consolidated ISDPs. GitLab’s strategy includes an on-par vision to deliver an excellent developer experience without sacrificing security or compliance. Its innovation is also good, going beyond traditional developers to include AI/ML engineering. GitLab is an open core product that not only invests heavily in the open source software (OSS) community but also enables its customers to contribute to the product, earning it

high scores for community. Pricing is completely transparent and includes a free tier. Its roadmap gets leading scores and includes enhanced supply chain security, enhanced UI, granular security and compliance controls, and pipeline security — all things enterprises need.

GitLab's platform cohesion received high scores for simplicity of setup and operation as both an on-premises product and a software-as-a-service (SaaS) product. Though GitLab comes with its own GitLab version control system (VCS), it supports complete integration with all major Git providers, earning it high scores for version control. GitLab gets high scores for DevSecOps by providing OOTB security tools for static application security testing (SAST), dynamic application security testing (DAST), and Secret Detection. GitLab is one of the few ISDP vendors to incorporate a test automation dashboard, earning a top score. Dev experience is on par, as are administration and management, deployment automation, and release readiness. Build automation, governance and compliance, and infrastructure as code (IaC) were on par with others. One reference customer commented, "The CI/CD experience using secrets, environments, runners, and SAST/DAST/license scans/etc. is unparalleled in my opinion." GitLab is great for enterprises wishing to consolidate their best-of-breed toolchain into one high-performing ISDP.

Strong Performers

- **CloudBees delivers on its core promise of CI/CD, but platform cohesion needs work.** CloudBees is primarily associated with Jenkins; this well-known open source CI tool that it donated to the Continuous Delivery Foundation is one of the original best-of-breed tools that accompanied the beginning of the DevOps era more than 10 years ago. CloudBees leverages open source Jenkins and augments that with enterprise-scale deployment automation, release readiness, and value stream management (VSM) analytics to provide an enterprise-class ISDP. It got high scores for community, with significant contributions to open source projects and cultivation of its own user community. Its strategy features a solid vision and one of the larger partner ecosystems in this market. Innovation was on par and includes efforts to improve codification of policies. A planned SaaS offering should address its platform cohesion weakness and will help CloudBees serve clients that want a more integrated and maintenance-free ISDP.

CloudBees deployment automation is above par, covering blue/green, canary, and powerful feature management capabilities that are third-party add-ons for most

other platforms. CloudBees release readiness, governance, and compliance are also top-notch with complete integration with popular IT service management (ITSM) tools such as ServiceNow that bring those UIs directly into the ISDP experience. Build automation requires a bit of effort to manage fleets of build runners, and pipeline security needs to go beyond branch security to earn higher scores. When asked what they liked about working with CloudBees, reference customers had good things to say: “The vendor relationship has been a true partnership. It’s listening to our internal customers.” CloudBees is a great fit for enterprise platform teams that value open source and open choice.

- **Microsoft differentiates with innovation, but governance and compliance need attention.** Microsoft acquired GitHub, the industry’s largest open source registry, in 2018. Since that time, Microsoft’s Azure DevOps and GitHub have slowly been moving toward a unified offering. However, both the product vision and the roadmap could have been clearer on how this will shape future offerings, resulting in an on-par score. Innovation strategy is tops because GitHub Copilot has set new standards for the development tools industry and planned investments will enable it to continue this leadership position. Its partner ecosystem also scored high, with Microsoft having one of the most extensive. GitHub also supports numerous open source projects, and its online community is one of the biggest, earning a top score. The roadmap heavily weighted developer productivity but was missing key enterprise needs for governance.

Microsoft’s developer experience gets top scores for GitHub Actions, which turned GitHub into a CI leader almost overnight; Codespaces, a virtual dev environment that requires no setup by developers to get started; and Copilot, the AI-powered coding assistant. It also received top scores for OOTB security tools such as Dependabot. Support for APIs and third-party tools was also top scoring, with numerous ways to incorporate not only software but also actions. Pipeline security and platform cohesion were on par, while administration and management were missing the granular permissions of those scoring higher. Microsoft data and analytics were mostly operational, resulting in a below-par score. Deployment automation was also below par — it lacked blue/green and canary deployment scenarios. Reference customers were mostly positive about Microsoft, noting, “Generally, adoption of standard features is quick and doesn’t require a great deal of time to become productive.” Microsoft is great for organizations looking to boost developer effectiveness with cutting-edge tools.

- **Atlassian gets top scores for cohesiveness, but pipeline security capabilities are lacking.** Atlassian is best known for Jira, one of the most popular agile planning

tools on the market serving 80% of the Fortune 500. Atlassian's product strategy is on par and strives to serve two audiences, teams of developers and platform teams, by building out a data layer to connect the tools together. Its partner ecosystem is top-notch. Innovation is on par and includes platform capabilities and a data lake component to capture and aggregate data for greater analytical insight. The roadmap reflects enterprise concerns with plans to aggregate third-party security data, automatically infuse issue tickets with context, provide greater VSM insights, and expand support plans for DevOps and platform teams.

Atlassian's Confluence, which enables teams to create custom developer onboarding content and a hub for team collaboration, enhances the developer experience. This plus integrated development environment plug-ins give Atlassian a solid on-par score for developer experience. The OOTB build and CI were on par, with reusable build templates and parallel build execution to reduce build times. Deployment automation was below par, as were capabilities to secure pipelines. OOTB security tools are few, earning a below-par score, because Atlassian prefers to leverage its partner ecosystem for those. A reference customer said, "We chose Atlassian for the product features and the maturity of the well-integrated suites of products and plug-ins, which help simplify using the well-designed solution." Atlassian is great fit for software teams and enterprise platform teams that are looking for a standard ISDP approach but want to leverage best-of-breed CD and security components.

- **Harness ISDP delivers on platform cohesion, but community contributions are nascent.** Harness has a SaaS ISDP with core CI/CD modules and cloud cost management capabilities. Its vision is to simplify software development with an end-to-end software delivery platform that offers a modern developer experience and automated governance and guardrails. The strategy has mostly on-par elements such as a growing partner ecosystem and investments in innovation. The only weak area was the smaller-than-average community program — those scoring higher had greater support for internal and external communities. The roadmap includes support for database development, test optimization, end-to-end visibility, and infusion of AI/ML throughout the platform.

Harness gets top scores for platform cohesion, delivering a great OOTB experience for first-time users. It also earned top scores in build automation that includes pipelines as code, reusable pipelines, resource management, support for test automation, and intelligent test automation that can drastically reduce build times. It supports multiple deployment strategies OOTB, including blue/green and

canary. In addition, Harness can pinpoint code that's caused pipeline errors, helping speed remediation. Analytics was also top scoring and offered dev, process, and operational metrics. Customization of the UI and some ISDP components is possible and easy, but the limited scope of those changes resulted in a below-par score. Other areas for improvement include expanding the number of third-party integrations and its footprint in the user and open source community. Customers appreciate Harness' "willingness to engage directly with our development teams." Harness is great for development teams that want to spend less time managing tools and more time developing code.

- **AWS is strong in administration and management, but dev metrics are a miss.**

Amazon Web Services (AWS) pioneered infrastructure as a service (IaaS) and offers nearly every possible form of it. Its ISDP product includes CodeBuild, CodeStar, CloudFormation, and several other SaaS offerings that are integrated on demand when users create their account. The overall strategy score was solid. Innovation scored on par for current delivery and upcoming features for helping developers code faster. The vision was on par, focusing mainly on CodeCatalyst that will enable AWS to make its developer platform more accessible to a broader set of end users with different levels of technical ability. AWS actively supports its user community and open source projects, getting an on-par score for community. The roadmap is also on par and includes the aforementioned CodeCatalyst, a product currently in beta that offers a preassembled and more opinionated ISDP to help customers accelerate project setup and collaborate more fluidly.

AWS shines with excellent administration and management capabilities that enable user permissions to be managed down to the API level. AWS supports all major authentication protocols including emerging standards such as SCIM. AWS gets top scores for governance and compliance that include OOTB compliance templates. Platform cohesion and developer experience are on par. Build automation would have scored higher had it offered test execution optimization. AWS got low scores for analytics and reports; other vendors offered better dashboards with more dev-related metrics. Release readiness also scored low. Reference customer sentiment was mostly positive, saying, "Security is a first-class citizen." AWS is great for companies that want speed, scale, and security hosted in the same environment as their public cloud infrastructure.

- **CircleCI provides outstanding build automation, but its vision does not go beyond CI/CD.** CircleCI was one of the first vendors to offer CI as SaaS. Over time, it added CD and extensibility features to make a more well-rounded ISDP offering. CircleCI's on-par vision to speed time to value while simplifying the ISDP

experience is well articulated and strikes a pragmatic balance between individual developer concerns, the needs of platform teams, and support for enterprises in regulated industries. CircleCI's strategic vision is to remain a best-of-breed CI/CD tool vendor and not extend beyond that into other ISDP areas, resulting in a below-par score. The on-par roadmap aims at enterprise teams and includes features to help platform teams better manage an ISDP at scale with pervasive validation at each step of software delivery.

CircleCI has invested in optimized build processes, giving it a top score for build automation and CI. Its support for a wide range of platforms, including all public clouds and even mainframe, earns above-par scores. IaC capabilities were also top scoring. Product support leverages dedicated customer success engineers for each account, but average reference customer scores kept it on par. CircleCI user identities are derived from third-party VCS identities (e.g., those managed by GitHub or Bitbucket) and thus received a low score for administration and management. However, a new identity management system is on the roadmap to address this shortcoming. CircleCI has a good set of OOTB metrics and analytics, including DORA metrics, but the dashboards and capabilities were not as extensive as those scoring higher. One reference customer listed the following capabilities as most appreciated: "parallelization of workflows and rerunning only failed parts of workflows." CircleCI is a great fit for dev teams that value team autonomy and superfast build automation.

- **Codefresh has differentiating CI/CD but needs to improve governance and compliance.** Codefresh provides high-scoring CI/CD capabilities that emphasize the GitOps methodology, which requires all changes to infrastructure to be stored and managed as code in a Git repository prior to deployment. This methodology gained popularity from the Argo CD open source project, which powers Codefresh. Codefresh is also a registered project maintainer and contributor, which factored into its on-par community score. The Codefresh vision and roadmap focus on improvements to the developer experience and deployment capabilities such as Kubernetes context injection, improved disk utilization, and better support for teams. Innovation is confined to Argo and GitOps improvements, earning on-par scores. The one below-par score was for its smaller-than-average partner ecosystem.

Codefresh earned above-par scores for build automation, CI, and deployment automation featuring unique pipeline debugging capabilities that enable users to step in and out of containerized build steps, which no other vendor demonstrated.

Product support was strong, earning on-par scores. IaC provides strong support for Kubernetes configurations, but support for other legacy platforms lacks, keeping this score at par. Developer experience is on par; vendors scoring higher had integrated knowledge management. Other on-par scores include pipeline security, administration and management, and analytics that incorporate OOTB DORA metrics. Codefresh earned below-par scores for governance and compliance (providing limited support for ITSM integration) and support for functional test automation. When asked what was best about Codefresh, one customer said, “easy, standardized deployments for a hybrid universe.” Codefresh is a great fit for organizations that have adopted GitOps and place high value on developer autonomy.

Contenders

- **Flexagon has excellent customer support, but developer experience was below par.** Flexagon builds an end-to-end ISDP that supports traditional forms of software and infrastructure (e.g., Java and Linux) as well as business platforms such as Salesforce, Oracle, and SAP. The product vision is clear and focuses on delivering enterprise value through security and lowering risk. Planned updates include managed Oracle Cloud services, SAP blueprints, and a compliance and policy engine. Innovation is below par, investment in leading-edge areas is lacking, and the partner ecosystem was small, compared with others in this evaluation. Flexagon pricing features multiple tiers and full transparency of pricing options. Flexagon got above-par scores for test automation because it was one of the few vendors that featured native functional test automation capabilities and support.

Flexagon offers good platform cohesion with its offering that includes everything from agile planning to release automation in one package. It would have scored higher if not for some complexity in setting up and operating fleets of build runners. Flexagon got high scores for version control, with its support for the largest number of VCSes including Git and non-Git systems, and environment management that enabled teams to easily operate and manage any number of environments to support engineering teams. Deployment automation, release readiness, and platform support were also solid. Dev experience was not a focus area and resulted in a low score for that capability. Reference customers gave Flexagon perfect high scores for customer support, with one stating, “Flexagon continues to knock our socks off.” Flexagon is a great fit for enterprises looking for DevOps capabilities that span both open systems and business systems.

- **VMware delivers strong pipeline security but is light on data and analytics.**

VMware is best known for its infrastructure virtualization software. Tanzu Application Platform is VMware's ISDP and has a vision that is broad and incorporates elements of sociological factors to help developers be happier and more productive. VMware scored on par for community with strong support for open source tools and its user community. Innovation includes investments in autoremediation and AI/ML, earning an on-par score. The on-par roadmap will help users simplify the product lifecycle, improve developer collaboration, provide cost governance, and have more secure supply chains.

VMware pipeline security is one of the best in the evaluation with signed pipelines, signed commits, and additional security capabilities throughout the pipeline. Governance and compliance were on par, with OOTB compliance policies. Platform cohesion was also good with minimal setup required. Developer experience included a built-in knowledge management system to help onboard new developers and scored on par. Deployment automation included blue/green and canary deployment strategies, earning an on-par score. Administration and management were below par because they were missing features of those scoring higher. Data and analytics were few, and platform support seemed limited to Kubernetes workloads. One reference customer remarked, "The vendor is always willing to accept and act on feedback." VMware is a great fit for enterprises operating on Kubernetes and places high value on pipeline security.

- **OpenText has top-rated platform support, but DevSecOps capabilities score**

low. OpenText is best known for its portfolio of software development tools ranging from portfolio to test automation. OpenText's ValueEdge is an open platform that bundles these capabilities together in one ISDP. Its roadmap is on par, featuring elements that will better connect business and development. Innovation features AI/ML infused in risk management and testing, and OpenText boasts a healthy partner ecosystem. Both areas scored on par, with good evidence of progress from its AI/ML investments. But its vision focused more on business concerns than stronger CI/CD capabilities. In addition, pricing is opaque, compared with those scoring higher.

OpenText has one of the strongest governance and compliance offerings of this evaluation because nearly everything connects to change management systems. It has superior support for test automation capabilities that include autoclassification of test failures. In addition to providing its own VCS, users can connect with virtually any other VCS, earning above-par scores for that capability. Support for a

multitude of platforms including mainframe was a plus. Administration, management, and developer experience were on par, as were data and analytics that featured DORA and flow metrics. Deployment automation was also on par with blue/green and canary deployment support. Pipeline security lagged others evaluated. IaC, an area of growing importance to developers, also scored low. OpenText reference customers were positive about their experience, with one stating, “We don’t have any challenges at this stage.” OpenText is a great fit for enterprises with varied platforms in need of strong ties to test automation and compliance.

- **JFrog provides differentiated pipeline security capabilities, but the core ISDP falls short.** JFrog is best known for its flagship product, Artifactory, a binary artifact repository with a name that is so well known that it is synonymous with artifact repositories in general. JFrog’s strategy includes strong on-par scores for innovation based on investments in helping secure enterprise software supply chains. JFrog has a strong user community and actively participates in open source projects. Pricing flexibility was also on par, offering several payment tiers. JFrog scored below par for vision because it seemed overly focused on security at the expense of other ISDP capabilities. The roadmap was also below par due to its limited scope.

JFrog has superior version control due to the strength of Artifactory. Pipeline security is also top-notch with signed artifacts and signed pipelines that provide some of the strongest supply chain security measures available. The JFrog platform has good cohesion and requires few steps to set up and get started. Administration and management are on par and offer LDAP, OAuth, and support for SCIM to manage users at scale. Xray, a security product that scans artifacts, comes bundled and offers container and image scanning along with OSS supply chain management. To make this a more competitive ISDP offering, JFrog needs to invest in release readiness, data, and analytics and expand policy as code to all aspects of the ISDP while doing more to support test automation. Customers like how they are treated by JFrog, saying: It’s a “partner, not a vendor — they bring this mentality to everything.” JFrog is great for enterprises that place high value on supply chain security.

Challengers

- **IBM UrbanCode Deploy offers solid build and deployment capabilities but looks dated.** IBM UrbanCode offers UrbanCode Build and UrbanCode Deploy as two components that work together to deliver an enterprise-scale ISDP. IBM has a

solid development community because IBM Garage has thousands of contributors. IBM's partner ecosystem is also solidly on par with a number of top systems integrators listed as partners. The strategy has several weak elements including a roadmap that emphasizes maintenance versus new features, and innovation proof points are lacking.

IBM has solid build automation, deployment automation, release readiness, and support for a large variety of target platforms, which are core ISDP capabilities valued by enterprise dev teams. IBM's UrbanCode Velocity module offers strong analytics and graphics, earning it on-par scores for analytics, and there's a healthy number of third-party integrations for an on-par score in that category. But platform cohesion suffers from a complicated installation process, administration and management offer limited capabilities such as LDAP, and the developer experience is bare bones. Pipelines are present but don't offer the same level of capability as more modern tools. IBM is a good fit for enterprises that need to build an ISDP bridge between legacy and modern infrastructure. IBM declined to participate in the full Forrester Wave evaluation process.

Evaluation Overview

We grouped our evaluation criteria into three high-level categories:

- **Current offering.** Each vendor's position on the vertical axis of the Forrester Wave graphic indicates the strength of its current offering. Key criteria for these solutions include platform capabilities to facilitate management at scale, CI/CD to perform core ISDP functions, extensibility to enable platform operators to incorporate third-party tools as needed, and integrated systems to secure the software delivery pipeline.
- **Strategy.** Placement on the horizontal axis indicates the strength of the vendors' strategies. We evaluated vision, roadmap, and innovation to determine which vendor was placing the best bets to fulfill long-term product promises.
- **Market presence.** Represented by the size of the markers on the graphic, our market presence scores reflect each vendor's ISDP revenue and number of enterprise customers.

Vendor Inclusion Criteria

Each of the vendors we included in this assessment has:

- **Integrated CI/CD (e.g., deployment and release automation) capabilities.** The vendor provides integrated CI/CD capabilities OOTB that enable teams to rapidly establish build and deploy processes with minimal configuration.
- **Extensibility via APIs or plug-ins.** The vendor provides APIs to support third-party tool integration to extend the platform to incorporate performance, test automation, and security.
- **A system for centralized administration.** A single dashboard helps platform administrators operate and maintain an ISDP at enterprise scale.
- **Strong client interest.** Analyst inquiry logs indicate the vendor is top of mind among Forrester's end-user clientele with regard to ISDPs.
- **General-purpose programming languages and platforms.** The vendor supports build and deployment automation for open source programming languages like C# and Java as well as deployment to general-purpose runtime platforms such as Linux or Windows. It is also not exclusively tied to any particular business platform such as Salesforce or SAP.
- **High market performance.** The vendor's market performance demonstrates that it is a significant player in the ISDP market.

Supplemental Material

Online Resource

We publish all our Forrester Wave scores and weightings in an Excel file that provides detailed product evaluations and customizable rankings; download this tool by clicking the link at the beginning of this report on Forrester.com. We intend these scores and default weightings to serve only as a starting point and encourage readers to adapt the weightings to fit their individual needs.

The Forrester Wave Methodology

A Forrester Wave is a guide for buyers considering their purchasing options in a technology marketplace. To offer an equitable process for all participants, Forrester follows [The Forrester Wave™ Methodology](#) to evaluate participating vendors.

In our review, we conduct primary research to develop a list of vendors to consider for the evaluation. From that initial pool of vendors, we narrow our final list based on the inclusion criteria. We then gather details of product and strategy through a detailed questionnaire, demos/briefings, and customer reference surveys/interviews. We use those inputs, along with the analyst's experience and expertise in the marketplace, to

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score vendors, using a relative rating system that compares each vendor against the others in the evaluation.

We include the Forrester Wave publishing date (quarter and year) clearly in the title of each Forrester Wave report. We evaluated the vendors participating in this Forrester Wave using materials they provided to us by March 2023 and did not allow additional information after that point. We encourage readers to evaluate how the market and vendor offerings change over time.

In accordance with [our vendor review policy](#), Forrester asks vendors to review our findings prior to publishing to check for accuracy. Vendors marked as nonparticipating vendors in the Forrester Wave graphic met our defined inclusion criteria but declined to participate in or contributed only partially to the evaluation. We score these vendors in accordance with [our vendor participation policy](#) and publish their positioning along with those of the participating vendors.

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WAVE REPORT

The Forrester WaveTM: Integrated Software Delivery Platforms, Q2 2023

The 13 Providers That Matter Most And How They
Stack Up

May 31, 2023

By Christopher Condo with Chris Gardner, Sara Sjoblom, Kara Hartig

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Summary

In our 26-criterion evaluation of integrated software delivery platform (ISDP) providers, we identified the most significant ones and researched, analyzed, and scored them. This report shows how each provider measures up and helps modern application development leaders and DevOps platform team leaders select the right one for their needs.

Additional resources are available in the [online version](#) of this report.

ISDPs Provide A Strong Platform Foundation But Still Need Best-Of-Breed Tools

In our reports, [The Future Of DevOps](#) and [The Forrester Guide To DevOps](#), Forrester interviewed dozens of enterprise software development leaders from diverse companies ranging from Netflix to TransUnion to understand the current state of DevOps, its relevance, and next steps enterprise leaders were taking to further advance their practices. In addition, we asked which tools they were using, how effective the tools were, which were using ISDPs, and how well those ISDPs were performing. We learned that many enterprises have shifted to platform teams to support their DevOps efforts and many platform leaders are evaluating and using ISDPs, but few are using all of their ISDP vendor's capabilities. They explained that while these ISDP tools hold great potential for a complete end-to-end toolchain, many individual capabilities fell short of enterprise needs, forcing these leaders to source those capabilities from separate best-of-breed vendors.

Our reference customer survey backs this up by indicating that enterprises are using (on average) only 53% of the capabilities they are paying for, with the remaining capabilities either being delivered by a separate vendor or simply not used. This means that ISDP vendors are not only competing with each other but also competing with the best-of-breeds, which is why the bar has risen so high for this evaluation. ISDP vendors recognize these challenges and are responding with a number of innovative solutions that will make them more attractive than a traditional DIY toolchain but open and extensible enough that end users can add the tools they want without sacrificing capability. The challenge is striking the right balance to create overall value.

As a result of these trends, ISDP customers should look for providers that:

- **Offer a strong core of CI/CD capabilities.** Continuous integration (CI) and continuous delivery (CD) are the core capabilities that the rest of the software delivery toolchain is built around. Several vendors offered build optimization algorithms that slash build and unit test times by intelligently building and testing only the code that changed. Deployment automation is also a differentiating capability, with only a few vendors offering advanced capabilities out of the box (OOTB) such as blue/green, canary, and feature management deployment strategies.
- **Provide connected platforms that best-of-breed tools can augment.** ISDPs provide engineering leverage in the form of preintegrated toolchains that enable work to easily flow from one process to the next, where administration occurs at

the platform level versus the tool level, and reduced context switching improves the developer experience. Often, platform leaders want to override the default capability with third-party tools that they believe offer superior capability, which can lead to a bit of a balancing act. That's why some ISDP vendors are heavily investing in improvements to their end-to-end platform, while others are banking on investments in third-party connectivity and partnerships.

- **Help enterprise leaders shift from DevOps to DevSecOps.** Forrester fields calls every day from enterprise development leaders that are transforming their DevOps practice into a development, security, and operations (DevSecOps) practice to make security a first-class concern for the entire development organization. ISDP vendors are responding with capabilities that better secure software delivery pipelines, including integrated secrets management, signed pipelines, and signed artifacts. Reference customers noted that not all of these tools are direct replacements for best-of-breed (best-of-breed security tools often having a significant advantage over bundled ones), but end users like them as forms of additional assurance and coverage.

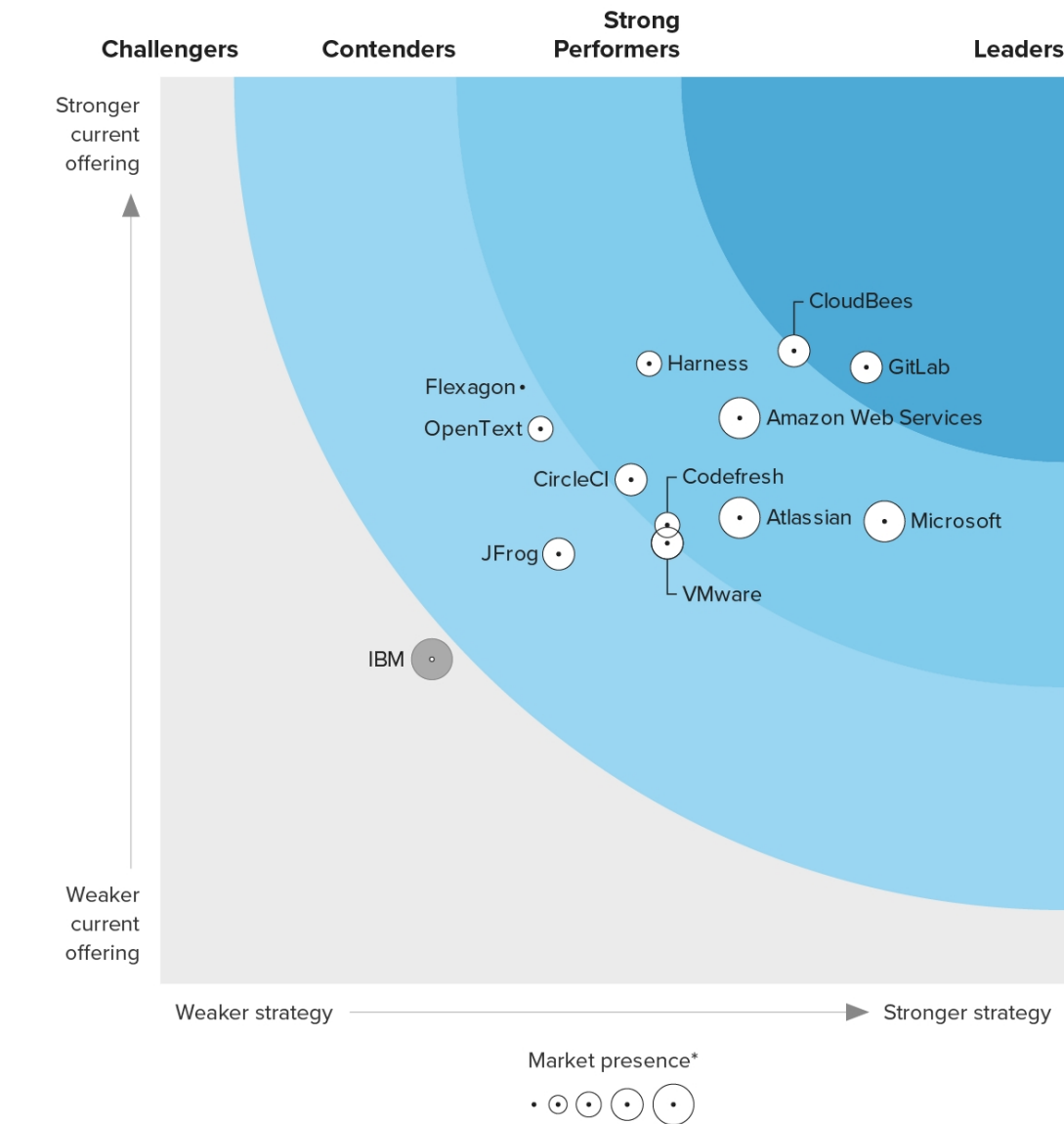
Evaluation Summary

The Forrester Wave™ evaluation highlights Leaders, Strong Performers, Contenders, and Challengers. It's an assessment of the top vendors in the market; it doesn't represent the entire vendor landscape. You'll find more information about this market in Forrester's [Now Tech: Integrated Software Delivery Platforms, Q2 2022](#).

We intend this evaluation to be a starting point only and encourage clients to view product evaluations and adapt criteria weightings using the Excel-based vendor comparison tool (see Figures 1 and 2). Click the link at the beginning of this report on [Forrester.com](#) to download the tool.

Figure 1
Forrester Wave™: Integrated Software Delivery Platforms, Q2 2023

THE FORRESTER WAVE™
Integrated Software Delivery Platforms
Q2 2023



*A gray bubble or open dot indicates a nonparticipating vendor.

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Figure 2**Forrester Wave™: Integrated Software Delivery Platforms Scorecard, Q2 2023**

	Forrester's weighting	Amazon Web Services	Atlassian	CircleCI	CloudBees	Codefresh	Flexagon	GitLab
Current offering	50%	2.57	3.12	2.78	3.49	2.53	3.29	3.40
Platform capabilities	20%	3.50	3.00	2.00	2.50	3.00	2.50	3.50
CI/CD capabilities	25%	2.00	2.60	3.00	4.00	4.00	3.20	3.40
Development, security, and operations	10%	1.00	3.00	2.00	2.00	2.00	3.00	4.00
Extensibility	10%	3.00	3.40	3.00	3.40	1.00	3.00	3.00
Governance and compliance	10%	3.00	5.00	3.00	5.00	1.00	3.00	3.00
Infrastructure management	10%	2.20	3.80	3.80	5.00	1.80	3.40	3.00
Product support	10%	3.00	3.00	3.00	3.00	3.00	5.00	3.00
Support for test automation	5%	3.00	1.00	3.00	3.00	1.00	5.00	5.00
Strategy	50%	3.20	3.20	2.60	3.50	2.80	2.00	3.90
Vision	20%	3.00	3.00	1.00	3.00	3.00	3.00	3.00
Roadmap	20%	3.00	3.00	3.00	3.00	3.00	3.00	5.00
Partner ecosystem	10%	5.00	5.00	1.00	5.00	1.00	1.00	3.00
Innovation	25%	3.00	3.00	3.00	3.00	3.00	1.00	3.00
Community	15%	3.00	3.00	3.00	5.00	3.00	1.00	5.00
Pricing flexibility and transparency	10%	3.00	3.00	5.00	3.00	3.00	3.00	5.00
Market presence	0%	5.00	4.50	3.50	3.50	2.50	1.00	3.50
ISDP revenue	50%	5.00	5.00	4.00	4.00	2.00	1.00	4.00
Number of enterprise customers	50%	5.00	4.00	3.00	3.00	3.00	1.00	3.00

All scores are based on a scale of 0 (weak) to 5 (strong).

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	Forrester's weighting	Harness	IBM*	JFrog	Microsoft	OpenText	VMware
Current offering	50%	3.42	1.79	2.37	2.55	3.06	2.43
Platform capabilities	20%	4.00	1.50	2.00	3.00	2.50	2.00
CI/CD capabilities	25%	4.20	2.80	2.40	2.40	3.40	2.40
Development, security, and operations	10%	3.00	1.00	4.00	4.00	1.00	3.00
Extensibility	10%	1.00	1.40	4.20	4.20	3.40	3.00
Governance and compliance	10%	3.00	3.00	1.00	1.00	5.00	3.00
Infrastructure management	10%	4.20	1.00	1.00	1.80	2.20	1.80
Product support	10%	3.00	1.00	3.00	1.00	3.00	3.00
Support for test automation	5%	3.00	1.00	1.00	3.00	5.00	1.00
Strategy	50%	2.70	1.50	2.20	4.00	2.10	2.80
Vision	20%	3.00	1.00	1.00	3.00	1.00	3.00
Roadmap	20%	3.00	1.00	1.00	3.00	3.00	3.00
Partner ecosystem	10%	3.00	3.00	3.00	5.00	3.00	3.00
Innovation	25%	3.00	1.00	3.00	5.00	3.00	3.00
Community	15%	1.00	3.00	3.00	5.00	1.00	3.00
Pricing flexibility and transparency	10%	3.00	1.00	3.00	3.00	1.00	1.00
Market presence	0%	3.00	4.50	4.00	5.00	2.50	4.00
ISDP revenue	50%	3.00	5.00	4.00	5.00	4.00	4.00
Number of enterprise customers	50%	3.00	4.00	4.00	5.00	1.00	4.00

All scores are based on a scale of 0 (weak) to 5 (strong).

*Indicates a nonparticipating vendor

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Vendor Offerings

Forrester evaluated the offerings listed below (see Figure 3).

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Figure 3
Evaluated Vendors And Product Information

Vendor	Product evaluated
Amazon Web Services	AWS Code Services
Atlassian	Atlassian Open DevOps
CircleCI	CircleCI
CloudBees	CloudBees Platform
Codefresh	Codefresh
Flexagon	Flexagon FlexDeploy
GitLab	GitLab
Harness	Harness Software Delivery Platform
IBM	IBM UrbanCode Deploy
JFrog	JFrog Software Supply Chain Platform
Microsoft	Microsoft Azure DevOps and GitHub Enterprise
OpenText (formerly Micro Focus)	Micro Focus ValueEdge
VMware	VMware Tanzu Application Platform

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Vendor Profiles

Our analysis uncovered the following strengths and weaknesses of individual vendors.

Leaders

- **GitLab leads with a balance between solid platform capabilities and overall strategy.** GitLab is known for being the vendor that made end-to-end OOTB toolchains hip again and has led the industry toward consolidated ISDPs. GitLab’s strategy includes an on-par vision to deliver an excellent developer experience without sacrificing security or compliance. Its innovation is also good, going beyond traditional developers to include AI/ML engineering. GitLab is an open core product that not only invests heavily in the open source software (OSS) community but also enables its customers to contribute to the product, earning it

high scores for community. Pricing is completely transparent and includes a free tier. Its roadmap gets leading scores and includes enhanced supply chain security, enhanced UI, granular security and compliance controls, and pipeline security — all things enterprises need.

GitLab's platform cohesion received high scores for simplicity of setup and operation as both an on-premises product and a software-as-a-service (SaaS) product. Though GitLab comes with its own GitLab version control system (VCS), it supports complete integration with all major Git providers, earning it high scores for version control. GitLab gets high scores for DevSecOps by providing OOTB security tools for static application security testing (SAST), dynamic application security testing (DAST), and Secret Detection. GitLab is one of the few ISDP vendors to incorporate a test automation dashboard, earning a top score. Dev experience is on par, as are administration and management, deployment automation, and release readiness. Build automation, governance and compliance, and infrastructure as code (IaC) were on par with others. One reference customer commented, "The CI/CD experience using secrets, environments, runners, and SAST/DAST/license scans/etc. is unparalleled in my opinion." GitLab is great for enterprises wishing to consolidate their best-of-breed toolchain into one high-performing ISDP.

Strong Performers

- **CloudBees delivers on its core promise of CI/CD, but platform cohesion needs work.** CloudBees is primarily associated with Jenkins; this well-known open source CI tool that it donated to the Continuous Delivery Foundation is one of the original best-of-breed tools that accompanied the beginning of the DevOps era more than 10 years ago. CloudBees leverages open source Jenkins and augments that with enterprise-scale deployment automation, release readiness, and value stream management (VSM) analytics to provide an enterprise-class ISDP. It got high scores for community, with significant contributions to open source projects and cultivation of its own user community. Its strategy features a solid vision and one of the larger partner ecosystems in this market. Innovation was on par and includes efforts to improve codification of policies. A planned SaaS offering should address its platform cohesion weakness and will help CloudBees serve clients that want a more integrated and maintenance-free ISDP.

CloudBees deployment automation is above par, covering blue/green, canary, and powerful feature management capabilities that are third-party add-ons for most

other platforms. CloudBees release readiness, governance, and compliance are also top-notch with complete integration with popular IT service management (ITSM) tools such as ServiceNow that bring those UIs directly into the ISDP experience. Build automation requires a bit of effort to manage fleets of build runners, and pipeline security needs to go beyond branch security to earn higher scores. When asked what they liked about working with CloudBees, reference customers had good things to say: “The vendor relationship has been a true partnership. It’s listening to our internal customers.” CloudBees is a great fit for enterprise platform teams that value open source and open choice.

- **Microsoft differentiates with innovation, but governance and compliance need attention.** Microsoft acquired GitHub, the industry’s largest open source registry, in 2018. Since that time, Microsoft’s Azure DevOps and GitHub have slowly been moving toward a unified offering. However, both the product vision and the roadmap could have been clearer on how this will shape future offerings, resulting in an on-par score. Innovation strategy is tops because GitHub Copilot has set new standards for the development tools industry and planned investments will enable it to continue this leadership position. Its partner ecosystem also scored high, with Microsoft having one of the most extensive. GitHub also supports numerous open source projects, and its online community is one of the biggest, earning a top score. The roadmap heavily weighted developer productivity but was missing key enterprise needs for governance.

Microsoft’s developer experience gets top scores for GitHub Actions, which turned GitHub into a CI leader almost overnight; Codespaces, a virtual dev environment that requires no setup by developers to get started; and Copilot, the AI-powered coding assistant. It also received top scores for OOTB security tools such as Dependabot. Support for APIs and third-party tools was also top scoring, with numerous ways to incorporate not only software but also actions. Pipeline security and platform cohesion were on par, while administration and management were missing the granular permissions of those scoring higher. Microsoft data and analytics were mostly operational, resulting in a below-par score. Deployment automation was also below par — it lacked blue/green and canary deployment scenarios. Reference customers were mostly positive about Microsoft, noting, “Generally, adoption of standard features is quick and doesn’t require a great deal of time to become productive.” Microsoft is great for organizations looking to boost developer effectiveness with cutting-edge tools.

- **Atlassian gets top scores for cohesiveness, but pipeline security capabilities are lacking.** Atlassian is best known for Jira, one of the most popular agile planning

tools on the market serving 80% of the Fortune 500. Atlassian's product strategy is on par and strives to serve two audiences, teams of developers and platform teams, by building out a data layer to connect the tools together. Its partner ecosystem is top-notch. Innovation is on par and includes platform capabilities and a data lake component to capture and aggregate data for greater analytical insight. The roadmap reflects enterprise concerns with plans to aggregate third-party security data, automatically infuse issue tickets with context, provide greater VSM insights, and expand support plans for DevOps and platform teams.

Atlassian's Confluence, which enables teams to create custom developer onboarding content and a hub for team collaboration, enhances the developer experience. This plus integrated development environment plug-ins give Atlassian a solid on-par score for developer experience. The OOTB build and CI were on par, with reusable build templates and parallel build execution to reduce build times. Deployment automation was below par, as were capabilities to secure pipelines. OOTB security tools are few, earning a below-par score, because Atlassian prefers to leverage its partner ecosystem for those. A reference customer said, "We chose Atlassian for the product features and the maturity of the well-integrated suites of products and plug-ins, which help simplify using the well-designed solution." Atlassian is great fit for software teams and enterprise platform teams that are looking for a standard ISDP approach but want to leverage best-of-breed CD and security components.

- **Harness ISDP delivers on platform cohesion, but community contributions are nascent.** Harness has a SaaS ISDP with core CI/CD modules and cloud cost management capabilities. Its vision is to simplify software development with an end-to-end software delivery platform that offers a modern developer experience and automated governance and guardrails. The strategy has mostly on-par elements such as a growing partner ecosystem and investments in innovation. The only weak area was the smaller-than-average community program — those scoring higher had greater support for internal and external communities. The roadmap includes support for database development, test optimization, end-to-end visibility, and infusion of AI/ML throughout the platform.

Harness gets top scores for platform cohesion, delivering a great OOTB experience for first-time users. It also earned top scores in build automation that includes pipelines as code, reusable pipelines, resource management, support for test automation, and intelligent test automation that can drastically reduce build times. It supports multiple deployment strategies OOTB, including blue/green and

canary. In addition, Harness can pinpoint code that's caused pipeline errors, helping speed remediation. Analytics was also top scoring and offered dev, process, and operational metrics. Customization of the UI and some ISDP components is possible and easy, but the limited scope of those changes resulted in a below-par score. Other areas for improvement include expanding the number of third-party integrations and its footprint in the user and open source community. Customers appreciate Harness' "willingness to engage directly with our development teams." Harness is great for development teams that want to spend less time managing tools and more time developing code.

- **AWS is strong in administration and management, but dev metrics are a miss.**

Amazon Web Services (AWS) pioneered infrastructure as a service (IaaS) and offers nearly every possible form of it. Its ISDP product includes CodeBuild, CodeStar, CloudFormation, and several other SaaS offerings that are integrated on demand when users create their account. The overall strategy score was solid. Innovation scored on par for current delivery and upcoming features for helping developers code faster. The vision was on par, focusing mainly on CodeCatalyst that will enable AWS to make its developer platform more accessible to a broader set of end users with different levels of technical ability. AWS actively supports its user community and open source projects, getting an on-par score for community. The roadmap is also on par and includes the aforementioned CodeCatalyst, a product currently in beta that offers a preassembled and more opinionated ISDP to help customers accelerate project setup and collaborate more fluidly.

AWS shines with excellent administration and management capabilities that enable user permissions to be managed down to the API level. AWS supports all major authentication protocols including emerging standards such as SCIM. AWS gets top scores for governance and compliance that include OOTB compliance templates. Platform cohesion and developer experience are on par. Build automation would have scored higher had it offered test execution optimization. AWS got low scores for analytics and reports; other vendors offered better dashboards with more dev-related metrics. Release readiness also scored low. Reference customer sentiment was mostly positive, saying, "Security is a first-class citizen." AWS is great for companies that want speed, scale, and security hosted in the same environment as their public cloud infrastructure.

- **CircleCI provides outstanding build automation, but its vision does not go beyond CI/CD.** CircleCI was one of the first vendors to offer CI as SaaS. Over time, it added CD and extensibility features to make a more well-rounded ISDP offering. CircleCI's on-par vision to speed time to value while simplifying the ISDP

experience is well articulated and strikes a pragmatic balance between individual developer concerns, the needs of platform teams, and support for enterprises in regulated industries. CircleCI's strategic vision is to remain a best-of-breed CI/CD tool vendor and not extend beyond that into other ISDP areas, resulting in a below-par score. The on-par roadmap aims at enterprise teams and includes features to help platform teams better manage an ISDP at scale with pervasive validation at each step of software delivery.

CircleCI has invested in optimized build processes, giving it a top score for build automation and CI. Its support for a wide range of platforms, including all public clouds and even mainframe, earns above-par scores. IaC capabilities were also top scoring. Product support leverages dedicated customer success engineers for each account, but average reference customer scores kept it on par. CircleCI user identities are derived from third-party VCS identities (e.g., those managed by GitHub or Bitbucket) and thus received a low score for administration and management. However, a new identity management system is on the roadmap to address this shortcoming. CircleCI has a good set of OOTB metrics and analytics, including DORA metrics, but the dashboards and capabilities were not as extensive as those scoring higher. One reference customer listed the following capabilities as most appreciated: "parallelization of workflows and rerunning only failed parts of workflows." CircleCI is a great fit for dev teams that value team autonomy and superfast build automation.

- **Codefresh has differentiating CI/CD but needs to improve governance and compliance.** Codefresh provides high-scoring CI/CD capabilities that emphasize the GitOps methodology, which requires all changes to infrastructure to be stored and managed as code in a Git repository prior to deployment. This methodology gained popularity from the Argo CD open source project, which powers Codefresh. Codefresh is also a registered project maintainer and contributor, which factored into its on-par community score. The Codefresh vision and roadmap focus on improvements to the developer experience and deployment capabilities such as Kubernetes context injection, improved disk utilization, and better support for teams. Innovation is confined to Argo and GitOps improvements, earning on-par scores. The one below-par score was for its smaller-than-average partner ecosystem.

Codefresh earned above-par scores for build automation, CI, and deployment automation featuring unique pipeline debugging capabilities that enable users to step in and out of containerized build steps, which no other vendor demonstrated.

Product support was strong, earning on-par scores. IaC provides strong support for Kubernetes configurations, but support for other legacy platforms lacks, keeping this score at par. Developer experience is on par; vendors scoring higher had integrated knowledge management. Other on-par scores include pipeline security, administration and management, and analytics that incorporate OOTB DORA metrics. Codefresh earned below-par scores for governance and compliance (providing limited support for ITSM integration) and support for functional test automation. When asked what was best about Codefresh, one customer said, “easy, standardized deployments for a hybrid universe.” Codefresh is a great fit for organizations that have adopted GitOps and place high value on developer autonomy.

Contenders

- **Flexagon has excellent customer support, but developer experience was below par.** Flexagon builds an end-to-end ISDP that supports traditional forms of software and infrastructure (e.g., Java and Linux) as well as business platforms such as Salesforce, Oracle, and SAP. The product vision is clear and focuses on delivering enterprise value through security and lowering risk. Planned updates include managed Oracle Cloud services, SAP blueprints, and a compliance and policy engine. Innovation is below par, investment in leading-edge areas is lacking, and the partner ecosystem was small, compared with others in this evaluation. Flexagon pricing features multiple tiers and full transparency of pricing options. Flexagon got above-par scores for test automation because it was one of the few vendors that featured native functional test automation capabilities and support.

Flexagon offers good platform cohesion with its offering that includes everything from agile planning to release automation in one package. It would have scored higher if not for some complexity in setting up and operating fleets of build runners. Flexagon got high scores for version control, with its support for the largest number of VCSes including Git and non-Git systems, and environment management that enabled teams to easily operate and manage any number of environments to support engineering teams. Deployment automation, release readiness, and platform support were also solid. Dev experience was not a focus area and resulted in a low score for that capability. Reference customers gave Flexagon perfect high scores for customer support, with one stating, “Flexagon continues to knock our socks off.” Flexagon is a great fit for enterprises looking for DevOps capabilities that span both open systems and business systems.

- **VMware delivers strong pipeline security but is light on data and analytics.**

VMware is best known for its infrastructure virtualization software. Tanzu Application Platform is VMware's ISDP and has a vision that is broad and incorporates elements of sociological factors to help developers be happier and more productive. VMware scored on par for community with strong support for open source tools and its user community. Innovation includes investments in autoremediation and AI/ML, earning an on-par score. The on-par roadmap will help users simplify the product lifecycle, improve developer collaboration, provide cost governance, and have more secure supply chains.

VMware pipeline security is one of the best in the evaluation with signed pipelines, signed commits, and additional security capabilities throughout the pipeline. Governance and compliance were on par, with OOTB compliance policies. Platform cohesion was also good with minimal setup required. Developer experience included a built-in knowledge management system to help onboard new developers and scored on par. Deployment automation included blue/green and canary deployment strategies, earning an on-par score. Administration and management were below par because they were missing features of those scoring higher. Data and analytics were few, and platform support seemed limited to Kubernetes workloads. One reference customer remarked, "The vendor is always willing to accept and act on feedback." VMware is a great fit for enterprises operating on Kubernetes and places high value on pipeline security.

- **OpenText has top-rated platform support, but DevSecOps capabilities score**

low. OpenText is best known for its portfolio of software development tools ranging from portfolio to test automation. OpenText's ValueEdge is an open platform that bundles these capabilities together in one ISDP. Its roadmap is on par, featuring elements that will better connect business and development. Innovation features AI/ML infused in risk management and testing, and OpenText boasts a healthy partner ecosystem. Both areas scored on par, with good evidence of progress from its AI/ML investments. But its vision focused more on business concerns than stronger CI/CD capabilities. In addition, pricing is opaque, compared with those scoring higher.

OpenText has one of the strongest governance and compliance offerings of this evaluation because nearly everything connects to change management systems. It has superior support for test automation capabilities that include autoclassification of test failures. In addition to providing its own VCS, users can connect with virtually any other VCS, earning above-par scores for that capability. Support for a

multitude of platforms including mainframe was a plus. Administration, management, and developer experience were on par, as were data and analytics that featured DORA and flow metrics. Deployment automation was also on par with blue/green and canary deployment support. Pipeline security lagged others evaluated. IaC, an area of growing importance to developers, also scored low. OpenText reference customers were positive about their experience, with one stating, “We don’t have any challenges at this stage.” OpenText is a great fit for enterprises with varied platforms in need of strong ties to test automation and compliance.

- **JFrog provides differentiated pipeline security capabilities, but the core ISDP falls short.** JFrog is best known for its flagship product, Artifactory, a binary artifact repository with a name that is so well known that it is synonymous with artifact repositories in general. JFrog’s strategy includes strong on-par scores for innovation based on investments in helping secure enterprise software supply chains. JFrog has a strong user community and actively participates in open source projects. Pricing flexibility was also on par, offering several payment tiers. JFrog scored below par for vision because it seemed overly focused on security at the expense of other ISDP capabilities. The roadmap was also below par due to its limited scope.

JFrog has superior version control due to the strength of Artifactory. Pipeline security is also top-notch with signed artifacts and signed pipelines that provide some of the strongest supply chain security measures available. The JFrog platform has good cohesion and requires few steps to set up and get started. Administration and management are on par and offer LDAP, OAuth, and support for SCIM to manage users at scale. Xray, a security product that scans artifacts, comes bundled and offers container and image scanning along with OSS supply chain management. To make this a more competitive ISDP offering, JFrog needs to invest in release readiness, data, and analytics and expand policy as code to all aspects of the ISDP while doing more to support test automation. Customers like how they are treated by JFrog, saying: It’s a “partner, not a vendor — they bring this mentality to everything.” JFrog is great for enterprises that place high value on supply chain security.

Challengers

- **IBM UrbanCode Deploy offers solid build and deployment capabilities but looks dated.** IBM UrbanCode offers UrbanCode Build and UrbanCode Deploy as two components that work together to deliver an enterprise-scale ISDP. IBM has a

solid development community because IBM Garage has thousands of contributors. IBM's partner ecosystem is also solidly on par with a number of top systems integrators listed as partners. The strategy has several weak elements including a roadmap that emphasizes maintenance versus new features, and innovation proof points are lacking.

IBM has solid build automation, deployment automation, release readiness, and support for a large variety of target platforms, which are core ISDP capabilities valued by enterprise dev teams. IBM's UrbanCode Velocity module offers strong analytics and graphics, earning it on-par scores for analytics, and there's a healthy number of third-party integrations for an on-par score in that category. But platform cohesion suffers from a complicated installation process, administration and management offer limited capabilities such as LDAP, and the developer experience is bare bones. Pipelines are present but don't offer the same level of capability as more modern tools. IBM is a good fit for enterprises that need to build an ISDP bridge between legacy and modern infrastructure. IBM declined to participate in the full Forrester Wave evaluation process.

Evaluation Overview

We grouped our evaluation criteria into three high-level categories:

- **Current offering.** Each vendor's position on the vertical axis of the Forrester Wave graphic indicates the strength of its current offering. Key criteria for these solutions include platform capabilities to facilitate management at scale, CI/CD to perform core ISDP functions, extensibility to enable platform operators to incorporate third-party tools as needed, and integrated systems to secure the software delivery pipeline.
- **Strategy.** Placement on the horizontal axis indicates the strength of the vendors' strategies. We evaluated vision, roadmap, and innovation to determine which vendor was placing the best bets to fulfill long-term product promises.
- **Market presence.** Represented by the size of the markers on the graphic, our market presence scores reflect each vendor's ISDP revenue and number of enterprise customers.

Vendor Inclusion Criteria

Each of the vendors we included in this assessment has:

- **Integrated CI/CD (e.g., deployment and release automation) capabilities.** The vendor provides integrated CI/CD capabilities OOTB that enable teams to rapidly establish build and deploy processes with minimal configuration.
- **Extensibility via APIs or plug-ins.** The vendor provides APIs to support third-party tool integration to extend the platform to incorporate performance, test automation, and security.
- **A system for centralized administration.** A single dashboard helps platform administrators operate and maintain an ISDP at enterprise scale.
- **Strong client interest.** Analyst inquiry logs indicate the vendor is top of mind among Forrester's end-user clientele with regard to ISDPs.
- **General-purpose programming languages and platforms.** The vendor supports build and deployment automation for open source programming languages like C# and Java as well as deployment to general-purpose runtime platforms such as Linux or Windows. It is also not exclusively tied to any particular business platform such as Salesforce or SAP.
- **High market performance.** The vendor's market performance demonstrates that it is a significant player in the ISDP market.

Supplemental Material

Online Resource

We publish all our Forrester Wave scores and weightings in an Excel file that provides detailed product evaluations and customizable rankings; download this tool by clicking the link at the beginning of this report on Forrester.com. We intend these scores and default weightings to serve only as a starting point and encourage readers to adapt the weightings to fit their individual needs.

The Forrester Wave Methodology

A Forrester Wave is a guide for buyers considering their purchasing options in a technology marketplace. To offer an equitable process for all participants, Forrester follows [The Forrester Wave™ Methodology](#) to evaluate participating vendors.

In our review, we conduct primary research to develop a list of vendors to consider for the evaluation. From that initial pool of vendors, we narrow our final list based on the inclusion criteria. We then gather details of product and strategy through a detailed questionnaire, demos/briefings, and customer reference surveys/interviews. We use those inputs, along with the analyst's experience and expertise in the marketplace, to

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score vendors, using a relative rating system that compares each vendor against the others in the evaluation.

We include the Forrester Wave publishing date (quarter and year) clearly in the title of each Forrester Wave report. We evaluated the vendors participating in this Forrester Wave using materials they provided to us by March 2023 and did not allow additional information after that point. We encourage readers to evaluate how the market and vendor offerings change over time.

In accordance with [our vendor review policy](#), Forrester asks vendors to review our findings prior to publishing to check for accuracy. Vendors marked as nonparticipating vendors in the Forrester Wave graphic met our defined inclusion criteria but declined to participate in or contributed only partially to the evaluation. We score these vendors in accordance with [our vendor participation policy](#) and publish their positioning along with those of the participating vendors.

Integrity Policy

We conduct all our research, including Forrester Wave evaluations, in accordance with the [integrity policy](#) posted on our website.



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